



EPCglobal Tag Data Translation (TDT) 1.0
Ratified Standard Specification
January 21, 2006

Copyright notice

© 2006, EPCglobal Inc.

All rights reserved. Unauthorized reproduction, modification, and/or use of this document is not permitted. Requests for permission to reproduce should be addressed to epcglobal@epcglobalinc.org.

Disclaimer

EPCglobal Inc.TM is providing this document as a service to interested industries. This document was developed through a consensus process of interested parties. Although efforts have been to assure that the document is correct, reliable, and technically accurate, EPCglobal Inc. makes NO WARRANTY, EXPRESS OR IMPLIED, THAT THIS DOCUMENT IS CORRECT, WILL NOT REQUIRE MODIFICATION AS EXPERIENCE AND TECHNOLOGICAL ADVANCES DICTATE, OR WILL BE SUITABLE FOR ANY PURPOSE OR WORKABLE IN ANY APPLICATION, OR OTHERWISE. Use of this document is with the understanding that EPCglobal Inc. has no liability for any claim to the contrary, or for any damage or loss of any kind or nature.

30 **Table of Contents**

31	Terminology.....	5
32	Status of this document.....	5
33	1. Introduction.....	6
34	1.1. Overview	6
35	1.2. Tag Data Translation Charter	7
36	1.3. Tag Data Translation Concept.....	7
37	1.4. Role within the EPC Network Architecture	9
38	1.5. Tag Data Translation Process.....	12
39	1.6. Expressing different representations of EPC.....	15
40	1.6.1. Patterns (Regular Expressions)	15
41	1.6.2. Grammar (Augmented Backus-Naur Form [ABNF])	15
42	1.6.3. Rules for obtaining additional fields	15
43	1.7. Translation Process Steps	16
44	2. Tag Data Standards	16
45	2.1. Overview	16
46	2.2. Many Schemes, 4 Levels within each scheme and multiple options for each level.....	18
47	3. TDT Markup and Logical Process.....	20
48	3.1. TDT Master Index file.....	20
49	3.2. TDT Markup.....	21
50	3.3. Definition of Formats via Regular Expression Patterns and ABNF Grammar	23
51	3.4. Determination of the inbound representation	23
52	3.5. Specification of the outbound representation	24
53	3.6. Specifying supplied parameter values	24
54	3.7. Validation of values for fields and fields derived via rules.....	26
55	3.8. Restricting and checking decimal ranges for values of fields	26
56	3.9. Restricting and checking character ranges for values of fields	27
57	3.10. Padding of fields.....	28
58	3.11. Compaction and Compression of fields.....	31
59	3.12. Names of fields used within the TDSv1.1r1.27 schemes	31
60	3.13. Rules and Derived Fields.....	32

61	3.14. Core Functions.....	34
62	4. TDT Markup - Elements and Attributes	38
63	4.1. Root Element	38
64	4.1.1. Attributes.....	38
65	4.1.2. Elements.....	38
66	4.2. Scheme Element	38
67	4.2.1. Attributes.....	38
68	4.2.2. Elements	39
69	4.3. Level Element.....	39
70	4.3.1. Attributes.....	39
71	4.3.2. Elements	40
72	4.4. Option Element.....	40
73	4.4.1. Attributes.....	40
74	4.4.2.....	40
75	4.4.3. Elements	41
76	4.5. Field Element.....	41
77	4.6. Attributes	41
78	4.7. Rule Element	42
79	4.7.1. Attributes.....	42
80	5. Translation Process	44
81	5.1. Tag Data Translation Software - Reference Implementation.....	46
82	6. Application Programming Interface	46
83	6.1. Client API.....	47
84	6.2. Maintenance API	48
85	7. TDT Schema Definition.....	49
86	8. TDT Markup V1.0	55
87	8.1. GID-96 General Identifier	55
88	8.2. SGTIN-64 Serialized Global Trade Item Number for 64-bit EPCs	56
89	8.3. SGTIN-96 Serialized Global Trade Item Number for 96-bit EPCs	61
90	8.4. SSCC-64 Serial Shipping Container Code for 64-bit EPCs.....	66
91	8.5. SSCC-96 Serial Shipping Container Code for 96-bit EPCs.....	70
92	8.6. SGLN-64 Serialized Global Location Number for 64-bit EPCs	73
93	8.7. SGLN-96 Serialized Global Location Number for 96-bit EPCs	78

94	8.8. GRAI-64	Global Returnable Asset Identifier for 64-bit EPCs.....	82
95	8.9. GRAI-96	Global Returnable Asset Identifier for 96-bit EPCs.....	87
96	8.10. GIAI-64	Global Individual Asset Identifier for 64-bit EPCs	91
97	8.11. GIAI-96	Global Individual Asset Identifier for 96-bit EPCs	94
98	8.12. USDOD-64	DEPARTMENT OF DEFENCE for 64-bit EPCs.....	98
99	8.13. USDOD-96	DEPARTMENT OF DEFENCE for 96-bit EPCs.....	99
100	9. Glossary (non-normative)		100
101	10. References.....		106
102			

Terminology

Within this specification, the terms SHALL, SHALL NOT, SHOULD, SHOULD NOT, MAY, NEED NOT, CAN, and CANNOT are to be interpreted as specified in Annex G of the ISO/IEC Directives, Part 2, 2001, 4th edition [ISODir2]. When used in this way, these terms will always be shown in ALL CAPS; when these words appear in ordinary typeface they are intended to have their ordinary English meaning.

The **Courier** font is used to indicate the names of XML elements and attributes and names of variable fields within the Tag Data Translation markup.

All sections of this document are normative, except where explicitly noted as non-normative.

Status of this document

This section describes the status of this document at the time of its publication. Other documents may supersede this document. The latest status of this document series is maintained at EPCglobal. This document is the version called the Ratified Standard Specification and was Ratified by the EPCglobal Board of Governors..

Comments on this document should be sent to the EPCglobal Software Action Tag Data Translation Working Group mailing list,
mailto:sag_tagdatatranslation@epclinklist.epcglobalinc.org.

1. Introduction

1.1. Overview

The Electronic Product Code (EPC) is a globally unique identifier that is designed to allow the automatic identification of objects anywhere.

The EPC Tag Data Standards (TDS) specification indicates how legacy coding systems such as the GS1 (formerly EAN.UCC) family of codes (GTIN, GLN, SSCC, GRAI, GIAI) should be embedded within the Electronic Product Code (EPC).

In the future, other industry sectors such as automotive, aviation and defence may call for the embedding of their existing numbering schemes directly within the EPC, especially for applications where they deem that it is impractical to map the original general-purpose General Identifier (GID) scheme of EPC to the corresponding identifier in their legacy coding system via a simple network lookup.

The EPC Tag Data Standards (TDS) specification also describes in terms of human-readable encoding and decoding rules for each coding scheme, how to translate between three representations of the electronic product code (EPC), namely the binary format and two formats of uniform resource identifiers (URI), one for tag-encoding and another for pure identity.

The binary format is used to store the EPC identifier in the memory of the RFID tag. EPC Tag Data Standards v1.1 defines binary formats consisting of either 64 bits or 96 bits. The binary format consists of a header (which indicates the coding scheme and version - usually the first 8 bits, although a 2-bit header is defined for SGTIN-64), a fast filter value (which can be used for distinguishing between different packaging levels), as well as fields indicating the company responsible for the object, the object class and a unique serial number.

The URI (or strictly speaking URN) representations are intended for communicating and storing EPCs in information systems, databases and applications, in order to insulate them from knowledge about the physical nature of the tag, so that although 64 bit tags may differ from 96 bit tags in the choice of literal binary header values and the number of bits allocated to each element or field within the EPC, the URN format does not require the information systems to know about these details; the URN can be just a pure identifier.

The tag-encoding URI provides a 1-1 mapping with the binary number recorded in the physical tag and as such indicates the bit-length of the tag and may also include an additional field (usually 3 bits) which is reserved for fast filtering purposes, e.g. to distinguish between various packaging levels for trade items. The tag-encoding URI is therefore intended for low-level applications which need to write EPCs to tags or physically sort items based on packaging level.

The pure-identity URI format isolates the application software from details of the bit-length of the tags or any fast filtering values, so that tags of different bit-lengths which code for the same unique object will result in an identical pure-identity URI, even though their tag-encoding URIs and binary representations may differ. This means that when a

162 *manufacturer switches from using 64-bit tags to 96-bit tags or longer for tagging a*
163 *particular product, the pure-identity URI representation of the EPC will appear the same*
164 *(except for different serial numbers for different instances of the product).*

165 This EPC Tag Data Translation (TDT) specification is concerned with a machine-
166 readable version of the EPC Tag Data Standards specification. The machine-readable
167 version can be readily used for validating EPC formats as well as translating between the
168 different levels of representation in a consistent way. This specification describes how to
169 interpret the machine-readable version. It contains details of the structure and elements
170 of the machine-readable markup files and provides guidance on how it might be used in
171 automatic translation or validation software, whether standalone or embedded in other
172 systems.

173 **1.2. Tag Data Translation Charter**

174 The three objectives in the charter of the Tag Data Translation working group are:

- 175 • To develop the necessary specifications to express the current TDS encoding and
176 decoding rules in an unambiguous machine-readable format; this will allow any
177 component in the EPC Network technology stack to automatically translate between
178 the binary and tag-encoding URI and pure-identity URI formats of the EPC as
179 appropriate. The motivation is to allow components flexibility in how they receive or
180 transmit EPCs, to reduce potential 'impedance mismatches' at interfaces in the EPC
181 Network technology stack. Reference implementations of software that demonstrate
182 these capabilities will also be developed.
- 183 • To provide documentation of the TDS encodings in such a way that the current prose
184 based documentation can be supplemented by the more structured machine-readable
185 formats.
- 186 • To ensure that automated tag data translation processes can continue to function and
187 also handle additional numbering schemes, which might be embedded within the EPC
188 in the future. By aiming for a future-proof mechanism which allows for smooth
189 upgrading to handle longer tags (e.g. 256 bits) and the incorporation of additional
190 encoding/decoding rules for other coding systems, we expect to substantially reduce
191 the marginal cost of redeveloping and upgrading software as the industry domains
192 covered by the EPC expand in the future. We envisage that data which specifies the
193 new rules for additional coding schemes will be readily available for download in
194 much the same way as current anti-virus software can keep itself up to date by
195 periodically downloading new definition files from an authoritative source.

196 **1.3. Tag Data Translation Concept**

197 The Tag Data Translation process translates one representation of EPC into another
198 representation, within a particular coding scheme. For example, it could translate from
199 the binary format for a GTIN on a 64-bit tag to a pure-identity URI representation of the
200 same identifier, although it could not translate a SSCC into a SGTIN or vice versa.

201 The Tag Data Translation concept is illustrated in Figure 1.

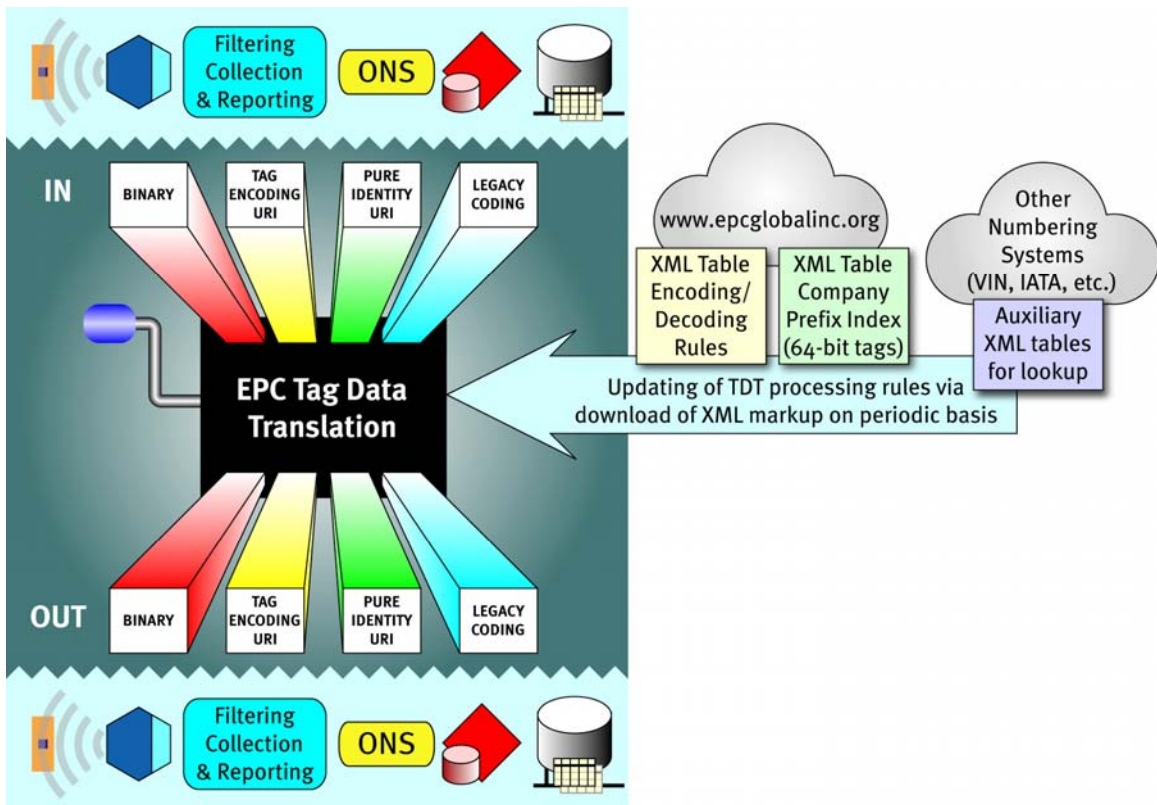


Figure 1 - Tag Data Translation - Concept

Tag Data Translation capabilities may be implemented at any level of the EPC Network stack, from readers, through filtering middleware, as a pre-resolver to the Object Name Service (ONS), as well as by applications and networked databases complying with the EPCIS interface. Tag Data Translation converts between different levels of representation of the EPC and may make use of external tables, such as the Company Prefix Index lookup table for 64-bit tags. It is envisaged that Tag Data Translation software will be able to keep itself up-to-date by periodically checking for and downloading TDT markup files, although a continuous network connection should not be required for performing translations or validations, since the TDT markup files and any auxiliary tables can be cached between periodic checks; in this way a generic translation mechanism can be extensible to further coding schemes or variations for longer tag lengths, which may be introduced in the future.

Although the TDT markup files are made available in XML format, this does not impose a requirement for all levels of the EPC Network technology stack to implement XML parsers, nor to obtain the TDT markup files directly from the authoritative source, such as EPCglobal and possibly other numbering authorities who may develop TDT markup files for other coding schemes / other industry sectors. For example, the manufacturer of an RFID reader may regularly check for and obtain the current TDT markup files, then use data binding software to convert these into hierarchical software data objects, which could be saved more compactly as serialized objects accessible from the particular programming language in which their reader software/firmware is written. The reader

226 *manufacturer could make these serialized objects available for download to owners of*
227 *their products – or bundle them with firmware updates, thus eliminating the need for*
228 *either embedded or real-time parsing of the TDT markup files in their original XML*
229 *format at the reader level.*

230 **1.4. Role within the EPC Network Architecture**

231 In the EPC Network Architecture as depicted in Figure 2 below, the green bars denote
232 interfaces governed by EPCglobal standards, while the blue boxes denote roles played by
233 hardware and/or software components of the system.

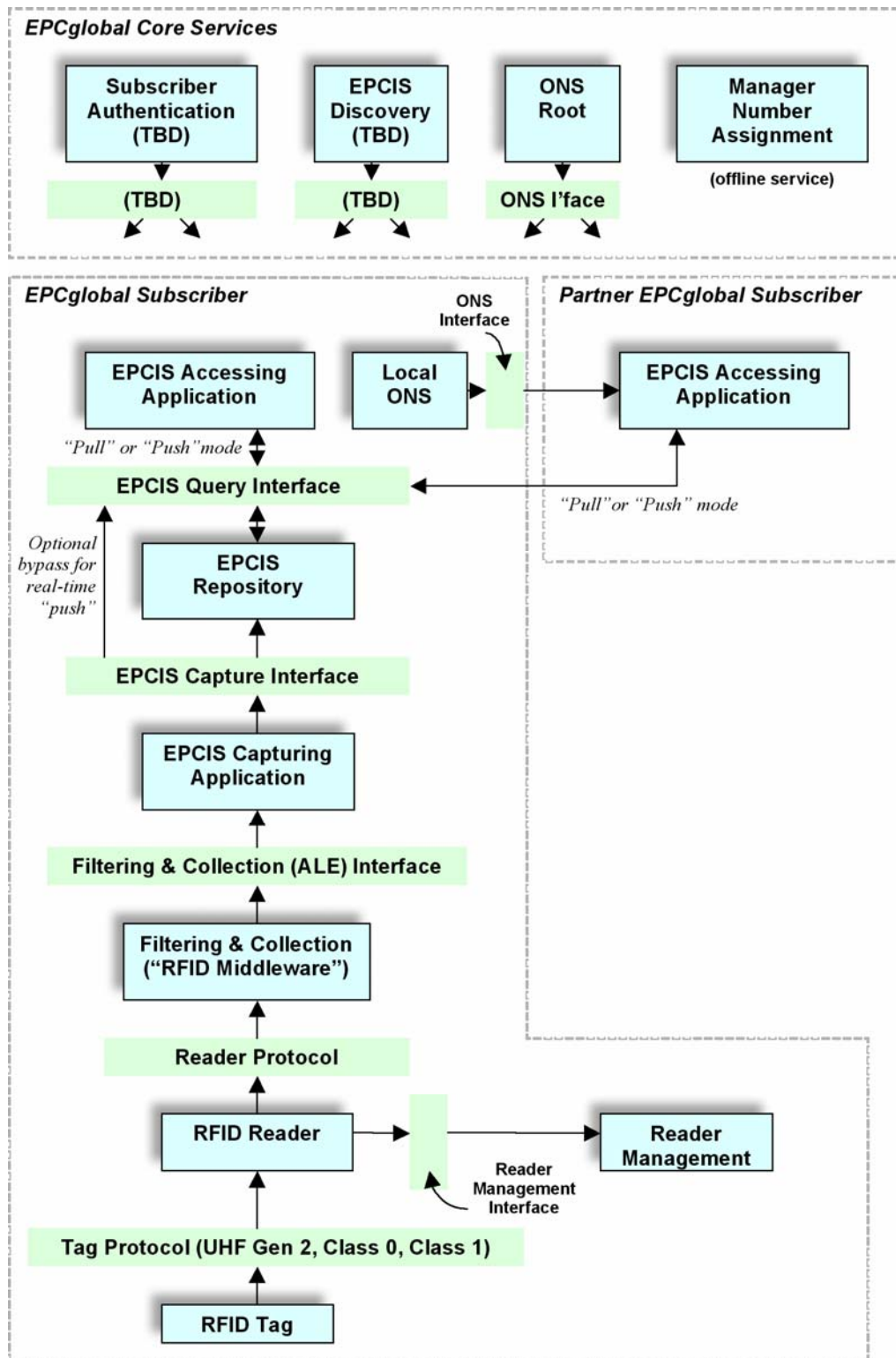


Figure 2 - EPC Network Architecture diagram

237 Table 1 describes the key elements of the EPC Network and the potential usages for the
 238 Tag Data Translation process for encoding and decoding tag data standards.

EPC Network Standards	Description	TDT Role	Potential TDT Usage
Reader Protocol	Defines the control and delivery of raw tag reads from Readers to F&C Middleware	Yes	Conversion upon 'impedance mismatch' of EPC representation
Application Level Events (ALE) Filtering & Collection	API for software that filters and collects raw tag reads, over time intervals delimited by event cycles as defined by applications such as the EPCIS Capturing Application	Yes	Conversion of other EPC representations into URI format for reports Assistance with converting declarative URI patterns into combinations of bit-mask
EPCIS Capturing Application	Software that supervises the operation of the lower EPC network elements and coordinates with enterprise level business events	Yes	Conversion upon 'impedance mismatch' of EPC representation
ONS	ONS is a network service layered over the existing Domain Name System that is used to lookup authoritative pointers to EPCIS-enabled Repositories and other EPC-related information services, given an EPC Manager Number or full Electronic Product Code	No	TDT could output a format which is the hostname for DNS type 35 lookup, in order to perform an ONS query ONS/TDS work groups would need to define this for all coding schemes specified in TDS
EPCIS Service Repository	Networked database or information system providing query/update access to EPC-related data	No	In underlying databases, EPCs might be stored in other formats (e.g. GTIN+serial, separately – or hexadecimal). TDT can convert these to URI formats
EPCIS Enabled Application	Application software responsible for carrying out overall enterprise business processes, such as warehouse management, shipping and receiving	No	Conversion upon 'impedance mismatch' of EPC representation
Trading Partner Application	Trading Partner software that performs the role of an EPCIS Accessing Application.	No	Conversion upon 'impedance mismatch' of EPC representation

239

Table 1 – Potential role for Tag Data Translation throughout the EPC Network

The majority of the EPC Network components require the ability to consistently translate between binary data on tags and URI formats for information systems. However, it should be noted that levels of the stack above the Reader Protocol interface should normally be using the URI representation rather than the binary representation. This also enforces a need for a standard translation mechanism across the entire EPC network so that the translation process and resulting data is consistent and valid.

1.5. Tag Data Translation Process

The fundamental concept of Tag Data Translation is to automatically convert one representation of an EPC (whether binary, tag-encoding URI, pure-identity URI) or a serialized legacy code – and convert it into another representation as required.

This is illustrated in Figure 3

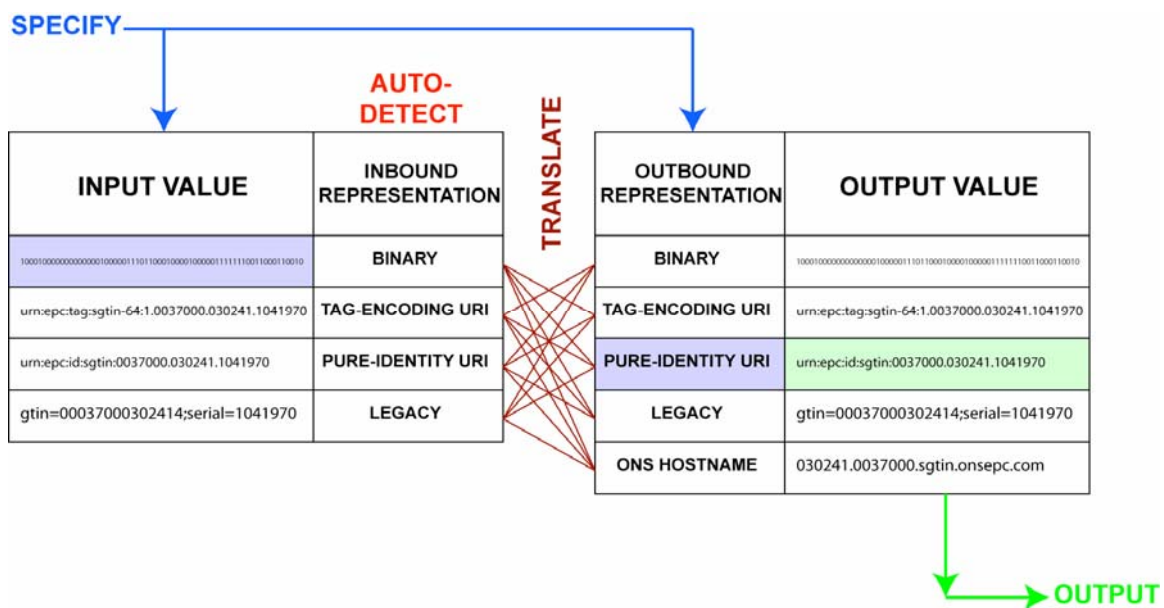


Figure 3 - Tag Data Translation process with examples of different representations.

The Tag Data Translation process takes an input value in a particular representation (binary / tag-encoding URI / pure-identity URI). We refer to the representation in which the input value is expressed as the inbound representation. In the conversion process, the desired outbound representation is also specified by the client requesting the translation. The Tag Data Translation process then returns an output value that is the input value after translation from the inbound representation to the outbound representation.

In practice, some representations contain more information than others. For example, the binary and tag-encoding URI representations also contain information about the number of bits used to store the EPC identifier on a physical RFID tag. They also contain information about a fast filter value, which can be used to discriminate between different packaging levels for trade items.

The serialized legacy codes contain the essential information (company, [object class,] serial number) for an EPC – but often they do not clearly indicate the boundary between the company identifier and the object class identifier – so additional information needs to be supplied, such as the length of the company identifier, from which the boundary can be determined..

This means that as well as providing an input value and a required outbound representation, there are cases where additional parameters need to be supplied. This is illustrated in Figure 4

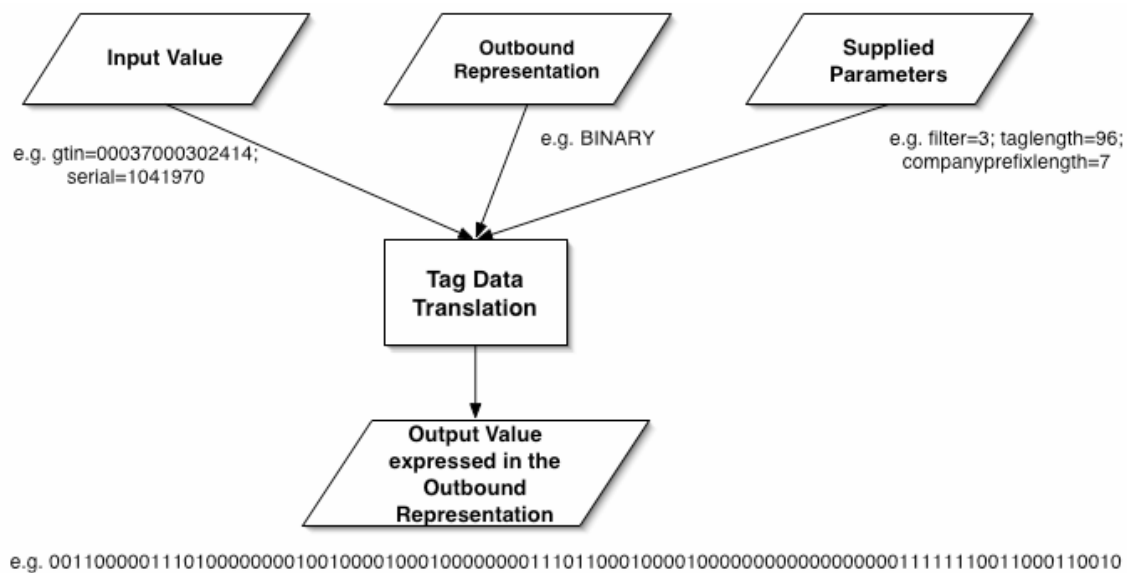
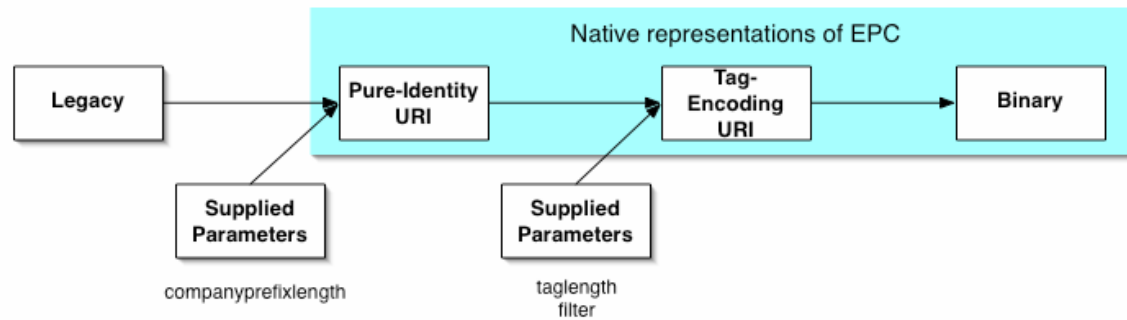


Figure 4 - Flowchart showing input and output parameters to a Tag Data Translation process.

In the context of Tag Data Translation, we refer to encoding as any conversion of the format in the direction of the binary representation, whereas decoding is any conversion away from the binary representation. This is illustrated in Figure 5.

ENCODING



DECODING

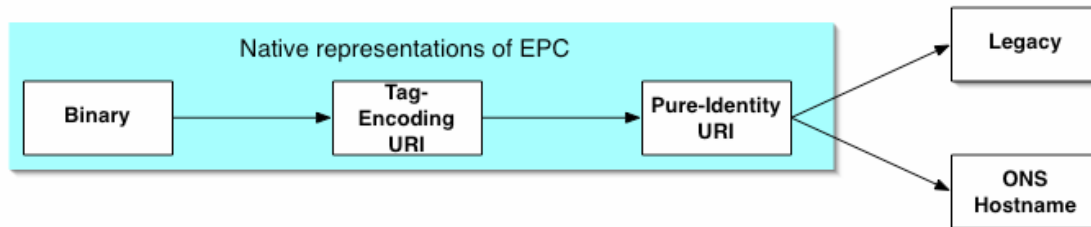


Figure 5 - Encoding and Decoding between different representations of an EPC. Note that when encoding, additional parameters need to be supplied.

In Figure 5 above, there are actually two distinct groups of supplied parameters – those such as `companyprefixlength` which are required for parsing the input value when it is a legacy code – and others such as `filter` and `taglength`, which are required to format the output for certain levels of representation, such as binary or tag-encoding URI. In order to assist Tag Data Translation software in checking that all the required information has been supplied to perform a translation, the `<level>` elements of the Tag Data Translation markup files may contain the attribute `requiredParsingParameters` to indicate which parameters are required for parsing input values from that level and `requiredFormattingParameters` to indicate which parameters are required for formatting the output at that outbound representation level. Further details on these attributes appear in Chapter 4, which describes the TDT markup files.

A desirable feature of a Tag Data Translation process is the ability to automatically detect both the coding scheme and the inbound representation of the input value. This is particularly important when multiple tags are being read – when potentially several different coding schemes could all be used together and read simultaneously.

For example, a shipment arriving on a pallet may consist of a number of cases tagged with SGTIN identifiers and a returnable pallet identified by a GRAI identifier but also carrying an SSCC identifier to identify the shipment as a whole. If a portal reader at a

dock door simply returns a number of binary EPCs, it is helpful to have translation software which can automatically detect which binary values correspond to which coding scheme, rather than requiring that the coding scheme and inbound representation are specified in addition to the input value.

1.6. Expressing different representations of EPC

1.6.1. Patterns (Regular Expressions)

Given an input value, regular expression patterns may be used to match and extract groups of characters, digits or bits from the input value, in order that their values may later be used for constructing the output value in the desired outbound representation, after suitable manipulation, such as binary – decimal conversion, padding etc. We refer to these variable parts as 'fields'. Examples of fields include the Company Prefix (which usually identifies the manufacturer), the Serial Number, Fast Filter value etc.

1.6.2. Grammar (Augmented Backus-Naur Form [ABNF])

An Augmented Backus-Naur Form (ABNF) grammar may be used to express how the output is reassembled from a sequence of literal values such as URN prefixes and fixed binary headers with the variable components, i.e. the values of the various fields. For the grammar attributes of the TDT markup files, in accordance with the ABNF grammar conventions, fixed literal strings SHALL be single-quoted, whereas unquoted strings SHALL indicate that the value of the field named by the unquoted string SHOULD BE inserted in place of the unquoted string.

1.6.3. Rules for obtaining additional fields

However, not all fields that are required for formatting the output value are obtained directly from pattern-matching of the inbound representation. Sometimes additional fields are required to be known. For example, when translating a SGTIN-64 from binary to legacy codes, it will be possible to extract a Company Prefix Index, Item Reference and Serial Number from pattern-matching on the binary input – but the outbound representation needs other fields such as Company Prefix, Check Digit, Indicator Digit, which SHOULD be derived from the fields extracted from the inbound representation. For this reason, the TDT markup files also include sequences of rules, mainly within the legacy codes and binary levels. The rules express how such additional fields may be calculated or obtained via functions operating on fields whose values are already known.

Furthermore, there are some fields that cannot even be derived from fields whose values are already known and which SHALL therefore be specified independently as supplied parameters. For example, when translating a legacy GTIN value together with a serial number into the binary representation, it is necessary to specify independently which length of tag to use (whether 64-bit or 96-bit) and also the fast filter value to be used. Such supplied parameters would be specified in addition to specifying the input value and the desired outbound representation. As illustrated in Figure 5, additional parameters SHOULD be supplied together with the input value when performing encoding. For decoding, it is generally not necessary to supply any additional parameters.

1.7. Translation Process Steps

There are five fundamental steps to a translation:

1. Use of the prefix matches and regular expression patterns to automatically detect the inbound representation and coding scheme of the supplied input value
2. Using the regular expression pattern to extract values of fields from the input value
3. Manipulation, (string manipulation, binary – decimal/alphanumeric conversion, padding etc.) of values of those fields in order to translate from the inbound representation to the outbound representation
4. Using the rules to calculate any additional fields required for the output
5. Using the ABNF grammar to format the required fields in the appropriate output representation

Note that the `prefixMatch` attribute in the TDT markup files is provided to allow optimization of software implementations to perform auto-detection of input representation more efficiently. Where multiple `option` elements are specified within a particular `level` element, each will generally have a `pattern` attribute with a subtly different regular expression as its value. The `prefixMatch` attribute of the enclosing `level` element expresses an initial prefix of these patterns which is common to all of the nested options. Optimized software need not test each nested option for a pattern match if the value of the `prefixMatch` attribute fails to match at the start of the input value. Only for those levels where the `prefixMatch` attribute matches at the start of the string should the patterns of the nested options be considered for matching.

Note that in the TDT markup files, the `prefixMatch` attribute SHALL be expressed as a substring to match at the beginning of the input value. The `prefixMatch` attribute SHOULD NOT be expressed in the TDT markup files as a regular expression value, since a simple string match should suffice. Software implementations MAY convert the `prefixMatch` attribute string value into a regular expression, if preferred, for example by prefixing with a leading caret `['^']` symbol (to require a match at the start of the string) and by escaping certain characters as required, e.g. escaping the dot character as `'\.'` or `'\\.'`

2. Tag Data Standards

2.1. Overview

In the EPC Tag Data Standards specification, the intention is that the Electronic Product Code (EPC[®]) may be stored in binary representation on the physical tag, but that it is communicated within and between information systems in URI format, of which two are defined: the tag-encoding URI which contains all of the same information present in the binary representation of the physical tag and a pure-identity URI to be used by higher-level applications which are not concerned with the nature of the physical tag in which

the EPC was encoded. We therefore have three representations of the Electronic Product Code, namely binary, tag-encoding URI and pure-identity URI.

Furthermore, the EPC Tag Data Standards specification (v1.1) describes how a number of the GS1 (formerly EAN.UCC) coding schemes (GTIN, SSCC, GLN, GRAI and GIAI) should be embedded within the EPC for 64-bit and 96-bit tags. The Electronic Product Code (EPC) is intended to enable unique identification of any object anywhere automatically. While some of the existing coding schemes (SSCC, GRAI and GIAI) are already fully serialised, others (GTIN, GLN) are extended with a serial number for use with the EPC and denoted SGTIN, SGLN.

Although technically the serialised GS1 codes are not themselves a representation of the EPC, they can be encoded into- and decoded from the three representations of EPC, as described in the EPC Tag Data Standards specification – so for this reason we consider four representation levels for a EPC Tag Data Translation process as illustrated in Table 2.

E N C O D E 	5	Hostname for DNS type 35 query in order to perform an ONS lookup	Output-only format	 D E C O D E
	4	Serialised legacy coding (SGTIN,SSCC,SGLN,GRAI,GIAI)	Constrained by specifications of legacy coding schemes Does not express tag length, filter value	
	3	Pure-identity URI format of EPC		
	2	Tag-encoding URI format of EPC	Constrained by number of bits available in physical tag. Expresses tag length, filter value	
	1	Binary representation of EPC		

Table 2 - Four Levels involved in the Translation Process

As Table 2 indicates, the four 'levels' involved in the translation process are not completely equivalent. There is a one-to-one mapping between the pair of levels numbered 1 and 2 (binary and tag-encoding URI) and between the pair of levels numbered 3 and 4 (pure-identity URI and serialized legacy code). The levels 3 and 4 lack the information present in levels 1 and 2 about tag length and fast filtering value. This is illustrated in more detail in Figure 6 below. Note that for convenience, it may prove useful to include a fifth 'level' of representation, corresponding to the hostname for which a DNS Type 35 (NAPTR) query should be performed in order to effect an ONS lookup. This is not strictly an equivalent level of representation of EPC, since ONS v1.0 does not currently provide serial-level pointers for all coding schemes. It is therefore an output-only format and not a valid input format for encoding purposes. For this reason, only an ABNF grammar would be defined for formatting the output in the ONS hostname representation – and no regular expression would be defined for parsing the ONS hostname representation as input. i.e. in the TDT markup files, the `pattern` attribute

SHALL always be absent from the `level` element representing the ONS hostname format. This SHALL indicate to translation software that any auto-detection of the inbound representation SHALL NOT consider the ONS hostname representation as a valid input.

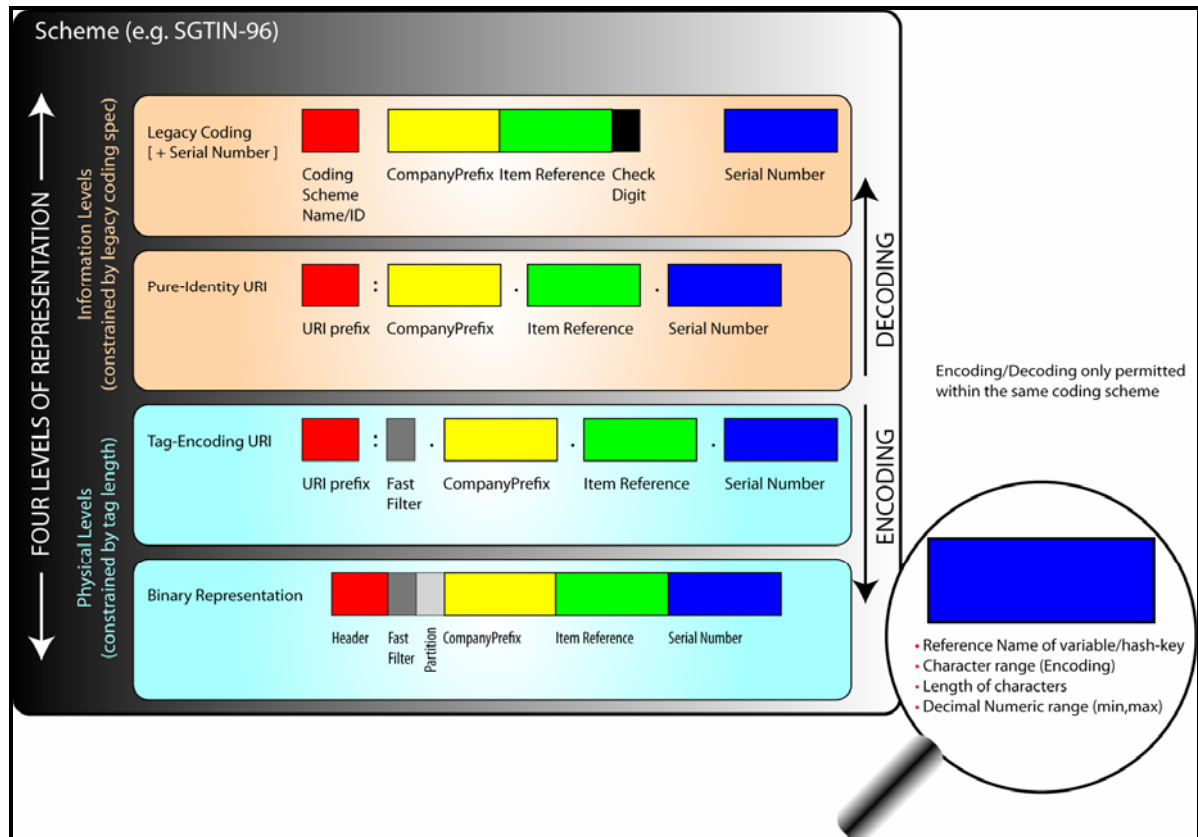


Figure 6 - Comparison of the data elements present in each level of each scheme

2.2. Many Schemes, 4 Levels within each scheme and multiple options for each level

We refer to each coding system (SGTIN, SSCC, SGLN, GRAI, GIAI and GID) as a scheme. For the purposes of Tag Data Translation for use with Generation 1 tags, EPC Tag Data Standards v1.1 defines separately a 64-bit scheme and a 96-bit scheme for each of these, with the exception of the original GID, for which only a 96-bit scheme is defined. Within each scheme, there are the four standard levels of representation as previously described – and for some schemes, a fifth level may also be defined where the structure of the ONS hostname is defined for that scheme. At the time of writing, this has only been defined for SGTIN in the ONS specification.

Furthermore, the GS1 legacy coding schemes use a GS1 (formerly EAN.UCC) Company Prefix of variable length, between 6 and 12 decimal digits. The TDS specification takes two different approaches to handling this in the 64-bit and 96-bit schemes. For the 64-bit schemes, an integer-based Company Prefix Index is encoded into the binary representation, in order to accommodate a larger range of numbers for the Item Reference

and Serial Number partitions. The Company Prefix is obtained from the encoded Company Prefix Index by lookup in a table and it is always the Company Prefix that appears in the URI formats. For the 96-bit schemes, a 3-bit field (the partition value) following the fast filter value within the binary representation is used to indicate the length of the Company Prefix, in the range 6-12 digits, denoted by binary partition values 000 – 110. The bit-length partitions allocated to the Company Prefix and Item Reference fields varies accordingly as described in EPC Tag Data Standards.

One option would be to use a separate lookup table for the partition values as described in the TDS specification. However, since the correspondence between the partition value and the length of the Company Prefix is common to all the GS1 schemes and the partition table is static in nature, we propose a more pragmatic approach and instead embed 7 variants ('Options') of the coding structure within each level, with the appropriate Option being selected either by matching a hard-coded partition value from the inbound data (where this is supplied in binary representation) – or from the length of the Company Prefix (which SHALL be supplied independently if encoding from the legacy coding). This approach also allows the TDT markup files to specify the length and minimum and maximum values for each field, which will often vary, depending on which Option was selected – i.e. depending on the length of the Company Prefix used.

For each option, the representation of the EPC is expressed as both a regular expression pattern to match the inbound representation against, and as an Augmented Backus-Naur Form (ABNF) grammar for formatting the outbound representation.

The regular expression patterns and ABNF grammar are therefore subtly different for each of the options within a particular level – usually in the literal values of the bits for the partition value and lengths of digits or bits for each of the subsequent partitions (where delimiters such as a period '.' separate these partitions) – or in the case of the legacy codes and binary representation, the way in which groups of digits or bits are grouped within the regular expression pattern. This approach facilitates the automatic detection of the boundary between company prefix and item reference simply by regular expression pattern matching.

Within each option, the various fields matched using the regular expression are specified, together with any constraints which may apply to them (e.g. maximum and minimum values), as well as information about how they should be properly formatted in both binary and non-binary (i.e. information about the number of characters or bits, when a certain length is required, as well as information about any padding conventions which are to be used (e.g. left-pad with '0' to reach the required length of a particular field). The concept of multiple options within each level of each scheme is illustrated in Figure 7.

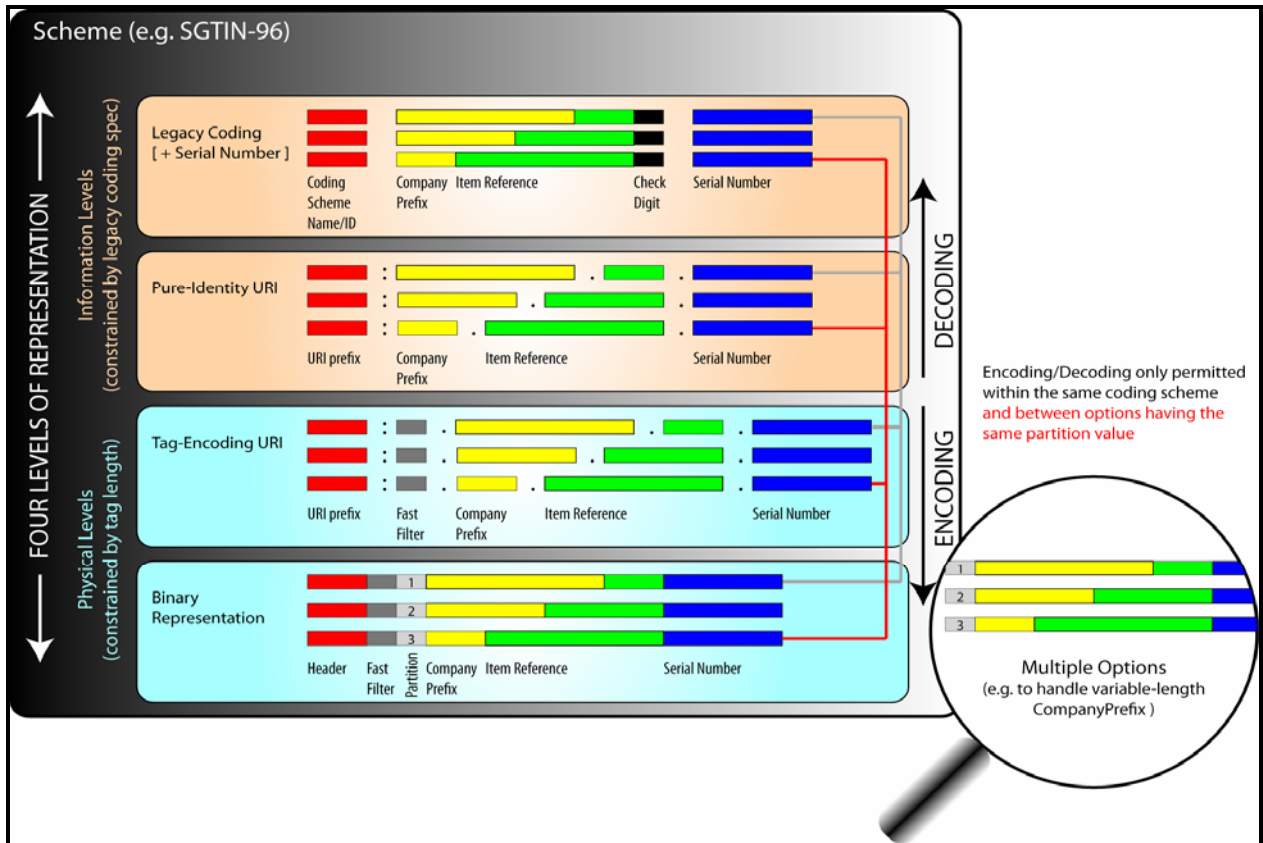


Figure 7 - Depiction of multiple options within each level to handle variable-length Company Prefixes.

3. TDT Markup and Logical Process

The key element of the above architecture is the collection of TDT markup files, which enables encoding and decoding between various levels of representation for each particular coding scheme. This generic design requires open and highly flexible representation of rules for translation software to encode/decode based on the input value. The TDT markup language is a machine-readable XML format expressing the encoding/decoding and validation rules for various identifiers / coding schemes defined in the TDS specification. The TDT markup SHALL be created and maintained by EPCglobal for all the identities defined by the EPC Tag Data Standard specification.

This chapter provides a descriptive explanation of how to interpret the TDT Markup files in the context of a Tag Data Translation process. Chapter 4 provides a formal explanation of the elements and attributes of the TDT markup files.

3.1. TDT Master Index file

It is envisaged that separate TDT markup files or instance documents will be maintained for each coding scheme (i.e. separate files for SGTIN-64, SGTIN-96, SSCC-64, SSCC-96, GID-96, etc.) and referenced from a main index document also in XML, which lists all supported coding schemes. Version control could be achieved via version numbers,

493 timestamps of updates and digests (such as MD5 digests) for each of the per-scheme
494 instance documents referenced from the index document. Furthermore, the relevant
495 numbering authority (such as EPCglobal) should digitally sign the relevant sections of the
496 index document for the coding schemes over which they have authority. In this way, the
497 integrity of the encoding/decoding instructions would be assured to have been thoroughly
498 checked by that numbering authority and to be authoritative.

499 **3.2. TDT Markup**

500 The key elements of the TDT markup are shown in Figure 8.

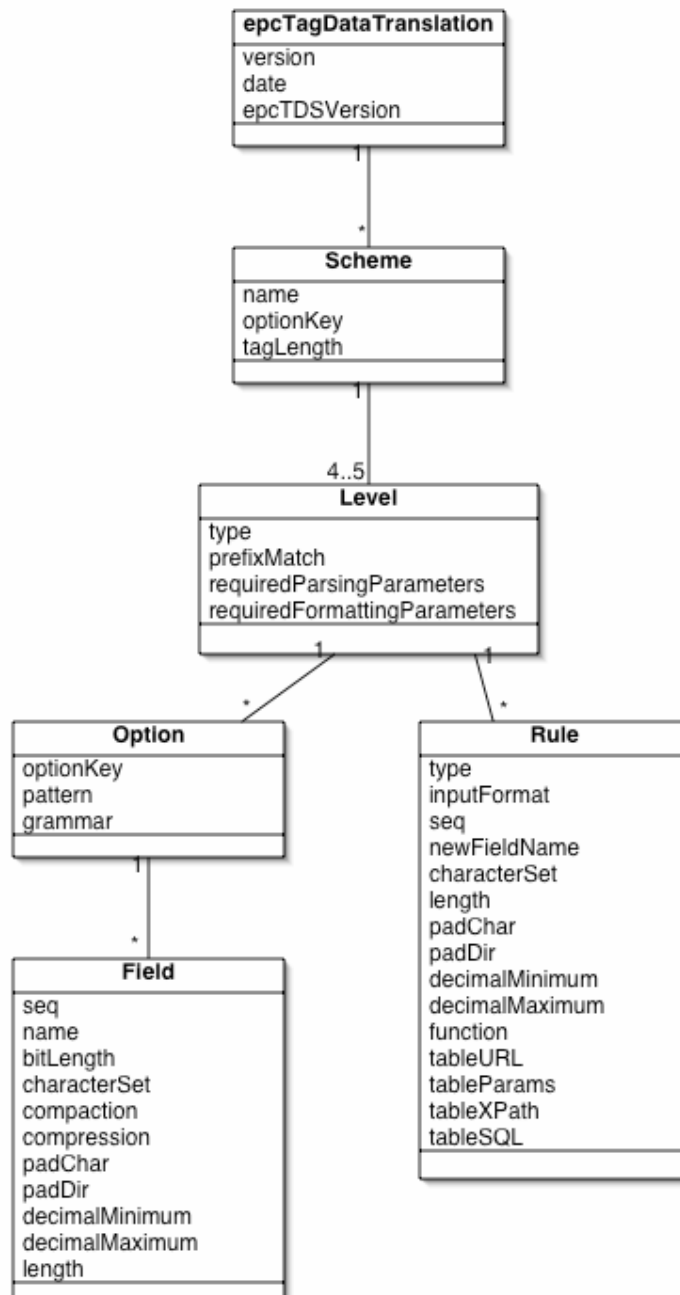


Figure 8 - Tag Data Translation Markup Language schema as a UML diagram

3.3. Definition of Formats via Regular Expression Patterns and ABNF Grammar

The TDT specification uses regular expression patterns and Augmented Backus-Naur Form (ABNF) grammar expressions to express the structure of the EPC in various levels of representation.

The regular expression patterns are primarily intended to be used to match the input value and extract values of particular fields via groups of bits, digits and characters which are indicated within the conventional round bracket parentheses used in regular expressions.

The regular expression patterns provided in the TDT markup files SHALL be written according to the Perl-Compliant Regular Expressions, with support for zero-length negative lookahead.

It is not sufficient to use the XSD regexp type as documented at <http://www.w3.org/TR/xmlschema-2/> because it is sometimes useful to be able to use a negative lookahead '?' construct within the regular expressions. The implementations of regular expressions in Perl, Java, C#, .NET all allow for negative lookahead.

The ABNF grammar form allows us to express the outbound string as a concatenation of fixed literal values and fields whose values are variables determined during the translation process. In the ABNF grammar, the fixed literal values are enclosed in single quotes, while the names of the variable elements are unquoted, indicating that their values should be substituted for the names at this position in the grammar. All elements of the grammar are separated by space characters. We use the Augmented Backus-Naur Form (ABNF) for the grammar rather than simple Backus-Naur Form (BNF) in order to improve readability because the latter requires the use of angle brackets around the names of variable fields, which would need to be escaped to < and > respectively for use in an XML document.

The child 'Field' elements within each option allow the constraints and formatting conventions for each individual field to be specified unambiguously, for the purposes of error-checking and validation of EPCs.

The use of regular expression patterns, ABNF grammar and separate nested (child) field elements with attributes for each of the fields allows for the constraints (minimum, maximum values, character set, required field length etc.) to be specified independently for each field, providing flexibility in the URI formats, so that for example an alphanumeric serial number field could co-exist alongside a decimal Company Prefix field, as would be required to support the full range of possible GRAI codes for a future tag with a larger number of bits devoted to the EPC identifier.

3.4. Determination of the inbound representation

A desirable feature of any Tag Data Translation software is the ability to automatically detect the format of the inbound string received, whether in binary, tag-encoding URI, pure-identity URI or legacy coding. Furthermore, the coding scheme should also be detected. The tag-length SHALL either be determined from the input value (i.e. given a

binary string or tag-encoding URI), – or otherwise, where the input value does not indicate a particular tag-length (e.g. pure-identity URI, serialized legacy code), the intended tag-length of the output SHALL be specified additionally via the supplied parameters when the input value is either a pure-identity URI or a serialized legacy code, neither of which specify the tag-length themselves. It is important that this initial matching can be done quickly without having to try matching against all possible patterns for all possible schemes, tag lengths and lengths of the Company Prefix.

For this reason the Tag Data Translation markup files specify a prefix-match for each level of each scheme, which SHALL match from the beginning of the input value. If the prefix-match matches, then the translation software can iterate in further detail through the full regular expression patterns for each of the options to extract parameter values – otherwise it should immediately skip to try the next possible prefix-match to test for a different scheme or different level of representation, without needing to try all the options nested within each of these, since all of the nested regular expression patterns share the same prefix-match.

Note that under UHF Generation 2, the bits stored in the EPC memory bank consist of 16 protocol-control (PC) bits, [of which the last eight bits will normally contain either the 8-bit EPC header or an 8-bit Application Family Identifier (AFI) allocated by ISO], followed by the remainder of the EPC. The Tag Data Standards work group is currently developing a revised specification dealing with Generation 2 and upwards. The consequence for Tag Data Translation is that it may be necessary to specify independently whether input in binary or hexadecimal format was read from a Class 0/Class 1 tag – or whether it was read from a Generation 2 tag. At the time of writing, the draft Reader Protocol specification is only concerned with Class 0/1 Generation 1 tags.

3.5. Specification of the outbound representation

The Tag Data Translation process only permits encoding or decoding between different representations of the same scheme. i.e. it is neither possible nor meaningful to translate a GTIN into an SSCC – but within any given scheme, it is possible to translate between the four levels of representation, namely binary, tag-encoding URI, pure-identity URI and legacy coding.

With this constraint, it should be possible for Tag Data Translation software to perform a conversion so long as the input value and the outbound representation level are specified.

In addition, it would be a trivial matter for Tag Data Translation software to also provide an output format which is the hostname for which a type 35 ('NAPTR') DNS lookup should be made in order to effect an ONS query. Note that this is an output-only representation, as indicated in Table 2.

3.6. Specifying supplied parameter values

Decoding from the binary level through the tag-encoding URI, pure-identity URI and finally to the legacy coding strings only ever involves a potential loss of information. With the exception of the lookup table mapping Company Prefix Index to Company

Prefix for the 64-bit tags, it is not necessary to specify supplied parameters when decoding, since the binary and tag-encoding formats already contain more information than is required for the pure-identity or legacy coding formats.

Encoding often requires additional information to be supplied independently of the inbound string. Examples of additional information include:

- Independent knowledge of the length of the Company Prefix
- Intended length of the physical tag (64-bit, 96-bit ...) to be encoded
- Fast filter values (e.g. to specify the packaging type – item/case/pallet)

It should be possible to provide these supplied parameters to Tag Data Translation software. In all the cases above, this may simply populate an internal key-value lookup table or associative array with parameter values additional to those that are automatically extracted from parsing the inbound string using the matching groups of characters within the appropriate matching regular expression pattern.

Note that for legacy codes such as GTIN and GLN, which are extended with serial numbers for EPC use, the serial number SHALL NOT be passed via the supplied parameters. Instead, the serial number SHALL be passed as part of the input value. In this way, either the GTIN or GLN and the serial number CAN be obtained as the output value because the same grammar is used for both input and output. This is important because the Tag Data Translation Application Programming Interface (API) defined in Chapter 6 of this document provides no direct access to the private values of intermediate variables or fields used within the translation process. Table 3 shows examples of how the input value should be formatted for serialized legacy codes. Note that SSCC, GRAI and GIAI are already intrinsically serialized and should not be appended with `';serial=...'`.

Coding Scheme	Example string format for input of legacy codes
SGTIN	gtin=00037000302414;serial=10419703
SSCC	sscc=000370003024147856
SGLN	gln=0003700030241;serial=1041970
GRAI	grai=00037000302414274877906943
GIAI	giai=00370003024149267890123
GID	generalmanager=5;objectclass=17;serial=23
USDOD	cageordodaac=AB123;serial=3789156

Table 3 Example formats for supplying serialized legacy codes as the input value.

Note that in Tag Data Translation implementations, the values extracted from the inbound EPC representation SHALL always override the values extracted from the

supplied parameters; i.e. the parameter string may specify '`filter=5`' – but if the inbound EPC representation encodes a fast filter value of 3, then the value of 3 shall be used for the output since the value extracted from the input value overrides any values supplied via the supplied parameters.

Although many programming languages support the concept of an associative array as a data type, these are not generally portable across different languages in the way that data types such as integer and string are. For this reason, the associative array of key-value pairs for the supplied parameters SHALL be passed as a string format, using a semicolon `;` as the delimiter between multiple `key=value` pairs. A string in this format can be readily converted into an associative array in most modern programming languages, while remaining portable and language-unspecific.

3.7. Validation of values for fields and fields derived via rules

The `field` element and the `rule` element contain several attributes for validating and ensuring that the values for particular fields fall within valid ranges, both in terms of numeric ranges, as well as lengths of characters, allowed character ranges and the use of padding characters.

TDT markup files use such an explicit markup of the format and constraints of each field in order to provide for a great deal of future extensibility, particularly for encoding alphanumeric characters, already required by the US DOD formats.

3.8. Restricting and checking decimal ranges for values of fields

In some cases, the numeric range which can be expressed using the specified number of bits exceeds the maximum decimal value permitted by the corresponding legacy coding specifications.

For example, the serial number of an SSCC may be up to ten decimal digits – permitting the decimal numbers 1-9,999,999,999. This requires 34 bits to encode in binary. However, 34 bits would allow numbers in the range 0-17,179,869,183, although those between 10,000,000,000 and 17,179,869,183 are deemed not valid for use as the serial reference of an SSCC – and should result in an error if an attempt is made to encode these into an SSCC.

In order to prevent encoding of numbers outside the ranges permitted by the legacy coding specifications, the decimal minimum and decimal maximum limits of each field are indicated via the field attributes `decimalMinimum` and `decimalMaximum`. Where these attributes are omitted, no numeric (minimum,maximum) limits are specified and checking of numeric range NEED NOT be performed by TDT implementations. Otherwise, where numeric values are specified, the software should check that the value of the field lies within the inclusive range, i.e.

$$\text{decimalMinimum} \leq \text{field} \leq \text{decimalMaximum}$$

Values which fall outside of the specified range should throw an exception.

3.9. Restricting and checking character ranges for values of fields

The `characterSet` attribute of the `field` element indicates the allowed range of characters which may be present in that field. The range is expressed using the same square-bracket notation as for character ranges within regular expressions. The asterisk symbol following the closing square bracket indicates that 0 or more characters within this range are required to match the field in its entirety. Implementations may find it useful to add a leading caret (^) and a trailing dollar symbol (\$) to ensure that the `characterSet` matches the entire field. e.g. for `[0-7]*` in the TDT markup, TDT implementations may use `^[0-7]*$` as the regular expression pattern.

For example,

`[01]*` permits only characters '0' and '1'

`[0-7]*` permits only characters '0' thru '7' inclusive

`[0-9]*` permits only characters '0' thru '9' inclusive

`[0-9 A-Z\-]*` permits digits '0' thru '9', the SPACE character (ASCII 32) and upper-case letters 'A' thru 'Z' inclusive and the hyphen character.

The `characterSet` attribute allows checking that all of the characters fall within the permitted range. For example, if a user specifies a serial number for GRAI containing characters that are not wholly numeric, although the character ranges for GRAI-96 and GRAI-64 only permit wholly numeric serial numbers, i.e. characters in the range `[0-9]`, this should result in an error. Note however that an error might not be reported in the situation where a user attempts to encode an alphanumeric GRAI serial code onto a 96-bit tag in the case where the serial code supplied fortuitously happens not to contain any alphabetic characters.

Furthermore, a 256-bit GRAI EPC might exist with two headers – one for wholly numeric serial numbers, the other for alphabetic serial numbers. The presence of the `compaction` attribute SHALL indicate that a particular field is to be interpreted as the binary encoding of a character string; its absence SHALL indicate that the field should be interpreted as an integer value or all-numeric integer string, with leading pad characters if the `padChar` attribute is also present and the integer has fewer digits than the `length` attribute specifies.

Tag Data Translation software SHOULD NOT rely upon particular values of the `characterSet` attribute as an alternative to taking notice of the `compaction` attribute; certain coding schemes, such as the US DOD's CAGE code omit certain characters, such as the letter 'I' in order to reduce confusion with the digit '1', when the CAGE code is communicated in human-readable format – in this case, the `characterSet` attribute may look like `'[0-9 A-HJ-NP-Z]*'`, in which case a naïve search for 'A-Z' in the `characterSet` attribute would fail to match, even though the binary value SHOULD BE converted to a character string because the `compaction` attribute was present.

3.10. Padding of fields

Certain fields within either the binary representation, the URI representations and also the legacy codes require the padding of the value to a particular number of characters, digits or bits, in order to reach a particular length for that field.

The attributes `length`, `padDir` and `padChar` MAY appear within any `field` or `rule` element of the TDT markup files. Within each `field` element, all three SHALL either be present together – or all three SHALL be absent together. Within `rule` elements, there is no requirement for the `padDir` and `padChar` attributes to be present, even if the `length` attribute is specified; functions defined in rules may return a value which does not require further padding – in this case, the `length` attribute may be specified, merely in order to verify that the result is of the correct length of characters.

If the `padChar` attribute is absent, then the field value or new field value obtained via execution of the rule SHALL NOT be padded.

If the `padChar` attribute is present, then the non-binary field value or new non-binary field value obtained via execution of the rule SHALL be padded according to the values of the attributes `length`, `padDir` and `padChar`. At the binary level of representation, each field SHALL first be converted to binary, then padded with zero bits to reach the number of bits indicated by the `bitLength` attribute for that field.

The `padChar` attribute SHALL consist of a single character to be used for padding. Typically this is the '0' digit (ASCII character 48 [30 hex]). Other coding schemes MAY specify the space character (ASCII character 32 [20 hex]) or a different character to use. The character indicated by the `padChar` attribute SHALL only be used for padding the non-binary equivalents of fields. When fields are to be encoded into the binary format, they SHALL always be padded with zero ['0'] bits as required, in order to reach the number of bits indicated by the `bitLength` attribute for that field.

The `padDir` attribute SHALL take a string value of either 'LEFT' or 'RIGHT', indicating whether the pad characters should appear to the left or right of the unpadded value. The `padDir` attribute is 'LEFT' for all coding schemes specified in Tag Data Standards v1.1.

For all levels except the binary level, if a `field` or `rule` element contains a `padChar` attribute, then the field SHALL be padded with repetitions of the character indicated by the `padChar` attribute, in the direction indicated by `padDir` attribute so that the padded value of the field has the length of characters as specified by the `length` attribute. This applies at the validation, parsing, rule execution and formatting stages of the translation process.

For the binary level, each field element SHALL contain a `bitLength` attribute and MAY contain a `length` attribute.

At the binary level, when encoding (formatting) into binary, the field value SHALL first be converted to binary, then padded to the appropriate number of zero ['0'] bits indicated by the `bitLength` attribute.

If the `padChar` attribute is present, then upon decoding (parsing or extracting) a field from the input value, its non-binary equivalent SHALL afterwards be padded with

repetitions of the character indicated by the `padChar` attribute, in the direction indicated by `padDir` attribute so that the padded non-binary value of the field has the length of characters as specified by the `length` attribute.

An example is the GS1 Company Prefix field in a 96-bit EPC – upon conversion from binary to decimal, the `length` attribute indicates the required length, so that the original Company Prefix can be recovered as a numeric string with leading zeros, e.g. '0037000'.

If the `padChar` attribute is absent, then upon decoding a field from the input value, its non-binary equivalent SHALL NOT be padded.

An example is the Company Prefix Index field in a 64-bit EPC – upon conversion from binary to decimal, the absence of the `length` attribute indicates that the binary value should not be padded. Instead, it is converted to an unpadded integer value, which is used as the key for a `TABLELOOKUP` function specified in a `rule` of type 'EXTRACT', in order to look up the corresponding GS1 Company Prefix.

These rules about whether or not to pad are summarized in Table 4

		padChar ABSENT	padChar PRESENT
BINARY LEVEL	Parsing/Extracting From Binary (DECODING)	If compaction attribute is ABSENT, Convert field from binary to integer and DO NOT PAD. Integer should have no leading zeros. If compaction attribute is PRESENT, convert field from binary to character string using appropriate compaction scheme method and DO NOT PAD this further.	If compaction attribute is ABSENT, Convert field from binary to integer THEN PAD with padChar characters in padDir direction to reach a numeric string of length characters. If compaction attribute is PRESENT, convert field from binary to character string using appropriate compaction scheme method THEN PAD with padChar characters in padDir direction to reach a character string of length characters..
	Formatting Output To Binary (ENCODING)	CONVERT field to binary string (using appropriate compaction scheme method if compaction attribute is specified) THEN PAD binary field to the left with 'zero' bits to reach bitLength bits	PAD field with padChar characters in padDir direction to reach a character string of length characters THEN CONVERT field to binary string (using appropriate compaction scheme method if compaction attribute is specified) THEN PAD binary field to the left with 'zero' bits to reach bitLength bits
ALL OTHER LEVELS	Parsing/Extracting AND Formatting Output	DO NOT PAD	PAD with padChar characters in padDir direction to reach a string of length characters.

Table 4 – Summary of rules about whether or not to pad a field

For example, for a 96-bit SGTIN, for the field whose *name*="companyprefix", the *length* attribute is 7, the *padChar* attribute is '0' and the *padDir* attribute is 'LEFT' for the option where *optionKey* = 7. This means that a 24-bit binary value of '000000001001000010001000' read from the tag for the field named companyprefix should be converted to the integer 37000, then padded with the pad character '0' to 7 characters, yielding '0037000' as the numeric string value for this field.

For a SGLN where the length of the companyprefix is 12 digits, the location reference is a string of zero characters length. This may result in URIs which look strange because there is an empty string between two successive delimiters, e.g. '..' in a URL which looks like urn:epc:id:sgln:123456789012..12345

This is however correct – and it is incorrect to render the zero-length field as '0' between the period (.) delimiters because '0' is of length 1 character – not zero characters length as required by the *length* attribute of the appropriate <field> element.

3.11. Compaction and Compression of fields

When strings other than purely numeric strings are to be encoded in the binary level of representation, the *field* element contains two additional attributes, *compaction* and *compression*. Absence of the *compaction* attribute SHALL indicate that the binary value represents an integer or all-numeric string. Presence of the *compaction* attribute SHALL indicate that the binary value represents a character string encoded into binary using a per-character compaction method for economizing on the number of bits required. Allowed values are '5-bit', '6-bit', '7-bit' and '8-bit', referring to the compaction methods described in ISO 15962, in which the most significant 3/2/1/0 bits of the 8-bit ASCII byte for each character are truncated.

Note that a *compaction* value of '8-bit' SHALL be used to indicate that each successive eight bits should be interpreted as an 8-bit ASCII character, even though there is effectively no compaction or per-byte truncation involved, unlike the other values of the *compaction* attribute. The *compaction* values '16-bit' and '32-bit' are not used in the markup files for this version of the TDT specification – but are reserved in the TDT XSD schema and SHALL indicate 16-bit and 32-bit UNICODE representation where this is required in the future.

The *compression* attribute is intended for future use, to indicate a compression technique to be applied to the value as a whole, rather than on a per-character basis. Permitted values for the *compression* attribute are not currently defined in this version of the Tag Data Translation specification but those values defined in future may indicate compression techniques such as zip / gzip compression, Huffman encoding etc.

3.12. Names of fields used within the TDSv1.1r1.27 schemes

The names of fields appearing in the TDT markup files are completely arbitrary but by convention SHALL consist of lower case alphanumeric words with no spaces or hyphens. There are no reserved words and the use of a name within one coding scheme does not

793 imply any correlation with an identically named field within a different coding scheme;
 794 each coding scheme effectively has its own namespace for field names. Table 5 lists
 795 some field names that are used in the coding schemes for EPC Tag Data Standards
 796 v1.1r1.27

filter	fast filter value – decimal range 0-7
serial	serial number – decimal or alphanumeric
companyprefixlength	length of a GS1 company prefix as a number of characters – decimal e.g. company prefix '0037000' → companyprefixlength=7
taglength	64/96/256 etc. – number of bits for the EPC identifier
companyprefixindex	an integer-based lookup key for accessing the real Company Prefix – for use with 64-bit tags
itemref	Identifies the Object Type or SKU within a particular company for a GTIN
locationref	Identifies the Location within a company for a GLN
assetref	A serialised asset reference – for use with the GIAI
serialref	A serialised reference – e.g. for use with the SSCC

797 *Table 5 – Names of fields used within Tag Data Standards v1.1 r1.27*

798 **3.13. Rules and Derived Fields**

799 Certain fields required for formatting the outbound representation are not obtained simply
 800 from pattern matching of the inbound representation. A sequence of rules allows the
 801 additional fields to be derived from fields whose values are already known.

802 The reason why this is necessary is that there is often some manipulation of the legacy
 803 codes required in order to translate them into the pure-identity URI representation.
 804 Examples include string manipulation such as the relocation of the initial indicator digit
 805 or extension digit to the front of the item reference field – or for decoding, the re-
 806 calculation of the GS1 checksum – and appending this as the last digit of the legacy
 807 coding representation. Likewise, replacement of the Company Prefix Index integer by
 808 the corresponding GS1 Company Prefix is something that is not readily expressed simply
 809 via regular expressions. By working through an example for the GTIN, it is clear that
 810 although the processing steps are reversible between encoding into the pure-identity URI
 811 and decoding into the legacy codes, the way in which those steps are defined takes on an
 812 unsymmetrical appearance in the sequence of rules. An example illustrates this point:
 813

814 Decoding the GTIN (i.e. translating from pure-identity URI into legacy
815 coding)
816

```
817 • indicatordigit = SUBSTR(itemref,0,1);  
818 • itemrefremainder = SUBSTR(itemref,1);  
819 • gtinprefix =  
820   CONCAT(indicatordigit,companyprefix,itemrefremainder);  
821 • checkdigit = GS1CHECKSUM(gtinprefix);  
822
```

823 The above are all examples of rules to be executed at the 'EXTRACT' stage, i.e.
824 immediately after parsing the input value.
825

826 Encoding the GTIN (i.e. translating from legacy coding into pure-
827 identity URI)

828 (assumes companyprefixlength is passed as a supplied parameter)
829

```
830 • gtinprefixremainder=SUBSTR(gtin,1,12);  
831 • indicatordigit=SUBSTR(gtin,0,1);  
832 • itemrefremainder=SUBSTR(gtinprefixremainder,companyprefix  
833   length);  
834 • itemref=CONCAT(indicatordigit,itemrefremainder);  
835 • companyprefix=SUBSTR(gtinprefixremainder,0,companyprefixl  
836   ength);  
837
```

838 The above are all examples of rules to be executed at the 'FORMAT' stage, i.e. when
839 constructing the output value.
840

841 As the above examples show, the definitions of particular fields (e.g. itemrefremainder)
842 depends upon whether encoding or decoding is being performed (or equivalently,
843 whether the field is required for formatting the output value – or being extracted from the
844 input value), since each successive definition depends on prior execution of the
845 definitions preceding it, in the correct order, in order that all the required fields are
846 available.

847 The rules in the example above apply generally, with minor modifications to all of the
848 GS1 coding schemes covered in the TDS Specification v1.1r1.27. It is worth noting that

each of the above rule steps contains only one function or operation per step, which means that even a very simple parser can be used, without needing to deal with nesting of functions in parentheses.

3.14. Core Functions

The core functions which SHALL be supported by Tag Data Translation software in order to encode/decode the GS1 coding schemes are described in Table 6.

SUBSTR (string, offset)	the substring starting at <offset> (offset=0 is the first character of string)
SUBSTR (string, offset, length)	the substring starting at <offset> (offset=0 is the first character of string) and of <length> characters
CONCAT (string1, string2, string3,...)	concatenation of string parameters
LENGTH(string)	number of characters of a string
GS1CHECKSUM (string)	Computes the GS1 checksum digit given a string containing all the preceding digits
TABLELOOKUP (inval, tablename, incol, outcol)	Performs a lookup in table called tablename. Given an input value <inval>, look in table <tablename> to find a match in column names <incol> and return the corresponding value for the same row from output column <outcol>. The TABLELOOKUP function only indicates the logical lookup – not any bindings. The table URL is specified via a separate attribute <code>tableURL</code> and bindings to XPath or SQL expressions are specified via separate attributes <code>tableXPath</code> and <code>tableSQL</code>.

Table 6 - Basic built-in functions required to support encoding and decoding within the GS1 schemes currently covered by the TDS specification

In order to make full use of the Tag Data Translation markup files, implementations of translation software should provide equivalent functions in the programming language in which they are written, either by the use of native functions or custom-built methods, functions or subroutines.

In this version of Tag Data Translation, the requirement that implementations should be able to recalculate check digits only applies to the GS1 coding schemes, when output in the legacy format is required. Further details on calculation of the GS1 checksum can be

found at <http://www.gs1.org>. It should be noted that ISO 7064 provides a standard for more general-purpose calculation of check digits and that this may be considered in future versions of this specification.

It is important to note that modern programming languages (including Java, C++, C#, Visual Basic, Perl, Python) do not all share the same convention in the definitions of their native functions, especially for string functions. In some languages the first character of the string has an index 0, whereas in others, the first character has an index 1. Furthermore, many of the languages provide a substring function which takes two additional parameters as well as the string itself. Usually, the first of these is the start index, indicating the starting position where the substring should be extracted. However, some languages (e.g. Java, Python) define the last parameter as the end index, whereas others (C++, VB.Net, Perl) define it as the length of the substring, i.e. number of characters to be extracted. Table 7 indicates a number of language-specific equivalents for the three-parameter SUBSTR function in Table 6.

	SUBSTR(string, offset, length)	Notes
C++	String.substr(offset, length);	
C#	String.Substring(offset, length);	
Perl	substr(\$stringvariable, offset, length);	
Visual Basic	String.Substring(offset, length)	
Java	Java.lang.String String.substring(beginIndex, endIndex)	beginIndex = offset endIndex = offset+length
Python	String[start:end]	start = offset end = offset+length

Table 7 – Comparison of how substring functions are defined in a number of modern programming languages. The parameters offset and length are of integer type.

Note that for the case of rules which use the TABLELOOKUP function, additional attributes `tableURL` and `tableXPath` or `tableSQL` are provided. Tables may be provided in XML format or as comma-separated values (CSV) or tab-separated values (TSV), even though any Tag Data Translation software MAY internally store the table values in a different format altogether. For this reason, the binding to the original format is handled separately via the `tableURL` and `tableParams` and either `tableXPath` or `tableSQL` attributes, while the TABLELOOKUP function expresses the logical lookup, irrespective of the format in which any table is actually supplied.

893 As an example, consider the GS1 Company Prefix Index lookup tables for use with 64-bit
894 tags. An XML version and a comma-separated values (CSV) version are provided at
895 <http://www.onsepc.com>

896

897 For the XML version,
898 `tableURL="http://www.onsepc.com/ManagerTranslation.xml"` and
899 `tableXPath` and `tableParams` are one of the following pairs:

900

901 `tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"`

902 `tableParams="companyprefixindex"`

903 for the case where

904 `function="TABLELOOKUP(companyprefixindex, 'GEPC64Table', comp`
905 `anyprefixindex, companyprefix)"`

906 OR

907 `tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index"`

908 `tableParams="companyprefix"`

909 for the case where

910 `function="TABLELOOKUP(companyprefix, 'GEPC64Table', companypr`
911 `efix, companyprefixindex)"`

912

913 The first example pair is used to obtain the value of `companyprefix` given the value
914 of `index` (e.g. retrieve `companyprefix='0037000'` given
915 `companyprefixindex='1'`).

916 The second example pair is used to obtain the value of `companyprefixindex` given
917 the `company prefix` (e.g. retrieve `companyprefixindex='1'` given that
918 `companyprefix='0037000'`).

919 Note that `tableParams` may be a comma-separated string of either fieldnames (if
920 unquoted) or fixed literal values, if wholly numeric or single-quoted strings. The `$1` in
921 the `tableXPath` expressions indicates that the actual value of the field named by the
922 first parameter of `tableParams` string should be substituted into the `tableXPath`
923 expression at this point before passing the XPath expression to an XML DOM parser.

924 For example, if the value of `companyprefix` is `'0037000'`, then for the second example
925 pair, the value of `'0037000'` would be substituted in place of `'$1'` in `tableXPath` so that
926 it would be the following XPath expression:

927 `"/GEPC64Table/entry[@companyPrefix='0037000']/@index"`

928 which is actually passed to the XML DOM parser.

929

930 Where more than one parameter is listed in `tableParams`, `$2` indicates where to
931 substitute the second parameter, while `$3` indicates where to substitute the third
932 parameter, and so on.

933

934 A table supplied as comma-separated values (CSV) or tab-separated values (TSV), can be
935 readily converted to a relational database table with the appropriate column headings.

936 For the example of the Company Prefix Index table for 64-bit tags, the CSV version is
937 available from <http://www.onsepc.com/ManagerTranslation.csv>

938 In this case, the attribute

939

940 `tableURI= "http://www.onsepc.com/ManagerTranslation.csv"`

941

942 and the attributes `tableSQL` and `tableParams` may be one of the following pairs:

943

944 `tableSQL="SELECT companyPrefix from GEPC64Table WHERE`
945 `index='$1'"`

946 `tableParams="companyprefixindex"`

947 for the case where

948 `function="TABLELOOKUP(companyprefixindex, 'GEPC64Table', comp`
949 `anyprefixindex, companyprefix)"`

950 OR

951 `tableSQL="SELECT index from GEPC64Table WHERE`
952 `companyPrefix='$1'"`

953 `tableParams="companyprefix"`

954 for the case where

955 `function="TABLELOOKUP(companyprefix, 'GEPC64Table', companypr`
956 `efix, companyprefixindex)"`

957

958 Each of the two example pairs above corresponds to the respective pairs in the previous
959 examples for the `tableXPath` attributes. Likewise, the notation `$1`, `$2`, etc.
960 indicates where values of fields named by parameters from the `tableParams` string
961 should be substituted into the `tableSQL` expression before passing to the relational
962 database engine for execution.

4. TDT Markup - Elements and Attributes

4.1. Root Element

The epcTagDataTranslation element is the root element of the TDT definition.

4.1.1. Attributes

Name	Description	Example Values
version	TDT Definition version number	1.00
date	Creation Date	2005-03-07T11:33Z
epcTDSVersion	TDS Specification version	1.1r1.27

4.1.2. Elements

Name	Description
scheme	Please see scheme definition below for more details

4.2. Scheme Element

For every identifier / coding scheme as defined in the TDS specification, the Scheme element provides details of encoding/decoding rules and formats for use by Tag Data Translation software. In this version of the TDT specification, markup files are provided for the following identifiers: SGTIN-64, SGTIN-96, SSCC-64, SSCC-96, GRAI-64, GRAI-96, GIAI-64, GIAI-96, SGLN-64, SGLN-96 and GID-96.

4.2.1. Attributes

Name	Description	Example Values
name	Name of the coding scheme	SGTIN-64, SGTIN-96, SSCC-64, SSCC-96, GRAI-64, GRAI-96, GIAI-64, GIAI-96, SGLN-64, SGLN-96 and GID-96
optionKey	The name of a variable whose value determines which one of multiple options to select	companyprefixlength
tagLength	Tag length	64 or 96

975 4.2.2. Elements

Name	Description
Level	Contains option elements expressing a pattern, grammar and encoding/decoding rules for each level of representation

976 4.3. Level Element

977 This element provides a prefix match for each level of representation. Nested within the
 978 `level` element are `option` elements (which provide the `pattern` regular expressions
 979 for parsing the input into fields and ABNF `grammar` for formatting the output) and
 980 `rule` elements used for obtaining additional fields from functional operations on known
 981 fields.

982 4.3.1. Attributes

Name	Description	Example Values
<code>type</code>	Indicates level of representation	BINARY TAG_ENCODING PURE_IDENTITY LEGACY ONS_HOSTNAME
<code>prefixMatch</code>	Prefix value required for each encoding/decoding level	00001010 uri:epc:tag:sscc-64 uri:epc:id:sscc sscc=
<code>requiredParsingParameters</code>	Comma-delimited string listing names of fields whose values SHALL be specified in the list of <code>suppliedParameters</code> in order to parse the fields of an input value at this level	companyprefixlength
<code>requiredFormattingParameters</code>	Comma-delimited string listing names of fields whose values SHALL be specified in the list of <code>suppliedParameters</code> in order to format the outbound value at this level	filter,taglength

983 4.3.2. Elements

Name	Description
option	Contains patterns and grammar
rule	Contains rules required for determining values of additional variables required

984

985 4.4. Option Element

986 4.4.1. Attributes

Name	Description	Example Values
optionKey	A fixed value which the optionKey attribute of the Scheme element SHALL match if this option is to be considered	any string value but for GS1 legacy codes defined in TDS v1.1, the values '6','7','8','9','10','11','12'
pattern	A regular expression pattern to be used for parsing the input string and extracting the values for variable fields	00101111([01]{4})00100000([01]{40})([01]{36})
grammar	An ABNF grammar indicating how the output can be reassembled from a combination of literal values and substituted variables (fields)	'00101111' filter cageordodaac serial <i>N.B. single quoted string indicate fixed literal strings, unquoted strings indicate substitution of the correspondingly named field values</i>

987 4.4.2.

988 **4.4.3. Elements**

Name	Description
field	Provides information about each of the variables, e.g. (min, max) values, allowed character set, length, padding etc.

989 **4.5. Field Element**

990 **4.6. Attributes**

Name	Description	Example Values
seq	The sequence number for a particular sub-pattern matched from a regular expression – e.g. 1 denotes the first sub-pattern extracted	1, 2, 3...
name	The name of the variable (or field) – just a reference used to ensure that each field may be used to construct the output format	filter, companyprefix, itemref, serial, ...
decimalMinimum	Decimal minimum value allowed for this field	0
decimalMaximum	Decimal maximum value allowed for this field	9999999
length	Required length of this field in string characters.	7
bitLength	Required length of this field in bits. Omitted for the non-binary levels.	24
characterSet	Allowed character set for this field, expressed in regular expression character range notation	[0-9],[01]
padChar	Character to be used to pad to required value of fieldlength. Omitted if no padding is required for the non-binary form. Presence indicates padding is required for the non-binary form.	'0', ''
padDir	Direction to insert pad characters.	'LEFT', 'RIGHT'

991 4.7. Rule Element

992 4.7.1. Attributes

Name	Description	Example Values
type	Indicates at which stage of the process the definition should be evaluated	EXTRACT, FORMAT
inputFormat	Indicates whether the input parameter to the definition is in binary format or non-binary ('string') format	STRING, BINARY
seq	A sequence number to indicate the running order for definitions sharing the same mode value. The definitions should be run in order of ascending 'seq' value	1,2,3,4,5...
newFieldName	A name for the new field or variable whose value is determined by evaluating the definition	Any string consisting of alphanumeric characters and underscore
function	An expression indicating how the new field can be determined from a function of already-known fields	e.g. SUBSTR(itemref,0,1)
decimalMinimum	For numeric fields, the decimal minimum value allowed for this field	e.g. 0
decimalMaximum	For numeric fields, the decimal maximum value allowed for this field	e.g. 9999999
length	Required length of this field in string characters.	7
padChar	Character to be used to pad to required value of fieldlength. Omitted if no padding is required. Present if padding is required.	'0', ''
padDir	Direction to insert pad characters	'LEFT', 'RIGHT'

Name	Description	Example Values
characterSet	Allowed character set for this field, expressed in regular expression character range notation	[0-9],[01]
tableURL	A URL where the data table can be obtained	http://www.onsepc.com/ManagerTranslation.xml
tableXPath	An XPath expression for obtaining a particular attribute or element value from an XML table. The inline notation '\$1', '\$2' etc. indicates where the values of the first, second, etc. elements of the tableParams list should be substituted before passing to an XML parsing engine.	/GEPC64Table/entry[@index='\$1']/@companyPrefix
tableSQL	A SQL expression for obtaining a particular field from a relational database table. The inline notation '\$1', '\$2' etc. indicates where the values of the first, second, etc. elements of the tableParams list should be substituted before passing to a relational database query engine.	SELECT companyPrefix FROM GEPC64Table WHERE index='\$1'
tableParams	A comma-delimited string list of fieldnames whose actual values should be substituted into the tableXPath or tableSQL expressions	e.g. companyprefixindex

5. Translation Process

The execution of the rules in the TDT process takes place at two distinct processing stages, denoted 'FORMAT' and 'EXTRACT', as explained in Table 8:

Stage	Description
EXTRACT	Operates on fields after parsing of the inbound value
FORMAT	Operates on fields in order to prepare additional fields required by the grammar for formatting the output value.

Table 8 – The two stages for processing rules in Tag Data Translation

The rules for each scheme are within the context of a particular level of representation. The first block of rules, 'EXTRACT' are tied to the inbound representation level. The last block of rules, 'FORMAT' is tied to the outbound representation level. Each block may consist of zero or more `rule` elements. The rules within each block are executed in a strict order, as specified by an ascending integer-based sequence number, indicated by the attribute 'seq' of the `rule` element.

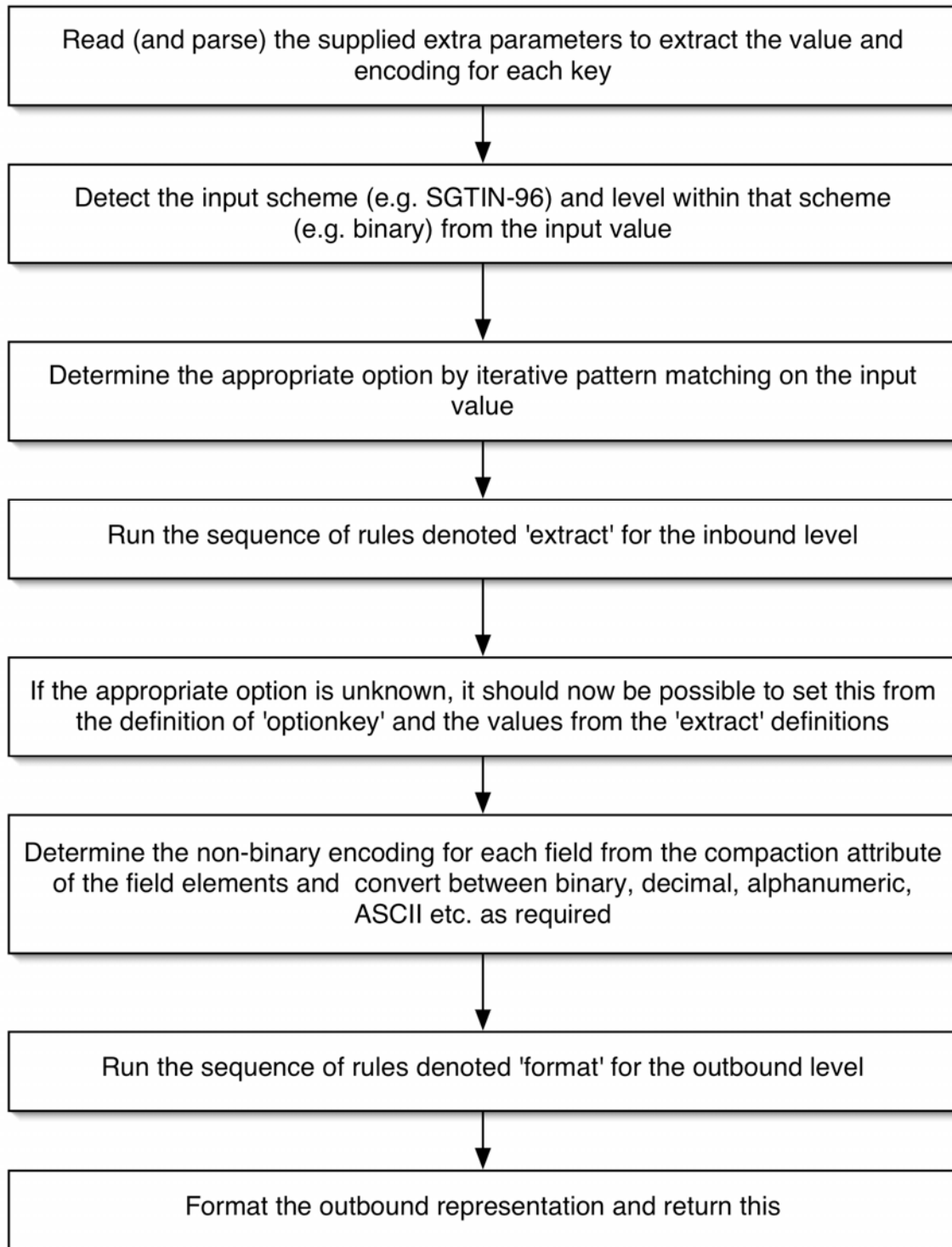


Figure 9 – Logical flowchart for a Tag Data Translation process

Figure 9 shows a flowchart indicating the key stages of operation of the translation process. These are:

- 1010 1. Read the input value and supplied parameters. Read the outbound level.
1011 2. Determine the coding scheme and inbound representation level
1012 (This may require `tagLength` to be specified as a supplied parameter)
1013 3. Determine the appropriate option value matching the input value string.
1014 4. Read the regular expression for pattern matching
1015 5. Populate any internal key-value lookup table or associative array with field values
1016 extracted from parsing of the input value string using the regular expression and from
1017 supplied parameters specified.
1018 6. Read the grammar for the outbound representation and check whether the key-value
1019 lookup table already contains all the field values needed for output. Supplied
1020 parameters required for formatting are indicated via the
1021 `requiredFormattingParameters` attribute of the `level` element.
1022 7. Check whether the values of the fields are within the ranges allowed for the output
1023 and throw errors if any fall outside the permitted numeric ranges, character sets or
1024 string lengths.
1025 8. If any of the field values are missing, parse the rules section of the TDT markup to
1026 determine how to obtain the missing required fields in the key-value lookup table –
1027 either by performing mathematical or string manipulations on other parameters, or
1028 using lookups in other tables based on known fields as lookup keys.
1029 9. Prepare the output string, using the key-value lookup table (which contains the values
1030 of the fields), the output grammar expression and taking into account the length of
1031 fields, and converting as necessary to/from binary.
1032 10. Return the output string and any error messages – e.g. undefined fields, fields out of
1033 numeric range, fields outside of character-set range

1034 **5.1. Tag Data Translation Software - Reference Implementation**

1035 A reference implementation may be a package / object class or subroutine, which may be
1036 used at any part of the EPC Network technology stack and integrated with existing
1037 software. Additionally, for educational and testing purposes, it will be useful to make a
1038 Tag Data Translation capability available as a standalone service, with interaction either
1039 via a web page form for a human operator or via a web service interface for automated
1040 use, enabling efficient batch conversions.

1041 **6. Application Programming Interface**

1042 There are essentially two interfaces to consider for Tag Data Translation software,
1043 namely a client-side interface, which provides conversion methods for users and a
1044 maintenance interface, which ensures that the translation software is kept up to date with
1045 the latest encoding/decoding definitions data.

6.1. Client API

```
public String translate(String epcIdentifier, String  
parameterList, String outputFormat)
```

Translates `epcIdentifier` from one representation into another within the same coding scheme.

Parameters:

`epcIdentifier` – The `epcIdentifier` to be converted. This should be expressed as a string, in accordance with one of the grammars or patterns in the TDT markup files, i.e. a binary string consisting of characters '0' and '1', a URI (either tag-encoding or pure-identity formats), or a serialized legacy code expressed as in Table 3.

`parameterList` – This is a parameter string containing key value pairs, using the semicolon ';' as delimiter between key=value pairs. For example, to convert a GTIN code the parameter string would look like the following:

`filter=3;companyprefixlength=7;taglength=96`

`outputFormat` – The output format into which the `epcIdentifier` SHALL be converted. The following are the formats supported:

1. BINARY
2. LEGACY
3. TAG_ENCODING
4. PURE_IDENTITY
5. ONS_HOSTNAME

Returns:

The converted value into one of the above formats as String.

Throws:

TDTTranslationException – Throws exceptions due to the following reason:

- 1.** `TDTFileNotFoundException` – Reports if the engine could not locate the configured definition file to compile.
- 2.** `TDTFieldBelowMinimum` - Reports a (numeric) Field that fell below the `decimalMinimum` value allowed by the TDT markup
- 3.** `TDTFieldAboveMaximum` - Reports a (numeric) Field that exceeded the `decimalMaximum` value allowed by the TDT markup

- 1083 **4.** `TDTFieldOutsideCharacterSet` - Reports a Field containing
1084 characters outside the `characterSet` range allowed by the TDT markup
- 1085 **5.** `TDTUndefinedField` - Reports a Field required for the output or an
1086 intermediate rule, whose value is undefined
- 1087 **6.** `TDTSchemeNotFound` - Reported if no matching Scheme can be found
1088 via `prefixMatch`
- 1089 **7.** `TDTLevelNotFound` - Reported if no matching Level can be found via
1090 `prefixMatch`
- 1091 **8.** `TDTOptionNotFound` - Reported if no matching Option can be found
1092 via the `optionKey` or via matching the `pattern`
- 1093 **9.** `TDTLookupFailed` - Reported if lookup in an external table failed to
1094 provide a value – reports table URI and path expression.
- 1095 **10.** `TDTNumericOverflow` - Reported when a numeric
1096 overflow occurs when handling numeric values such
1097 as serial number.

1098 **6.2. Maintenance API**

1099 `public void refreshTranslations()`

1100 Checks each subscription for any update, reloading new rules where necessary and forces
1101 the software to reload or recompile its internal representation of the encoding/decoding
1102 rules based on the current remaining subscriptions.

7. TDT Schema Definition

EpcTagDataTranslation.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
attributeFormDefault="unqualified" version="1.0">
  <xsd:annotation>
    <xsd:documentation><![CDATA[
      <epcglobal:copyright>Copyright ©2004 Epcglobal Inc., All Rights
      Reserved.</epcglobal:copyright>
      <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
      employees, or agents shall not be liable for any injury, loss, damages,
      financial or otherwise, arising from, related to, or caused by the use of this
      document. The use of said document shall constitute your express consent to
      the foregoing exculpation.</epcglobal:disclaimer>
      <epcglobal:specification>Tag Data Translation (TDT) version
      1.0</epcglobal:specification>
    ]]></xsd:documentation>
  </xsd:annotation>

  <xsd:include schemaLocation="LevelTypeList.xsd"/>
  <xsd:include schemaLocation="TagLengthList.xsd"/>
  <xsd:include schemaLocation="SchemeNameList.xsd"/>
  <xsd:include schemaLocation="InputFormatList.xsd"/>
  <xsd:include schemaLocation="ModeList.xsd"/>
  <xsd:include schemaLocation="CompactionMethodList.xsd"/>
  <xsd:include schemaLocation="PadDirectionList.xsd"/>

  <xsd:complexType name="Field">
    <xsd:attribute name="seq" type="xsd:integer" use="required"/>
    <xsd:attribute name="name" type="xsd:string" use="required"/>
    <xsd:attribute name="bitLength" type="xsd:integer" />
    <xsd:attribute name="characterSet" type="xsd:string" use="required"/>
    <xsd:attribute name="compaction" type="tdt:CompactionMethodList"/>
    <xsd:attribute name="compression" type="xsd:string"/>
    <xsd:attribute name="padChar" type="xsd:string"/>
    <xsd:attribute name="padDir" type="tdt:PadDirectionList"/>
    <xsd:attribute name="decimalMinimum" type="xsd:long"/>
    <xsd:attribute name="decimalMaximum" type="xsd:long"/>
    <xsd:attribute name="length" type="xsd:integer"/>
  </xsd:complexType>

  <xsd:complexType name="Option">
    <xsd:sequence>
      <xsd:element name="field" type="tdt:Field" maxOccurs="unbounded"/>
    </xsd:sequence>
    <xsd:attribute name="optionKey" type="xsd:string" use="required"/>
    <xsd:attribute name="pattern" type="xsd:string"/>
    <xsd:attribute name="grammar" type="xsd:string" use="required"/>
  </xsd:complexType>

  <xsd:complexType name="Rule">
    <xsd:attribute name="type" type="tdt:ModeList" use="required"/>
    <xsd:attribute name="inputFormat" type="tdt:InputFormatList"
use="required"/>
    <xsd:attribute name="seq" type="xsd:integer" use="required"/>
    <xsd:attribute name="newFieldName" type="xsd:string" use="required"/>
```

```

1162     <xsd:attribute name="characterSet" type="xsd:string" use="required"/>
1163     <xsd:attribute name="padChar" type="xsd:string"/>
1164     <xsd:attribute name="padDir" type="tdt:PadDirectionList"/>
1165     <xsd:attribute name="decimalMinimum" type="xsd:long"/>
1166     <xsd:attribute name="decimalMaximum" type="xsd:long"/>
1167     <xsd:attribute name="length" type="xsd:string"/>
1168     <xsd:attribute name="function" type="xsd:string" use="required"/>
1169     <xsd:attribute name="tableURL" type="xsd:string"/>
1170     <xsd:attribute name="tableParams" type="xsd:string"/>
1171     <xsd:attribute name="tableXPath" type="xsd:string"/>
1172     <xsd:attribute name="tableSQL" type="xsd:string"/>
1173 </xsd:complexType>
1174
1175 <xsd:complexType name="Level">
1176     <xsd:sequence>
1177         <xsd:element name="option" type="tdt:Option" minOccurs="1"
1178 maxOccurs="unbounded"/>
1179         <xsd:element name="rule" type="tdt:Rule" minOccurs="0"
1180 maxOccurs="unbounded"/>
1181     </xsd:sequence>
1182     <xsd:attribute name="type" type="tdt:LevelTypeList" use="required"/>
1183     <xsd:attribute name="prefixMatch" type="xsd:string" use="required"/>
1184     <xsd:attribute name="requiredParsingParameters" type="xsd:string"/>
1185     <xsd:attribute name="requiredFormattingParameters" type="xsd:string"/>
1186 </xsd:complexType>
1187
1188 <xsd:complexType name="Scheme">
1189     <xsd:sequence>
1190         <xsd:element name="level" type="tdt:Level" minOccurs="4" maxOccurs="5"/>
1191     </xsd:sequence>
1192     <xsd:attribute name="name" type="tdt:SchemeNameList" use="required"/>
1193     <xsd:attribute name="optionKey" type="xsd:string" use="required"/>
1194     <xsd:attribute name="tagLength" type="tdt:TagLengthList" use="required"/>
1195 </xsd:complexType>
1196
1197 <xsd:complexType name="EpcTagDataTranslation">
1198     <xsd:sequence>
1199         <xsd:element name="scheme" type="tdt:Scheme" maxOccurs="unbounded"/>
1200     </xsd:sequence>
1201     <xsd:attribute name="version" type="xsd:string" use="required"/>
1202     <xsd:attribute name="date" type="xsd:dateTime" use="required"/>
1203     <xsd:attribute name="epcTDSVersion" type="xsd:string" use="required"/>
1204 </xsd:complexType>
1205
1206     <xsd:element name="epcTagDataTranslation" type="tdt:EpcTagDataTranslation"/>
1207 </xsd:schema>
1208
1209

```

1210 LevelTypeList.xsd

```

1211 <?xml version="1.0" encoding="UTF-8"?>
1212 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1213 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1214 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1215 attributeFormDefault="unqualified" version="1.0">
1216     <xsd:annotation>
1217         <xsd:documentation><![CDATA[
1218             <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1219 Reserved.</epcglobal:copyright>
1220             <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1221 employees, or agents shall not be liable for any injury, loss, damages,
1222 financial or otherwise, arising from, related to, or caused by the use of this

```

```

1223 document. The use of said document shall constitute your express consent to
1224 the foregoing exculpation.</epcglobal:disclaimer>
1225     <epcglobal:specification>Tag Data Translation (TDT) version
1226 1.0</epcglobal:specification>
1227 ]]></xsd:documentation>
1228 </xsd:annotation>
1229 <xsd:simpleType name="LevelTypeList">
1230     <xsd:restriction base="xsd:string">
1231         <xsd:enumeration value="BINARY"/>
1232         <xsd:enumeration value="TAG_ENCODING"/>
1233         <xsd:enumeration value="PURE_IDENTITY"/>
1234         <xsd:enumeration value="LEGACY"/>
1235         <xsd:enumeration value="ONS_HOSTNAME"/>
1236     </xsd:restriction>
1237 </xsd:simpleType>
1238 </xsd:schema>
1239
1240

```

1241 TagLengthList.xsd

```

1242 <?xml version="1.0" encoding="UTF-8"?>
1243 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1244 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1245 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1246 attributeFormDefault="unqualified" version="1.0">
1247     <xsd:annotation>
1248         <xsd:documentation><![CDATA[
1249             <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1250 Reserved.</epcglobal:copyright>
1251             <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1252 employees, or agents shall not be liable for any injury, loss, damages,
1253 financial or otherwise, arising from, related to, or caused by the use of this
1254 document. The use of said document shall constitute your express consent to
1255 the foregoing exculpation.</epcglobal:disclaimer>
1256             <epcglobal:specification>Tag Data Translation (TDT) version
1257 1.0</epcglobal:specification>
1258         ]]></xsd:documentation>
1259     </xsd:annotation>
1260     <xsd:simpleType name="TagLengthList">
1261         <xsd:restriction base="xsd:string">
1262             <xsd:enumeration value="64"/>
1263             <xsd:enumeration value="96"/>
1264         </xsd:restriction>
1265     </xsd:simpleType>
1266 </xsd:schema>
1267
1268

```

1269 SchemeNameList.xsd

```

1270 <?xml version="1.0" encoding="UTF-8"?>
1271 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1272 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1273 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1274 attributeFormDefault="unqualified" version="1.0">
1275     <xsd:annotation>
1276         <xsd:documentation><![CDATA[
1277             <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1278 Reserved.</epcglobal:copyright>
1279             <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1280 employees, or agents shall not be liable for any injury, loss, damages,
1281 financial or otherwise, arising from, related to, or caused by the use of this

```

```

1282 document. The use of said document shall constitute your express consent to
1283 the foregoing exculpation.</epcglobal:disclaimer>
1284     <epcglobal:specification>Tag Data Translation (TDT) version
1285 1.0</epcglobal:specification>
1286 ]]></xsd:documentation>
1287 </xsd:annotation>
1288 <xsd:simpleType name="SchemeNameList">
1289     <xsd:restriction base="xsd:string">
1290         <xsd:enumeration value="SGTIN-64"/>
1291         <xsd:enumeration value="SGTIN-96"/>
1292         <xsd:enumeration value="SGLN-64"/>
1293         <xsd:enumeration value="SGLN-96"/>
1294         <xsd:enumeration value="GRAI-64"/>
1295         <xsd:enumeration value="GRAI-96"/>
1296         <xsd:enumeration value="GIAI-64"/>
1297         <xsd:enumeration value="GIAI-96"/>
1298         <xsd:enumeration value="SSCC-64"/>
1299         <xsd:enumeration value="SSCC-96"/>
1300         <xsd:enumeration value="GID-96"/>
1301         <xsd:enumeration value="USDOD-64"/>
1302         <xsd:enumeration value="USDOD-96"/>
1303     </xsd:restriction>
1304 </xsd:simpleType>
1305 </xsd:schema>
1306
1307

```

1308 InputFormatList.xsd

```

1309 <?xml version="1.0" encoding="UTF-8"?>
1310 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1311 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1312 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1313 attributeFormDefault="unqualified" version="1.0">
1314     <xsd:annotation>
1315         <xsd:documentation><![CDATA[
1316             <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1317             Reserved.</epcglobal:copyright>
1318             <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1319             employees, or agents shall not be liable for any injury, loss, damages,
1320             financial or otherwise, arising from, related to, or caused by the use of this
1321             document. The use of said document shall constitute your express consent to
1322             the foregoing exculpation.</epcglobal:disclaimer>
1323             <epcglobal:specification>Tag Data Translation (TDT) version
1324             1.0</epcglobal:specification>
1325             ]]></xsd:documentation>
1326         </xsd:annotation>
1327     <xsd:simpleType name="InputFormatList">
1328         <xsd:restriction base="xsd:string">
1329             <xsd:enumeration value="BINARY"/>
1330             <xsd:enumeration value="STRING"/>
1331         </xsd:restriction>
1332     </xsd:simpleType>
1333 </xsd:schema>
1334
1335

```

1336 ModeList.xsd

```

1337 <?xml version="1.0" encoding="UTF-8"?>
1338 <!-- Generated from annotated java -->
1339 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1340 xmlns:xsd="http://www.w3.org/2001/XMLSchema"

```

```

1341 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1342 attributeFormDefault="unqualified" version="1.0">
1343   <xsd:annotation>
1344     <xsd:documentation><![CDATA[
1345       <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1346       Reserved.</epcglobal:copyright>
1347       <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1348       employees, or agents shall not be liable for any injury, loss, damages,
1349       financial or otherwise, arising from, related to, or caused by the use of this
1350       document. The use of said document shall constitute your express consent to
1351       the foregoing exculpation.</epcglobal:disclaimer>
1352       <epcglobal:specification>Tag Data Translation (TDT) version
1353       1.0</epcglobal:specification>
1354     ]]></xsd:documentation>
1355   </xsd:annotation>
1356   <xsd:simpleType name="ModeList">
1357     <xsd:restriction base="xsd:string">
1358       <xsd:enumeration value="EXTRACT"/>
1359       <xsd:enumeration value="FORMAT"/>
1360     </xsd:restriction>
1361   </xsd:simpleType>
1362 </xsd:schema>
1363
1364

```

1365 CompactionMethodList.xsd

```

1366 <?xml version="1.0" encoding="UTF-8"?>
1367 <!-- Generated from annotated java -->
1368 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1369 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1370 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1371 attributeFormDefault="unqualified" version="1.0">
1372   <xsd:annotation>
1373     <xsd:documentation><![CDATA[
1374       <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1375       Reserved.</epcglobal:copyright>
1376       <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1377       employees, or agents shall not be liable for any injury, loss, damages,
1378       financial or otherwise, arising from, related to, or caused by the use of this
1379       document. The use of said document shall constitute your express consent to
1380       the foregoing exculpation.</epcglobal:disclaimer>
1381       <epcglobal:specification>Tag Data Translation (TDT) version
1382       1.0</epcglobal:specification>
1383     ]]></xsd:documentation>
1384   </xsd:annotation>
1385   <xsd:simpleType name="CompactionMethodList">
1386     <xsd:restriction base="xsd:string">
1387       <xsd:enumeration value="32-bit"/>
1388       <xsd:enumeration value="16-bit"/>
1389       <xsd:enumeration value="8-bit"/>
1390       <xsd:enumeration value="7-bit"/>
1391       <xsd:enumeration value="6-bit"/>
1392       <xsd:enumeration value="5-bit"/>
1393     </xsd:restriction>
1394   </xsd:simpleType>
1395 </xsd:schema>
1396

```

1397 PadDirectionList.xsd

```

1398 <?xml version="1.0" encoding="UTF-8"?>
1399 <!-- Generated from annotated java -->

```

```

1400 <xsd:schema targetNamespace="urn:epcglobal:tdt:xsd:1"
1401 xmlns:xsd="http://www.w3.org/2001/XMLSchema"
1402 xmlns:tdt="urn:epcglobal:tdt:xsd:1" elementFormDefault="unqualified"
1403 attributeFormDefault="unqualified" version="1.0">
1404   <xsd:annotation>
1405     <xsd:documentation><![CDATA[
1406       <epcglobal:copyright>Copyright ©2005 Epcglobal Inc., All Rights
1407       Reserved.</epcglobal:copyright>
1408       <epcglobal:disclaimer>EPCglobal Inc., its members, officers, directors,
1409       employees, or agents shall not be liable for any injury, loss, damages,
1410       financial or otherwise, arising from, related to, or caused by the use of this
1411       document. The use of said document shall constitute your express consent to
1412       the foregoing exculpation.</epcglobal:disclaimer>
1413       <epcglobal:specification>Tag Data Translation (TDT) version
1414       1.0</epcglobal:specification>
1415     ]]></xsd:documentation>
1416   </xsd:annotation>
1417   <xsd:simpleType name="PadDirectionList">
1418     <xsd:restriction base="xsd:string">
1419       <xsd:enumeration value="LEFT"/>
1420       <xsd:enumeration value="RIGHT"/>
1421     </xsd:restriction>
1422   </xsd:simpleType>
1423 </xsd:schema>
1424
1425
1426
1427

```

8. TDT Markup V1.0

The final draft Tag Data Translation markup files as defined in the TDS Version 1.1r1.27 will be made freely available at the following address:

<http://www.autoidlabs.org/Cambridge/TDT>

The markup files are also embedded in this appendix for ease of reference.

8.1. GID-96 General Identifier

```
<?xml version="1.0" encoding="UTF-8"?>
<epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
xmlns:xsi="http://www.w3.org/2001/XMLSchema"
xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
  <scheme name="GID-96" optionKey="1" tagLength="96" >
    <level type="BINARY" prefixMatch="00110101" requiredFormattingParameters="taglength" >
      <option optionKey="1" pattern="00110101([01]{28})([01]{24})([01]{36})"
grammar="'00110101' generalmanager objectclass serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[01]"
bitLength="28" name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[01]"
bitLength="24" name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[01]"
bitLength="36" name="serial"/>
      </option>
    </level>
    <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:gid-96"
requiredFormattingParameters="taglength" >
      <option optionKey="1" pattern="urn:epc:tag:gid-96:([0-9]*)\.([0-9]*)\.([0-9]*)"
grammar="'urn:epc:tag:gid-96:' generalmanager '.' objectclass '.' serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-9]"
name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]"
name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]"
name="serial"/>
      </option>
    </level>
    <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:gid" >
      <option optionKey="1" pattern="urn:epc:id:gid:([0-9]*)\.([0-9]*)\.([0-9]*)"
grammar="'urn:epc:id:gid:' generalmanager '.' objectclass '.' serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-9]"
name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]"
name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]"
name="serial"/>
      </option>
    </level>
    <level type="LEGACY" prefixMatch="generalmanager=" >
      <option optionKey="1" pattern="generalmanager=([0-9]*);objectclass=([0-9]*);serial=([0-9]*)"
grammar="'generalmanager=' generalmanager ';'objectclass='
objectclass ';'serial=' serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="268435455" characterSet="[0-9]"
name="generalmanager"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]"
name="objectclass"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]"
name="serial"/>
      </option>
    </level>
  </scheme>
</epcTagDataTranslation>
```

8.2. SGTIN-64 Serialized Global Trade Item Number for 64-bit EPCs

```
<?xml version="1.0" encoding="UTF-8"?>
<epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
xmlns:xsi="http://www.w3.org/2001/XMLSchema"
xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
  <scheme name="SGTIN-64" optionKey="companyprefixlength" tagLength="64">
    <level type="BINARY" prefixMatch="10" requiredFormattingParameters="filter,taglength" >
      <option optionKey="12" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitLength="14" name="companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
bitLength="20" length="1" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]*"
bitLength="25" name="serial"/>
      </option>
      <option optionKey="11" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitLength="14" name="companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
bitLength="20" length="2" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]*"
bitLength="25" name="serial"/>
      </option>
      <option optionKey="10" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitLength="14" name="companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*"
bitLength="20" length="3" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]*"
bitLength="25" name="serial"/>
      </option>
      <option optionKey="9" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitLength="14" name="companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*"
bitLength="20" length="4" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]*"
bitLength="25" name="serial"/>
      </option>
      <option optionKey="8" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
bitLength="14" name="companyprefixindex"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*"
bitLength="20" length="5" padChar="0" padDir="LEFT" name="itemref"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]*"
bitLength="25" name="serial"/>
      </option>
      <option optionKey="7" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
grammar="'10' filter companyprefixindex itemref serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
```



```

1556     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
1557 bitLength="14" name="companyprefixindex"/>
1558     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]"*
1559 bitLength="20" length="6" padChar="0" padDir="LEFT" name="itemref"/>
1560     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]"*
1561 bitLength="25" name="serial"/>
1562 </option>
1563 <option optionKey="6" pattern="10([01]{3})([01]{14})([01]{20})([01]{25})"
1564 grammar="'10' filter companyprefixindex itemref serial" >
1565     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
1566 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1567     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
1568 bitLength="14" name="companyprefixindex"/>
1569     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"*
1570 bitLength="20" length="7" padChar="0" padDir="LEFT" name="itemref"/>
1571     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[01]"*
1572 bitLength="25" name="serial"/>
1573 </option>
1574 <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="companyprefix"
1575 characterSet="[0-9]"*
1576 function="TABLELOOKUP(companyprefixindex,tdt64bitcpi,companyprefixindex,companyprefix)"
1577 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
1578 tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
1579 tableParams="companyprefixindex"/>
1580 <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="companyprefixlength"
1581 characterSet="[0-9]"* function="LENGTH(companyprefix)"/>
1582 <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="companyprefixindex"
1583 characterSet="[0-9]"*
1584 function="TABLELOOKUP(companyprefix,tdt64bitcpi,companyprefix,companyprefixindex)"
1585 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
1586 tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index" tableParams="companyprefix"/>
1587 </level>
1588 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgtin-64"
1589 requiredFormattingParameters="filter,taglength" >
1590     <option optionKey="12" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.\([0-9]{12})\.\([0-
1591 9]{1})\.\([0-9]*)" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1592 '.' serial" >
1593         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
1594 length="1" padChar="0" padDir="LEFT" name="filter"/>
1595         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
1596 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
1597         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
1598 length="1" padChar="0" padDir="LEFT" name="itemref"/>
1599         <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1600 name="serial"/>
1601     </option>
1602     <option optionKey="11" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.\([0-9]{11})\.\([0-
1603 9]{2})\.\([0-9]*)" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1604 '.' serial" >
1605         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
1606 length="1" padChar="0" padDir="LEFT" name="filter"/>
1607         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
1608 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
1609         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"*
1610 length="2" padChar="0" padDir="LEFT" name="itemref"/>
1611         <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1612 name="serial"/>
1613     </option>
1614     <option optionKey="10" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.\([0-9]{10})\.\([0-
1615 9]{3})\.\([0-9]*)" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1616 '.' serial" >
1617         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
1618 length="1" padChar="0" padDir="LEFT" name="filter"/>
1619         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
1620 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
1621         <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"*
1622 length="3" padChar="0" padDir="LEFT" name="itemref"/>
1623         <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1624 name="serial"/>
1625     </option>

```

```

1626     <option optionKey="9" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{9})\.[0-
1627 9]{4})\.[0-9]*" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1628 '.' serial" >
1629     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1630 length="1" padChar="0" padDir="LEFT" name="filter"/>
1631     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
1632 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
1633     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
1634 length="4" padChar="0" padDir="LEFT" name="itemref"/>
1635     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1636 name="serial"/>
1637 </option>
1638 <option optionKey="8" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{8})\.[0-
1639 9]{5})\.[0-9]*" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1640 '.' serial" >
1641     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1642 length="1" padChar="0" padDir="LEFT" name="filter"/>
1643     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
1644 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
1645     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
1646 length="5" padChar="0" padDir="LEFT" name="itemref"/>
1647     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1648 name="serial"/>
1649 </option>
1650 <option optionKey="7" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{7})\.[0-
1651 9]{6})\.[0-9]*" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1652 '.' serial" >
1653     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1654 length="1" padChar="0" padDir="LEFT" name="filter"/>
1655     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
1656 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
1657     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
1658 length="6" padChar="0" padDir="LEFT" name="itemref"/>
1659     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1660 name="serial"/>
1661 </option>
1662 <option optionKey="6" pattern="urn:epc:tag:sgtin-64:([0-7]{1})\.[0-9]{6})\.[0-
1663 9]{7})\.[0-9]*" grammar="urn:epc:tag:sgtin-64:' filter '.' companyprefix '.' itemref
1664 '.' serial" >
1665     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1666 length="1" padChar="0" padDir="LEFT" name="filter"/>
1667     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
1668 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
1669     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[0-9]"
1670 length="7" padChar="0" padDir="LEFT" name="itemref"/>
1671     <field seq="4" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1672 name="serial"/>
1673 </option>
1674 </level>
1675 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgtin" >
1676     <option optionKey="12" pattern="urn:epc:id:sgtin:([0-9]{12})\.[0-9]{1})\.[0-9]*"
1677 grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1678     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
1679 9]" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
1680     <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"
1681 length="1" padChar="0" padDir="LEFT" name="itemref"/>
1682     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1683 name="serial"/>
1684 </option>
1685     <option optionKey="11" pattern="urn:epc:id:sgtin:([0-9]{11})\.[0-9]{2})\.[0-9]*"
1686 grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1687     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
1688 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
1689     <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"
1690 length="2" padChar="0" padDir="LEFT" name="itemref"/>
1691     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"
1692 name="serial"/>
1693 </option>
1694     <option optionKey="10" pattern="urn:epc:id:sgtin:([0-9]{10})\.[0-9]{3})\.[0-9]*"
1695 grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >

```

```

1696     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
1697     length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
1698     <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*"
1699     length="3" padChar="0" padDir="LEFT" name="itemref"/>
1700     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1701     name="serial"/>
1702   </option>
1703   <option optionKey="9" pattern="urn:epc:id:sgtin:([0-9]{9})\.[0-9]{4}\.[0-9]*"
1704   grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1705     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
1706     length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
1707     <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*"
1708     length="4" padChar="0" padDir="LEFT" name="itemref"/>
1709     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1710     name="serial"/>
1711   </option>
1712   <option optionKey="8" pattern="urn:epc:id:sgtin:([0-9]{8})\.[0-9]{5}\.[0-9]*"
1713   grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1714     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
1715     length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
1716     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*"
1717     length="5" padChar="0" padDir="LEFT" name="itemref"/>
1718     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1719     name="serial"/>
1720   </option>
1721   <option optionKey="7" pattern="urn:epc:id:sgtin:([0-9]{7})\.[0-9]{6}\.[0-9]*"
1722   grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1723     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
1724     length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
1725     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
1726     length="6" padChar="0" padDir="LEFT" name="itemref"/>
1727     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1728     name="serial"/>
1729   </option>
1730   <option optionKey="6" pattern="urn:epc:id:sgtin:([0-9]{6})\.[0-9]{7}\.[0-9]*"
1731   grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
1732     <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
1733     length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
1734     <field seq="2" decimalMinimum="0" decimalMaximum="1048575" characterSet="[0-9]*"
1735     length="7" padChar="0" padDir="LEFT" name="itemref"/>
1736     <field seq="3" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1737     name="serial"/>
1738   </option>
1739 </level>
1740 <level type="LEGACY" prefixMatch="gtin="
1741 requiredParsingParameters="companyprefixlength" >
1742   <option optionKey="12" pattern="gtin=([0-9]{14});serial=([0-9]*" grammar="gtin='
1743   indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1744     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
1745   9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
1746     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1747     name="serial"/>
1748   </option>
1749   <option optionKey="11" pattern="gtin=([0-9]{14});serial=([0-9]*" grammar="gtin='
1750   indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1751     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
1752   9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
1753     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1754     name="serial"/>
1755   </option>
1756   <option optionKey="10" pattern="gtin=([0-9]{14});serial=([0-9]*" grammar="gtin='
1757   indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1758     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
1759   9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
1760     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]*"
1761     name="serial"/>
1762   </option>
1763   <option optionKey="9" pattern="gtin=([0-9]{14});serial=([0-9]*" grammar="gtin='
1764   indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1765     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
1766   9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>

```

```

1767     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1768     name="serial"/>
1769     </option>
1770     <option optionKey="8" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
1771     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1772     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
1773     9]"* length="14" padChar="0" padDir="LEFT" name="gtin"/>
1774     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1775     name="serial"/>
1776     </option>
1777     <option optionKey="7" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
1778     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1779     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
1780     9]"* length="14" padChar="0" padDir="LEFT" name="gtin"/>
1781     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1782     name="serial"/>
1783     </option>
1784     <option optionKey="6" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
1785     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
1786     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
1787     9]"* length="14" padChar="0" padDir="LEFT" name="gtin"/>
1788     <field seq="2" decimalMinimum="0" decimalMaximum="33554431" characterSet="[0-9]"*
1789     name="serial"/>
1790     </option>
1791     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder"
1792     characterSet="[0-9]"* length="12" function="SUBSTR(gtin,1,12)"/>
1793     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicatordigit"
1794     characterSet="[0-9]"* length="1" function="SUBSTR(gtin,0,1)"/>
1795     <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemrefremainder"
1796     characterSet="[0-9]"* function="SUBSTR(gtinprefixremainder,companyprefixlength)"/>
1797     <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
1798     characterSet="[0-9]"* function="CONCAT(indicatordigit,itemrefremainder)"/>
1799     <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="companyprefix"
1800     characterSet="[0-9]"* function="SUBSTR(gtinprefixremainder,0,companyprefixlength)"/>
1801     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicatordigit"
1802     characterSet="[0-9]"* length="1" function="SUBSTR(itemref,0,1)"/>
1803     <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemrefremainder"
1804     characterSet="[0-9]"* function="SUBSTR(itemref,1)"/>
1805     <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtinprefix"
1806     characterSet="[0-9]"* length="13"
1807     function="CONCAT(indicatordigit,companyprefix,itemrefremainder)"/>
1808     <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
1809     characterSet="[0-9]"* length="1" function="GS1CHECKSUM(gtinprefix)"/>
1810     </level>
1811     <level type="ONS_HOSTNAME" >
1812     <option optionKey="12" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1813     <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
1814     length="1" padChar="0" padDir="LEFT" name="itemref"/>
1815     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
1816     9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
1817     </option>
1818     <option optionKey="11" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1819     <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"*
1820     length="2" padChar="0" padDir="LEFT" name="itemref"/>
1821     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]"*
1822     length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
1823     </option>
1824     <option optionKey="10" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1825     <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"*
1826     length="3" padChar="0" padDir="LEFT" name="itemref"/>
1827     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]"*
1828     length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
1829     </option>
1830     <option optionKey="9" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1831     <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"*
1832     length="4" padChar="0" padDir="LEFT" name="itemref"/>
1833     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]"*
1834     length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
1835     </option>
1836     <option optionKey="8" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >

```

```

1837     <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*"
1838     length="5" padChar="0" padDir="LEFT" name="itemref"/>
1839     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*"
1840     length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
1841   </option>
1842   <option optionKey="7" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1843     <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
1844     length="6" padChar="0" padDir="LEFT" name="itemref"/>
1845     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
1846     length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
1847   </option>
1848   <option optionKey="6" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
1849     <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
1850     length="7" padChar="0" padDir="LEFT" name="itemref"/>
1851     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
1852     length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
1853   </option>
1854 </level>
1855 </scheme>
1856 </epcTagDataTranslation>

```

1857 8.3. SGTIN-96 Serialized Global Trade Item Number for 96-bit 1858 EPCs

```

1859 <?xml version="1.0" encoding="UTF-8"?>
1860 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
1861 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
1862 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
1863   <scheme name="SGTIN-96" optionKey="companyprefixlength" tagLength="96" >
1864     <level type="BINARY" prefixMatch="00110000"
1865     requiredFormattingParameters="filter,taglength" >
1866       <option optionKey="12" pattern="00110000([01]{3})000([01]{40})([01]{4})([01]{38})"
1867       grammar="'00110000' filter '000' companyprefix itemref serial" >
1868         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1869         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1870         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
1871         bitLength="40" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
1872         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
1873         bitLength="4" length="1" padChar="0" padDir="LEFT" name="itemref"/>
1874         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1875         bitLength="38" name="serial"/>
1876       </option>
1877       <option optionKey="11" pattern="00110000([01]{3})001([01]{37})([01]{7})([01]{38})"
1878       grammar="'00110000' filter '001' companyprefix itemref serial" >
1879         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1880         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1881         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
1882         bitLength="37" length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
1883         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
1884         bitLength="7" length="2" padChar="0" padDir="LEFT" name="itemref"/>
1885         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1886         bitLength="38" name="serial"/>
1887       </option>
1888       <option optionKey="10" pattern="00110000([01]{3})010([01]{34})([01]{10})([01]{38})"
1889       grammar="'00110000' filter '010' companyprefix itemref serial" >
1890         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1891         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1892         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
1893         bitLength="34" length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
1894         <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*"
1895         bitLength="10" length="3" padChar="0" padDir="LEFT" name="itemref"/>
1896         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1897         bitLength="38" name="serial"/>
1898       </option>
1899       <option optionKey="9" pattern="00110000([01]{3})011([01]{30})([01]{14})([01]{38})"
1900       grammar="'00110000' filter '011' companyprefix itemref serial" >
1901         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1902         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1903

```

```

1904     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
1905 bitLength="30" length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
1906     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*"
1907 bitLength="14" length="4" padChar="0" padDir="LEFT" name="itemref"/>
1908     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1909 bitLength="38" name="serial"/>
1910 </option>
1911 <option optionKey="8" pattern="00110000([01]{3})100([01]{27})([01]{17})([01]{38})"
1912 grammar="'00110000' filter '100' companyprefix itemref serial" >
1913     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1914 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1915     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
1916 bitLength="27" length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
1917     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*"
1918 bitLength="17" length="5" padChar="0" padDir="LEFT" name="itemref"/>
1919     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1920 bitLength="38" name="serial"/>
1921 </option>
1922 <option optionKey="7" pattern="00110000([01]{3})101([01]{24})([01]{20})([01]{38})"
1923 grammar="'00110000' filter '101' companyprefix itemref serial" >
1924     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1925 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1926     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
1927 bitLength="24" length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
1928     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*"
1929 bitLength="20" length="6" padChar="0" padDir="LEFT" name="itemref"/>
1930     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1931 bitLength="38" name="serial"/>
1932 </option>
1933 <option optionKey="6" pattern="00110000([01]{3})110([01]{20})([01]{24})([01]{38})"
1934 grammar="'00110000' filter '110' companyprefix itemref serial" >
1935     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
1936 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
1937     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
1938 bitLength="20" length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
1939     <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]*"
1940 bitLength="24" length="7" padChar="0" padDir="LEFT" name="itemref"/>
1941     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]*"
1942 bitLength="38" name="serial"/>
1943 </option>
1944 </level>
1945 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgtin-96"
1946 requiredFormattingParameters="filter,taglength" >
1947     <option optionKey="12" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{12}\.[0-
1948 9]{1})\.[0-9]*" grammar="urn:epc:tag:sgtin-96: filter '.' companyprefix '.' itemref
1949 '.' serial" >
1950         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
1951 length="1" padChar="0" padDir="LEFT" name="filter"/>
1952         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
1953 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
1954         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
1955 length="1" padChar="0" padDir="LEFT" name="itemref"/>
1956         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
1957 9]*" name="serial"/>
1958     </option>
1959     <option optionKey="11" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{11}\.[0-
1960 9]{2})\.[0-9]*" grammar="urn:epc:tag:sgtin-96: filter '.' companyprefix '.' itemref
1961 '.' serial" >
1962         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
1963 length="1" padChar="0" padDir="LEFT" name="filter"/>
1964         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-9]*"
1965 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
1966         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*"
1967 length="2" padChar="0" padDir="LEFT" name="itemref"/>
1968         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
1969 9]*" name="serial"/>
1970     </option>
1971     <option optionKey="10" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{10}\.[0-
1972 9]{3})\.[0-9]*" grammar="urn:epc:tag:sgtin-96: filter '.' companyprefix '.' itemref
1973 '.' serial" >

```

```

1974     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1975 length="1" padChar="0" padDir="LEFT" name="filter"/>
1976     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
1977 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
1978     <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"
1979 length="3" padChar="0" padDir="LEFT" name="itemref"/>
1980     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
1981 9]" name="serial"/>
1982 </option>
1983 <option optionKey="9" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{9})\.[0-
1984 9]{4})\.[0-9]*" grammar="urn:epc:tag:sgtin-96:' filter '.' companyprefix '.' itemref
1985 '.' serial" >
1986     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1987 length="1" padChar="0" padDir="LEFT" name="filter"/>
1988     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
1989 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
1990     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
1991 length="4" padChar="0" padDir="LEFT" name="itemref"/>
1992     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
1993 9]" name="serial"/>
1994 </option>
1995 <option optionKey="8" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{8})\.[0-
1996 9]{5})\.[0-9]*" grammar="urn:epc:tag:sgtin-96:' filter '.' companyprefix '.' itemref
1997 '.' serial" >
1998     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
1999 length="1" padChar="0" padDir="LEFT" name="filter"/>
2000     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
2001 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2002     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
2003 length="5" padChar="0" padDir="LEFT" name="itemref"/>
2004     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2005 9]" name="serial"/>
2006 </option>
2007 <option optionKey="7" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{7})\.[0-
2008 9]{6})\.[0-9]*" grammar="urn:epc:tag:sgtin-96:' filter '.' companyprefix '.' itemref
2009 '.' serial" >
2010     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
2011 length="1" padChar="0" padDir="LEFT" name="filter"/>
2012     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
2013 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2014     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
2015 length="6" padChar="0" padDir="LEFT" name="itemref"/>
2016     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2017 9]" name="serial"/>
2018 </option>
2019 <option optionKey="6" pattern="urn:epc:tag:sgtin-96:([0-7]{1})\.[0-9]{6})\.[0-
2020 9]{7})\.[0-9]*" grammar="urn:epc:tag:sgtin-96:' filter '.' companyprefix '.' itemref
2021 '.' serial" >
2022     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
2023 length="1" padChar="0" padDir="LEFT" name="filter"/>
2024     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
2025 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2026     <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2027 length="7" padChar="0" padDir="LEFT" name="itemref"/>
2028     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2029 9]" name="serial"/>
2030 </option>
2031 </level>
2032 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgtin" >
2033     <option optionKey="12" pattern="urn:epc:id:sgtin:([0-9]{12})\.[0-9]{1})\.[0-9]*"
2034 grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2035     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-
2036 9]" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2037     <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"
2038 length="1" padChar="0" padDir="LEFT" name="itemref"/>
2039     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2040 9]" name="serial"/>
2041 </option>
2042 <option optionKey="11" pattern="urn:epc:id:sgtin:([0-9]{11})\.[0-9]{2})\.[0-9]*"
2043 grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >

```

```

2044     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2045     length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2046     <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*"
2047     length="2" padChar="0" padDir="LEFT" name="itemref"/>
2048     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2049     9]*" name="serial"/>
2050     </option>
2051     <option optionKey="10" pattern="urn:epc:id:sgtin:([0-9]{10})\.[0-9]{3}\.[0-9]*"
2052     grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2053     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2054     length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2055     <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*"
2056     length="3" padChar="0" padDir="LEFT" name="itemref"/>
2057     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2058     9]*" name="serial"/>
2059     </option>
2060     <option optionKey="9" pattern="urn:epc:id:sgtin:([0-9]{9})\.[0-9]{4}\.[0-9]*"
2061     grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2062     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2063     length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2064     <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*"
2065     length="4" padChar="0" padDir="LEFT" name="itemref"/>
2066     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2067     9]*" name="serial"/>
2068     </option>
2069     <option optionKey="8" pattern="urn:epc:id:sgtin:([0-9]{8})\.[0-9]{5}\.[0-9]*"
2070     grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2071     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2072     length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2073     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*"
2074     length="5" padChar="0" padDir="LEFT" name="itemref"/>
2075     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2076     9]*" name="serial"/>
2077     </option>
2078     <option optionKey="7" pattern="urn:epc:id:sgtin:([0-9]{7})\.[0-9]{6}\.[0-9]*"
2079     grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2080     <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*"
2081     length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2082     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2083     length="6" padChar="0" padDir="LEFT" name="itemref"/>
2084     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2085     9]*" name="serial"/>
2086     </option>
2087     <option optionKey="6" pattern="urn:epc:id:sgtin:([0-9]{6})\.[0-9]{7}\.[0-9]*"
2088     grammar="urn:epc:id:sgtin:' companyprefix '.' itemref '.' serial" >
2089     <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2090     length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2091     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2092     length="7" padChar="0" padDir="LEFT" name="itemref"/>
2093     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2094     9]*" name="serial"/>
2095     </option>
2096     </level>
2097     <level type="LEGACY" prefixMatch="gtin="
2098     requiredParsingParameters="companyprefixlength" >
2099     <option optionKey="12" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="gtin='
2100     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
2101     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2102     9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2103     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2104     9]*" name="serial"/>
2105     </option>
2106     <option optionKey="11" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="gtin='
2107     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >
2108     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2109     9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2110     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2111     9]*" name="serial"/>
2112     </option>
2113     <option optionKey="10" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="gtin='
2114     indicatordigit companyprefix itemrefremainder checkdigit ';serial=' serial" >

```



```

2115     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2116 9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2117     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2118 9]*" name="serial"/>
2119     </option>
2120     <option optionKey="9" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
2121 indicator digit company prefix itemref remainder checkdigit ';serial=' serial" >
2122     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2123 9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2124     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2125 9]*" name="serial"/>
2126     </option>
2127     <option optionKey="8" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
2128 indicator digit company prefix itemref remainder checkdigit ';serial=' serial" >
2129     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2130 9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2131     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2132 9]*" name="serial"/>
2133     </option>
2134     <option optionKey="7" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
2135 indicator digit company prefix itemref remainder checkdigit ';serial=' serial" >
2136     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2137 9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2138     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2139 9]*" name="serial"/>
2140     </option>
2141     <option optionKey="6" pattern="gtin=([0-9]{14});serial=([0-9]*)" grammar="'gtin='
2142 indicator digit company prefix itemref remainder checkdigit ';serial=' serial" >
2143     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2144 9]*" length="14" padChar="0" padDir="LEFT" name="gtin"/>
2145     <field seq="2" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
2146 9]*" name="serial"/>
2147     </option>
2148     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="gtinprefixremainder"
2149 characterSet="[0-9]*" length="12" function="SUBSTR(gtin,1,12)"/>
2150     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="indicator digit"
2151 characterSet="[0-9]*" length="1" function="SUBSTR(gtin,0,1)"/>
2152     <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="itemref remainder"
2153 characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,company prefix length)"/>
2154     <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="itemref"
2155 characterSet="[0-9]*" function="CONCAT(indicator digit,itemref remainder)"/>
2156     <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="company prefix"
2157 characterSet="[0-9]*" function="SUBSTR(gtinprefixremainder,0,company prefix length)"/>
2158     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="indicator digit"
2159 characterSet="[0-9]*" length="1" function="SUBSTR(itemref,0,1)"/>
2160     <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="itemref remainder"
2161 characterSet="[0-9]*" function="SUBSTR(itemref,1)"/>
2162     <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="gtin prefix"
2163 characterSet="[0-9]*" length="13"
2164 function="CONCAT(indicator digit,company prefix,itemref remainder)"/>
2165     <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
2166 characterSet="[0-9]*" length="1" function="GS1CHECKSUM(gtinprefix)"/>
2167 </level>
2168 <level type="ONS_HOSTNAME" >
2169     <option optionKey="12" grammar="itemref '.' company prefix '.sgtin.id.onsepc.com'" >
2170     <field seq="1" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
2171 length="1" padChar="0" padDir="LEFT" name="itemref"/>
2172     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2173 9]*" length="12" padChar="0" padDir="LEFT" name="company prefix"/>
2174     </option>
2175     <option optionKey="11" grammar="itemref '.' company prefix '.sgtin.id.onsepc.com'" >
2176     <field seq="1" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*"
2177 length="2" padChar="0" padDir="LEFT" name="itemref"/>
2178     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*"
2179 length="11" padChar="0" padDir="LEFT" name="company prefix"/>
2180     </option>
2181     <option optionKey="10" grammar="itemref '.' company prefix '.sgtin.id.onsepc.com'" >
2182     <field seq="1" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*"
2183 length="3" padChar="0" padDir="LEFT" name="itemref"/>
2184     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-9]*"
2185 length="10" padChar="0" padDir="LEFT" name="company prefix"/>

```

```

2186     </option>
2187     <option optionKey="9" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
2188       <field seq="1" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*"
2189       length="4" padChar="0" padDir="LEFT" name="itemref"/>
2190       <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2191       length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2192     </option>
2193     <option optionKey="8" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
2194       <field seq="1" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*"
2195       length="5" padChar="0" padDir="LEFT" name="itemref"/>
2196       <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2197       length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2198     </option>
2199     <option optionKey="7" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
2200       <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2201       length="6" padChar="0" padDir="LEFT" name="itemref"/>
2202       <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2203       length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2204     </option>
2205     <option optionKey="6" grammar="itemref '.' companyprefix '.sgtin.id.onsepc.com'" >
2206       <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2207       length="7" padChar="0" padDir="LEFT" name="itemref"/>
2208       <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2209       length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2210     </option>
2211   </level>
2212 </scheme>
2213 </epcTagDataTranslation>

```

8.4. SSCC-64 Serial Shipping Container Code for 64-bit EPCs

```

2214 <?xml version="1.0" encoding="UTF-8"?>
2215 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
2216 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
2217 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
2218   <scheme name="SSCC-64" optionKey="companyprefixlength" tagLength="64" >
2219     <level type="BINARY" prefixMatch="00001000"
2220     requiredFormattingParameters="filter,taglength" >
2221       <option optionKey="12" pattern="00001000([01]{3})([01]{14})([01]{39})"
2222       grammar="'00001000' filter companyprefixindex serialref" >
2223         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2224         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2225         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
2226         bitLength="14" name="companyprefixindex"/>
2227         <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*"
2228         bitLength="39" length="5" padChar="0" padDir="LEFT" name="serialref"/>
2229       </option>
2230       <option optionKey="11" pattern="00001000([01]{3})([01]{14})([01]{39})"
2231       grammar="'00001000' filter companyprefixindex serialref" >
2232         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2233         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2234         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
2235         bitLength="14" name="companyprefixindex"/>
2236         <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*"
2237         bitLength="39" length="6" padChar="0" padDir="LEFT" name="serialref"/>
2238       </option>
2239       <option optionKey="10" pattern="00001000([01]{3})([01]{14})([01]{39})"
2240       grammar="'00001000' filter companyprefixindex serialref" >
2241         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2242         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2243         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
2244         bitLength="14" name="companyprefixindex"/>
2245         <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*"
2246         bitLength="39" length="7" padChar="0" padDir="LEFT" name="serialref"/>
2247       </option>
2248       <option optionKey="9" pattern="00001000([01]{3})([01]{14})([01]{39})"
2249       grammar="'00001000' filter companyprefixindex serialref" >
2250         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2251         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2252       </option>
2253     </level>

```

```

2254     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
2255 bitLength="14" name="companyprefixindex"/>
2256     <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"*
2257 bitLength="39" length="8" padChar="0" padDir="LEFT" name="serialref"/>
2258     </option>
2259     <option optionKey="8" pattern="00001000([01]{3})([01]{14})([01]{39})"
2260 grammar="'00001000' filter companyprefixindex serialref" >
2261     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
2262 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2263     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
2264 bitLength="14" name="companyprefixindex"/>
2265     <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]"*
2266 bitLength="39" length="9" padChar="0" padDir="LEFT" name="serialref"/>
2267     </option>
2268     <option optionKey="7" pattern="00001000([01]{3})([01]{14})([01]{39})"
2269 grammar="'00001000' filter companyprefixindex serialref" >
2270     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
2271 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2272     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
2273 bitLength="14" name="companyprefixindex"/>
2274     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"*
2275 bitLength="39" length="10" padChar="0" padDir="LEFT" name="serialref"/>
2276     </option>
2277     <option optionKey="6" pattern="00001000([01]{3})([01]{14})([01]{39})"
2278 grammar="'00001000' filter companyprefixindex serialref" >
2279     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
2280 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2281     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
2282 bitLength="14" name="companyprefixindex"/>
2283     <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[01]"*
2284 bitLength="39" length="11" padChar="0" padDir="LEFT" name="serialref"/>
2285     </option>
2286     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="companyprefix"
2287 characterSet="[0-9]"*
2288 function="TABLELOOKUP(companyprefixindex, tdt64bitcpi, companyprefixindex, companyprefix)"
2289 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
2290 tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
2291 tableParams="companyprefixindex"/>
2292     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="companyprefixlength"
2293 characterSet="[0-9]"* function="LENGTH(companyprefix)"/>
2294     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="companyprefixindex"
2295 characterSet="[0-9]"*
2296 function="TABLELOOKUP(companyprefix, tdt64bitcpi, companyprefix, companyprefixindex)"
2297 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
2298 tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index" tableParams="companyprefix"/>
2299     </level>
2300     <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sscc-64"
2301 requiredFormattingParameters="filter, taglength" >
2302     <option optionKey="12" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[([0-9]{12})\.[([0-
2303 9]{5})]" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2304     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2305 length="1" padChar="0" padDir="LEFT" name="filter"/>
2306     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
2307 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2308     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"*
2309 length="5" padChar="0" padDir="LEFT" name="serialref"/>
2310     </option>
2311     <option optionKey="11" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[([0-9]{11})\.[([0-
2312 9]{6})]" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2313     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2314 length="1" padChar="0" padDir="LEFT" name="filter"/>
2315     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
2316 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2317     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"*
2318 length="6" padChar="0" padDir="LEFT" name="serialref"/>
2319     </option>
2320     <option optionKey="10" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[([0-9]{10})\.[([0-
2321 9]{7})]" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2322     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2323 length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

2324     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2325 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2326     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2327 length="7" padChar="0" padDir="LEFT" name="serialref"/>
2328 </option>
2329 <option optionKey="9" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{9})\.[0-
2330 9]{8}" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2331     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
2332 length="1" padChar="0" padDir="LEFT" name="filter"/>
2333     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2334 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2335     <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*"
2336 length="8" padChar="0" padDir="LEFT" name="serialref"/>
2337 </option>
2338 <option optionKey="8" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{8})\.[0-
2339 9]{9}" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2340     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
2341 length="1" padChar="0" padDir="LEFT" name="filter"/>
2342     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*"
2343 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2344     <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2345 length="9" padChar="0" padDir="LEFT" name="serialref"/>
2346 </option>
2347 <option optionKey="7" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{7})\.[0-
2348 9]{10}" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2349     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
2350 length="1" padChar="0" padDir="LEFT" name="filter"/>
2351     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2352 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2353     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2354 length="10" padChar="0" padDir="LEFT" name="serialref"/>
2355 </option>
2356 <option optionKey="6" pattern="urn:epc:tag:sscc-64:([0-7]{1})\.[0-9]{6})\.[0-
2357 9]{11}" grammar="urn:epc:tag:sscc-64:' filter '.' companyprefix '.' serialref" >
2358     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
2359 length="1" padChar="0" padDir="LEFT" name="filter"/>
2360     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2361 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2362     <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]*"
2363 length="11" padChar="0" padDir="LEFT" name="serialref"/>
2364 </option>
2365 </level>
2366 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sscc" >
2367     <option optionKey="12" pattern="urn:epc:id:sscc:([0-9]{12})\.[0-9]{5}"
2368 grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2369     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
2370 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2371     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]*"
2372 length="5" padChar="0" padDir="LEFT" name="serialref"/>
2373 </option>
2374     <option optionKey="11" pattern="urn:epc:id:sscc:([0-9]{11})\.[0-9]{6}"
2375 grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2376     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
2377 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2378     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2379 length="6" padChar="0" padDir="LEFT" name="serialref"/>
2380 </option>
2381     <option optionKey="10" pattern="urn:epc:id:sscc:([0-9]{10})\.[0-9]{7}"
2382 grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2383     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
2384 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2385     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2386 length="7" padChar="0" padDir="LEFT" name="serialref"/>
2387 </option>
2388     <option optionKey="9" pattern="urn:epc:id:sscc:([0-9]{9})\.[0-9]{8}"
2389 grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2390     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
2391 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2392     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2393 length="8" padChar="0" padDir="LEFT" name="serialref"/>
2394 </option>

```

```

2395     <option optionKey="8" pattern="urn:epc:id:sscc:([0-9]{8})\.([0-9]{9})"
2396     grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2397     <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]*"
2398     length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2399     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
2400     length="9" padChar="0" padDir="LEFT" name="serialref"/>
2401     </option>
2402     <option optionKey="7" pattern="urn:epc:id:sscc:([0-9]{7})\.([0-9]{10})"
2403     grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2404     <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
2405     length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2406     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
2407     length="10" padChar="0" padDir="LEFT" name="serialref"/>
2408     </option>
2409     <option optionKey="6" pattern="urn:epc:id:sscc:([0-9]{6})\.([0-9]{11})"
2410     grammar="urn:epc:id:sscc:' companyprefix '.' serialref" >
2411     <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
2412     length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2413     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]*"
2414     length="11" padChar="0" padDir="LEFT" name="serialref"/>
2415     </option>
2416     </level>
2417     <level type="LEGACY" prefixMatch="sscc="
2418     requiredParsingParameters="companyprefixlength" >
2419     <option optionKey="12" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2420     companyprefix serialrefremainder checkdigit" >
2421     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2422     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2423     </option>
2424     <option optionKey="11" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2425     companyprefix serialrefremainder checkdigit" >
2426     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2427     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2428     </option>
2429     <option optionKey="10" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2430     companyprefix serialrefremainder checkdigit" >
2431     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2432     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2433     </option>
2434     <option optionKey="9" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2435     companyprefix serialrefremainder checkdigit" >
2436     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2437     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2438     </option>
2439     <option optionKey="8" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2440     companyprefix serialrefremainder checkdigit" >
2441     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2442     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2443     </option>
2444     <option optionKey="7" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2445     companyprefix serialrefremainder checkdigit" >
2446     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2447     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2448     </option>
2449     <option optionKey="6" pattern="sscc:([0-9]{18})" grammar="'sscc=' extensiondigit
2450     companyprefix serialrefremainder checkdigit" >
2451     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999"
2452     characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2453     </option>
2454     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccprefixremainder"
2455     characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
2456     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extensiondigit"
2457     characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
2458     <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialrefremainder"
2459     characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,companyprefixlength)"/>
2460     <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
2461     characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
2462     <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="companyprefix"
2463     characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,0,companyprefixlength)"/>
2464     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extensiondigit"
2465     characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>

```

```

2466     <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialrefremainder"
2467 characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
2468     <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
2469 characterSet="[0-9]*" length="17"
2470 function="CONCAT(extensiondigit,companyprefix,serialrefremainder)"/>
2471     <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
2472 characterSet="[0-9]*" length="1" function="GS1CHECKSUM(ssccprefix)"/>
2473   </level>
2474 </scheme>
2475 </epcTagDataTranslation>

```

8.5. SSCC-96 Serial Shipping Container Code for 96-bit EPCs

```

2476 <?xml version="1.0" encoding="UTF-8"?>
2477 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
2478 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
2479 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
2480   <scheme name="SSCC-96" optionKey="companyprefixlength" tagLength="96">
2481     <level type="BINARY" prefixMatch="00110001"
2482 requiredFormattingParameters="filter,taglength" >
2483       <option optionKey="12">
2484         pattern="00110001([01]{3})000([01]{40})([01]{18})000000000000000000000000"
2485         grammar="'00110001' filter '000' companyprefix serialref '000000000000000000000000'" >
2486           <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2487 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2488           <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2489 bitLength="40" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2490           <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*"
2491 bitLength="18" length="5" padChar="0" padDir="LEFT" name="serialref"/>
2492         </option>
2493       <option optionKey="11">
2494         pattern="00110001([01]{3})001([01]{37})([01]{21})000000000000000000000000"
2495         grammar="'00110001' filter '001' companyprefix serialref '000000000000000000000000'" >
2496           <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2497 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2498           <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2499 bitLength="37" length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2500           <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*"
2501 bitLength="21" length="6" padChar="0" padDir="LEFT" name="serialref"/>
2502         </option>
2503       <option optionKey="10">
2504         pattern="00110001([01]{3})010([01]{34})([01]{24})000000000000000000000000"
2505         grammar="'00110001' filter '010' companyprefix serialref '000000000000000000000000'" >
2506           <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2507 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2508           <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2509 bitLength="34" length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2510           <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]*"
2511 bitLength="24" length="7" padChar="0" padDir="LEFT" name="serialref"/>
2512         </option>
2513       <option optionKey="9">
2514         pattern="00110001([01]{3})011([01]{30})([01]{28})000000000000000000000000"
2515         grammar="'00110001' filter '011' companyprefix serialref '000000000000000000000000'" >
2516           <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2517 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2518           <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2519 bitLength="30" length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2520           <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2521 bitLength="28" length="8" padChar="0" padDir="LEFT" name="serialref"/>
2522         </option>
2523       <option optionKey="8">
2524         pattern="00110001([01]{3})100([01]{27})([01]{31})000000000000000000000000"
2525         grammar="'00110001' filter '100' companyprefix serialref '000000000000000000000000'" >
2526           <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
2527 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2528           <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2529 bitLength="27" length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2530           <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
2531 bitLength="31" length="9" padChar="0" padDir="LEFT" name="serialref"/>
2532         </option>
2533       </level>

```

```

2534     </option>
2535     <option optionKey="7"
2536 pattern="00110001([01]{3})101([01]{24})([01]{34})00000000000000000000"
2537 grammar="'00110001' filter '101' companyprefix serialref '00000000000000000000'" >
2538     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
2539 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2540     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]"*
2541 bitLength="24" length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2542     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"*
2543 bitLength="34" length="10" padChar="0" padDir="LEFT" name="serialref"/>
2544     </option>
2545     <option optionKey="6"
2546 pattern="00110001([01]{3})110([01]{20})([01]{38})00000000000000000000"
2547 grammar="'00110001' filter '110' companyprefix serialref '00000000000000000000'" >
2548     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
2549 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2550     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]"*
2551 bitLength="20" length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2552     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"*
2553 bitLength="38" length="11" padChar="0" padDir="LEFT" name="serialref"/>
2554     </option>
2555 </level>
2556 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sscc-96"
2557 requiredFormattingParameters="filter,taglength" >
2558     <option optionKey="12" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{12})\.[([0-
2559 9]{5})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >
2560     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2561 length="1" padChar="0" padDir="LEFT" name="filter"/>
2562     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
2563 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2564     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"*
2565 length="5" padChar="0" padDir="LEFT" name="serialref"/>
2566     </option>
2567     <option optionKey="11" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{11})\.[([0-
2568 9]{6})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >
2569     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2570 length="1" padChar="0" padDir="LEFT" name="filter"/>
2571     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
2572 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2573     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"*
2574 length="6" padChar="0" padDir="LEFT" name="serialref"/>
2575     </option>
2576     <option optionKey="10" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{10})\.[([0-
2577 9]{7})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >
2578     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2579 length="1" padChar="0" padDir="LEFT" name="filter"/>
2580     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
2581 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2582     <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
2583 length="7" padChar="0" padDir="LEFT" name="serialref"/>
2584     </option>
2585     <option optionKey="9" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{9})\.[([0-
2586 9]{8})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >
2587     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2588 length="1" padChar="0" padDir="LEFT" name="filter"/>
2589     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
2590 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2591     <field seq="3" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
2592 length="8" padChar="0" padDir="LEFT" name="serialref"/>
2593     </option>
2594     <option optionKey="8" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{8})\.[([0-
2595 9]{9})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >
2596     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2597 length="1" padChar="0" padDir="LEFT" name="filter"/>
2598     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
2599 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2600     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"*
2601 length="9" padChar="0" padDir="LEFT" name="serialref"/>
2602     </option>
2603     <option optionKey="7" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[([0-9]{7})\.[([0-
2604 9]{10})]" grammar="'urn:epc:tag:sscc-96:' filter '.' companyprefix '.' serialref" >

```

```

2605     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2606 length="1" padChar="0" padDir="LEFT" name="filter"/>
2607     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
2608 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2609     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"*
2610 length="10" padChar="0" padDir="LEFT" name="serialref"/>
2611 </option>
2612 <option optionKey="6" pattern="urn:epc:tag:sscc-96:([0-7]{1})\.[0-9]{6})\.[0-
2613 9]{11}" grammar="urn:epc:tag:sscc-96: filter '.' companyprefix '.' serialref" >
2614     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2615 length="1" padChar="0" padDir="LEFT" name="filter"/>
2616     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"*
2617 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2618     <field seq="3" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2619 length="11" padChar="0" padDir="LEFT" name="serialref"/>
2620 </option>
2621 </level>
2622 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sscc" >
2623     <option optionKey="12" pattern="urn:epc:id:sscc:([0-9]{12})\.[0-9]{5}"
2624 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2625     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
2626 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2627     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"*
2628 length="5" padChar="0" padDir="LEFT" name="serialref"/>
2629 </option>
2630     <option optionKey="11" pattern="urn:epc:id:sscc:([0-9]{11})\.[0-9]{6}"
2631 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2632     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2633 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2634     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"*
2635 length="6" padChar="0" padDir="LEFT" name="serialref"/>
2636 </option>
2637     <option optionKey="10" pattern="urn:epc:id:sscc:([0-9]{10})\.[0-9]{7}"
2638 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2639     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2640 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2641     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
2642 length="7" padChar="0" padDir="LEFT" name="serialref"/>
2643 </option>
2644     <option optionKey="9" pattern="urn:epc:id:sscc:([0-9]{9})\.[0-9]{8}"
2645 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2646     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"*
2647 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2648     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
2649 length="8" padChar="0" padDir="LEFT" name="serialref"/>
2650 </option>
2651     <option optionKey="8" pattern="urn:epc:id:sscc:([0-9]{8})\.[0-9]{9}"
2652 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2653     <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
2654 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2655     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"*
2656 length="9" padChar="0" padDir="LEFT" name="serialref"/>
2657 </option>
2658     <option optionKey="7" pattern="urn:epc:id:sscc:([0-9]{7})\.[0-9]{10}"
2659 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2660     <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
2661 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2662     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2663 length="10" padChar="0" padDir="LEFT" name="serialref"/>
2664 </option>
2665     <option optionKey="6" pattern="urn:epc:id:sscc:([0-9]{6})\.[0-9]{11}"
2666 grammar="urn:epc:id:sscc: companyprefix '.' serialref" >
2667     <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
2668 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2669     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
2670 length="11" padChar="0" padDir="LEFT" name="serialref"/>
2671 </option>
2672 </level>
2673 <level type="LEGACY" prefixMatch="sscc"
2674 requiredParsingParameters="companyprefixlength" >

```



```

2675     <option optionKey="12" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2676 companyprefix serialrefremainder checkdigit" >
2677     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2678 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2679     </option>
2680     <option optionKey="11" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2681 companyprefix serialrefremainder checkdigit" >
2682     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2683 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2684     </option>
2685     <option optionKey="10" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2686 companyprefix serialrefremainder checkdigit" >
2687     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2688 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2689     </option>
2690     <option optionKey="9" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2691 companyprefix serialrefremainder checkdigit" >
2692     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2693 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2694     </option>
2695     <option optionKey="8" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2696 companyprefix serialrefremainder checkdigit" >
2697     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2698 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2699     </option>
2700     <option optionKey="7" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2701 companyprefix serialrefremainder checkdigit" >
2702     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2703 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2704     </option>
2705     <option optionKey="6" pattern="sscc=([0-9]{18})" grammar="'sscc=' extensiondigit
2706 companyprefix serialrefremainder checkdigit" >
2707     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999"
2708 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="sscc"/>
2709     </option>
2710     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="ssccprefixremainder"
2711 characterSet="[0-9]*" length="16" function="SUBSTR(sscc,1,16)"/>
2712     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="extensiondigit"
2713 characterSet="[0-9]*" length="1" function="SUBSTR(sscc,0,1)"/>
2714     <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="serialrefremainder"
2715 characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,companyprefixlength)"/>
2716     <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="serialref"
2717 characterSet="[0-9]*" function="CONCAT(extensiondigit,serialrefremainder)"/>
2718     <rule type="EXTRACT" inputFormat="STRING" seq="5" newFieldName="companyprefix"
2719 characterSet="[0-9]*" function="SUBSTR(ssccprefixremainder,0,companyprefixlength)"/>
2720     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="extensiondigit"
2721 characterSet="[0-9]*" length="1" function="SUBSTR(serialref,0,1)"/>
2722     <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="serialrefremainder"
2723 characterSet="[0-9]*" function="SUBSTR(serialref,1)"/>
2724     <rule type="FORMAT" inputFormat="STRING" seq="3" newFieldName="ssccprefix"
2725 characterSet="[0-9]*" length="17"
2726 function="CONCAT(extensiondigit,companyprefix,serialrefremainder)"/>
2727     <rule type="FORMAT" inputFormat="STRING" seq="4" newFieldName="checkdigit"
2728 characterSet="[0-9]*" length="1" function="GS1CHECKSUM(ssccprefix)"/>
2729     </level>
2730 </scheme>
2731 </epcTagDataTranslation>
2732

```

2733 8.6. SGLN-64 Serialized Global Location Number for 64-bit 2734 EPCs

```

2735 <?xml version="1.0" encoding="UTF-8"?>
2736 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
2737 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
2738 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
2739   <scheme name="SGLN-64" optionKey="companyprefixlength" tagLength="64" >
2740     <level type="BINARY" prefixMatch="00001001"
2741     requiredFormattingParameters="filter,taglength" >

```

```

2742     <option optionKey="12" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2743     grammar="'00001001' filter companyprefixindex locationref serial" >
2744     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2745     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2746     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2747     bitLength="14" name="companyprefixindex"/>
2748     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2749     bitLength="20" length="0" padChar="0" padDir="LEFT" name="locationref"/>
2750     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2751     bitLength="19" name="serial"/>
2752     </option>
2753     <option optionKey="11" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2754     grammar="'00001001' filter companyprefixindex locationref serial" >
2755     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2756     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2757     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2758     bitLength="14" name="companyprefixindex"/>
2759     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2760     bitLength="20" length="1" padChar="0" padDir="LEFT" name="locationref"/>
2761     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2762     bitLength="19" name="serial"/>
2763     </option>
2764     <option optionKey="10" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2765     grammar="'00001001' filter companyprefixindex locationref serial" >
2766     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2767     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2768     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2769     bitLength="14" name="companyprefixindex"/>
2770     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2771     bitLength="20" length="2" padChar="0" padDir="LEFT" name="locationref"/>
2772     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2773     bitLength="19" name="serial"/>
2774     </option>
2775     <option optionKey="9" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2776     grammar="'00001001' filter companyprefixindex locationref serial" >
2777     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2778     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2779     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2780     bitLength="14" name="companyprefixindex"/>
2781     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2782     bitLength="20" length="3" padChar="0" padDir="LEFT" name="locationref"/>
2783     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2784     bitLength="19" name="serial"/>
2785     </option>
2786     <option optionKey="8" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2787     grammar="'00001001' filter companyprefixindex locationref serial" >
2788     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2789     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2790     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2791     bitLength="14" name="companyprefixindex"/>
2792     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2793     bitLength="20" length="4" padChar="0" padDir="LEFT" name="locationref"/>
2794     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2795     bitLength="19" name="serial"/>
2796     </option>
2797     <option optionKey="7" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2798     grammar="'00001001' filter companyprefixindex locationref serial" >
2799     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2800     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
2801     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"
2802     bitLength="14" name="companyprefixindex"/>
2803     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"
2804     bitLength="20" length="5" padChar="0" padDir="LEFT" name="locationref"/>
2805     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"
2806     bitLength="19" name="serial"/>
2807     </option>
2808     <option optionKey="6" pattern="00001001([01]{3})([01]{14})([01]{20})([01]{19})"
2809     grammar="'00001001' filter companyprefixindex locationref serial" >
2810     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
2811     bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

2812     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
2813 bitLength="14" name="companyprefixindex"/>
2814     <field seq="3" decimalMinimum="0" decimalMaximum="1048575" characterSet="[01]"*
2815 bitLength="20" length="6" padChar="0" padDir="LEFT" name="locationref"/>
2816     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]"*
2817 bitLength="19" name="serial"/>
2818   </option>
2819   <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="companyprefix"
2820 characterSet="[0-9]"*
2821 function="TABLELOOKUP(companyprefixindex,tdt64bitcpi,companyprefixindex,companyprefix)"
2822 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
2823 tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
2824 tableParams="companyprefixindex"/>
2825   <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="companyprefixlength"
2826 characterSet="[0-9]"* function="LENGTH(companyprefix)"/>
2827   <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="companyprefixindex"
2828 characterSet="[0-9]"*
2829 function="TABLELOOKUP(companyprefix,tdt64bitcpi,companyprefix,companyprefixindex)"
2830 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
2831 tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index" tableParams="companyprefix"/>
2832 </level>
2833 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgln-64"
2834 requiredFormattingParameters="filter,taglength" >
2835   <option optionKey="12" pattern="urn:epc:tag:sgln-64:([0-7]{1})\\.([0-9]{12})\\.([0-
2836 9]{0})\\.([0-9]*)" grammar="urn:epc:tag:sgln-64: filter '.' companyprefix '.'
2837 locationref '.' serial" >
2838     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2839 length="1" padChar="0" padDir="LEFT" name="filter"/>
2840     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-
2841 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2842     <field seq="3" decimalMinimum="0" decimalMaximum="1" characterSet="[0-9]"*
2843 length="1" padChar="0" padDir="LEFT" name="locationref"/>
2844     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"*
2845 name="serial"/>
2846   </option>
2847   <option optionKey="11" pattern="urn:epc:tag:sgln-64:([0-7]{1})\\.([0-9]{11})\\.([0-
2848 9]{1})\\.([0-9]*)" grammar="urn:epc:tag:sgln-64: filter '.' companyprefix '.'
2849 locationref '.' serial" >
2850     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2851 length="1" padChar="0" padDir="LEFT" name="filter"/>
2852     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2853 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2854     <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
2855 length="1" padChar="0" padDir="LEFT" name="locationref"/>
2856     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"*
2857 name="serial"/>
2858   </option>
2859   <option optionKey="10" pattern="urn:epc:tag:sgln-64:([0-7]{1})\\.([0-9]{10})\\.([0-
2860 9]{2})\\.([0-9]*)" grammar="urn:epc:tag:sgln-64: filter '.' companyprefix '.'
2861 locationref '.' serial" >
2862     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2863 length="1" padChar="0" padDir="LEFT" name="filter"/>
2864     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2865 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2866     <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"*
2867 length="2" padChar="0" padDir="LEFT" name="locationref"/>
2868     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"*
2869 name="serial"/>
2870   </option>
2871   <option optionKey="9" pattern="urn:epc:tag:sgln-64:([0-7]{1})\\.([0-9]{9})\\.([0-
2872 9]{3})\\.([0-9]*)" grammar="urn:epc:tag:sgln-64: filter '.' companyprefix '.'
2873 locationref '.' serial" >
2874     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
2875 length="1" padChar="0" padDir="LEFT" name="filter"/>
2876     <field seq="2" decimalMinimum="0" decimalMaximum="99999999999" characterSet="[0-9]"*
2877 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2878     <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"*
2879 length="3" padChar="0" padDir="LEFT" name="locationref"/>
2880     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"*
2881 name="serial"/>
2882   </option>

```

```

2883     <option optionKey="8" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{8})\.[0-
2884 9]{4})\.[0-9]*" grammar="urn:epc:tag:sgln-64:' filter '.' companyprefix '.'
2885 locationref '.' serial" >
2886     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
2887 length="1" padChar="0" padDir="LEFT" name="filter"/>
2888     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2889 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2890     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
2891 length="4" padChar="0" padDir="LEFT" name="locationref"/>
2892     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2893 name="serial"/>
2894 </option>
2895     <option optionKey="7" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{7})\.[0-
2896 9]{5})\.[0-9]*" grammar="urn:epc:tag:sgln-64:' filter '.' companyprefix '.'
2897 locationref '.' serial" >
2898     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
2899 length="1" padChar="0" padDir="LEFT" name="filter"/>
2900     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2901 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2902     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
2903 length="5" padChar="0" padDir="LEFT" name="locationref"/>
2904     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2905 name="serial"/>
2906 </option>
2907     <option optionKey="6" pattern="urn:epc:tag:sgln-64:([0-7]{1})\.[0-9]{6})\.[0-
2908 9]{6})\.[0-9]*" grammar="urn:epc:tag:sgln-64:' filter '.' companyprefix '.'
2909 locationref '.' serial" >
2910     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
2911 length="1" padChar="0" padDir="LEFT" name="filter"/>
2912     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
2913 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2914     <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
2915 length="6" padChar="0" padDir="LEFT" name="locationref"/>
2916     <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2917 name="serial"/>
2918 </option>
2919 </level>
2920     <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgln" >
2921     <option optionKey="12" pattern="urn:epc:id:sgln:([0-9]{12})\.[0-9]{0})\.[0-9]*"
2922 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2923     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
2924 9]" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
2925     <field seq="2" decimalMinimum="0" decimalMaximum="1" characterSet="[0-9]"
2926 length="0" padChar="0" padDir="LEFT" name="locationref"/>
2927     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2928 name="serial"/>
2929 </option>
2930     <option optionKey="11" pattern="urn:epc:id:sgln:([0-9]{11})\.[0-9]{1})\.[0-9]*"
2931 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2932     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
2933 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
2934     <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"
2935 length="1" padChar="0" padDir="LEFT" name="locationref"/>
2936     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2937 name="serial"/>
2938 </option>
2939     <option optionKey="10" pattern="urn:epc:id:sgln:([0-9]{10})\.[0-9]{2})\.[0-9]*"
2940 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2941     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
2942 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
2943     <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"
2944 length="2" padChar="0" padDir="LEFT" name="locationref"/>
2945     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2946 name="serial"/>
2947 </option>
2948     <option optionKey="9" pattern="urn:epc:id:sgln:([0-9]{9})\.[0-9]{3})\.[0-9]*"
2949 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2950     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
2951 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
2952     <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"
2953 length="3" padChar="0" padDir="LEFT" name="locationref"/>

```

```

2954     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2955     name="serial"/>
2956   </option>
2957   <option optionKey="8" pattern="urn:epc:id:sgln:([0-9]{8})\.([0-9]{4})\.([0-9]*)"
2958   grammar="'urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2959     <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2960     length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
2961     <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
2962     length="4" padChar="0" padDir="LEFT" name="locationref"/>
2963     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2964     name="serial"/>
2965   </option>
2966   <option optionKey="7" pattern="urn:epc:id:sgln:([0-9]{7})\.([0-9]{5})\.([0-9]*)"
2967   grammar="'urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2968     <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2969     length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
2970     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
2971     length="5" padChar="0" padDir="LEFT" name="locationref"/>
2972     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2973     name="serial"/>
2974   </option>
2975   <option optionKey="6" pattern="urn:epc:id:sgln:([0-9]{6})\.([0-9]{6})\.([0-9]*)"
2976   grammar="'urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
2977     <field seq="1" decimalMinimum="0" decimalMaximum="99999999" characterSet="[0-9]"
2978     length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
2979     <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
2980     length="6" padChar="0" padDir="LEFT" name="locationref"/>
2981     <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2982     name="serial"/>
2983   </option>
2984 </level>
2985 <level type="LEGACY" prefixMatch="gln" requiredParsingParameters="companyprefixlength"
2986 >
2987   <option optionKey="12" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
2988   companyprefix locationref checkdigit ';serial=' serial" >
2989     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2990 9]" length="13" padChar="0" padDir="LEFT" name="gln"/>
2991     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2992     name="serial"/>
2993   </option>
2994   <option optionKey="11" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
2995   companyprefix locationref checkdigit ';serial=' serial" >
2996     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
2997 9]" length="13" padChar="0" padDir="LEFT" name="gln"/>
2998     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
2999     name="serial"/>
3000   </option>
3001   <option optionKey="10" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3002   companyprefix locationref checkdigit ';serial=' serial" >
3003     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
3004 9]" length="13" padChar="0" padDir="LEFT" name="gln"/>
3005     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3006     name="serial"/>
3007   </option>
3008   <option optionKey="9" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3009   companyprefix locationref checkdigit ';serial=' serial" >
3010     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
3011 9]" length="13" padChar="0" padDir="LEFT" name="gln"/>
3012     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3013     name="serial"/>
3014   </option>
3015   <option optionKey="8" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3016   companyprefix locationref checkdigit ';serial=' serial" >
3017     <field seq="1" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
3018 9]" length="13" padChar="0" padDir="LEFT" name="gln"/>
3019     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3020     name="serial"/>
3021   </option>
3022   <option optionKey="7" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3023   companyprefix locationref checkdigit ';serial=' serial" >

```

```

3024     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3025 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3026     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3027 name="serial"/>
3028   </option>
3029   <option optionKey="6" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3030 companyprefix locationref checkdigit ';serial=' serial" >
3031     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3032 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3033     <field seq="2" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3034 name="serial"/>
3035   </option>
3036   <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnprefixremainder"
3037 characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)"/>
3038   <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="locationref"
3039 characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,companyprefixlength)"/>
3040   <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="companyprefix"
3041 characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,0,companyprefixlength)"/>
3042   <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
3043 characterSet="[0-9]*" length="12" function="CONCAT(companyprefix,locationref)"/>
3044   <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
3045 characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)"/>
3046 </level>
3047 </scheme>
3048 </epcTagDataTranslation>

```

3049 8.7. SGLN-96 Serialized Global Location Number for 96-bit 3050 EPCs

```

3051 <?xml version="1.0" encoding="UTF-8"?>
3052 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
3053 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
3054 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
3055   <scheme name="SGLN-96" optionKey="companyprefixlength" tagLength="96" >
3056     <level type="BINARY" prefixMatch="00110010"
3057 requiredFormattingParameters="filter,taglength" >
3058       <option optionKey="12" pattern="00110010([01]{3})000([01]{40})([01]{1})([01]{41})"
3059 grammar="'00110010' filter '000' companyprefix locationref serial" >
3060         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3061 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3062         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
3063 bitLength="40" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3064         <field seq="3" characterSet="[01]*" bitLength="1" length="0" padChar="0"
3065 padDir="LEFT" name="locationref"/>
3066         <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
3067 characterSet="[01]*" bitLength="41" name="serial"/>
3068       </option>
3069       <option optionKey="11" pattern="00110010([01]{3})001([01]{37})([01]{4})([01]{41})"
3070 grammar="'00110010' filter '001' companyprefix locationref serial" >
3071         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3072 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3073         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
3074 bitLength="37" length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3075         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
3076 bitLength="4" length="1" padChar="0" padDir="LEFT" name="locationref"/>
3077         <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
3078 characterSet="[01]*" bitLength="41" name="serial"/>
3079       </option>
3080       <option optionKey="10" pattern="00110010([01]{3})010([01]{34})([01]{7})([01]{41})"
3081 grammar="'00110010' filter '010' companyprefix locationref serial" >
3082         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3083 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3084         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]*"
3085 bitLength="34" length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3086         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
3087 bitLength="7" length="2" padChar="0" padDir="LEFT" name="locationref"/>
3088         <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551"
3089 characterSet="[01]*" bitLength="41" name="serial"/>
3090       </option>

```

```

3091 <option optionKey="9" pattern="00110010([01]{3})011([01]{30})([01]{11})([01]{41})"
3092 grammar="'00110010' filter '011' companyprefix locationref serial" >
3093 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3094 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3095 <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]"*
3096 bitLength="30" length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3097 <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]"*
3098 bitLength="11" length="3" padChar="0" padDir="LEFT" name="locationref"/>
3099 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551"
3100 characterSet="[01]"* bitLength="41" name="serial"/>
3101 </option>
3102 <option optionKey="8" pattern="00110010([01]{3})100([01]{27})([01]{14})([01]{41})"
3103 grammar="'00110010' filter '100' companyprefix locationref serial" >
3104 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3105 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3106 <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[01]"*
3107 bitLength="27" length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3108 <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]"*
3109 bitLength="14" length="4" padChar="0" padDir="LEFT" name="locationref"/>
3110 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551"
3111 characterSet="[01]"* bitLength="41" name="serial"/>
3112 </option>
3113 <option optionKey="7" pattern="00110010([01]{3})101([01]{24})([01]{17})([01]{41})"
3114 grammar="'00110010' filter '101' companyprefix locationref serial" >
3115 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3116 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3117 <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]"*
3118 bitLength="24" length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3119 <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]"*
3120 bitLength="17" length="5" padChar="0" padDir="LEFT" name="locationref"/>
3121 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551"
3122 characterSet="[01]"* bitLength="41" name="serial"/>
3123 </option>
3124 <option optionKey="6" pattern="00110010([01]{3})110([01]{20})([01]{21})([01]{41})"
3125 grammar="'00110010' filter '110' companyprefix locationref serial" >
3126 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3127 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3128 <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]"*
3129 bitLength="20" length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3130 <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]"*
3131 bitLength="21" length="6" padChar="0" padDir="LEFT" name="locationref"/>
3132 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551"
3133 characterSet="[01]"* bitLength="41" name="serial"/>
3134 </option>
3135 </level>
3136 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:sgln-96"
3137 requiredFormattingParameters="filter,taglength" >
3138 <option optionKey="12" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[([0-9]{12})\.[([0-
3139 9]{0})\.[([0-9]*)" grammar="'urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3140 locationref '.' serial" >
3141 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3142 length="1" padChar="0" padDir="LEFT" name="filter"/>
3143 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3144 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3145 <field seq="3" decimalMinimum="0" decimalMaximum="1" characterSet="[0-9]"*
3146 length="0" padChar="0" padDir="LEFT" name="locationref"/>
3147 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-
3148 9]"* name="serial"/>
3149 </option>
3150 <option optionKey="11" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[([0-9]{11})\.[([0-
3151 9]{1})\.[([0-9]*)" grammar="'urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3152 locationref '.' serial" >
3153 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3154 length="1" padChar="0" padDir="LEFT" name="filter"/>
3155 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
3156 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3157 <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
3158 length="1" padChar="0" padDir="LEFT" name="locationref"/>
3159 <field seq="4" decimalMinimum="0" decimalMaximum="219902325551" characterSet="[0-
3160 9]"* name="serial"/>
3161 </option>

```

```

3162     <option optionKey="10" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{10}\.[0-
3163 9]{2})\.[0-9]*" grammar="urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3164 locationref '.' serial" >
3165     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3166 length="1" padChar="0" padDir="LEFT" name="filter"/>
3167     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3168 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3169     <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"
3170 length="2" padChar="0" padDir="LEFT" name="locationref"/>
3171     <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3172 9]" name="serial"/>
3173 </option>
3174     <option optionKey="9" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{9}\.[0-
3175 9]{3})\.[0-9]*" grammar="urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3176 locationref '.' serial" >
3177     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3178 length="1" padChar="0" padDir="LEFT" name="filter"/>
3179     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3180 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3181     <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"
3182 length="3" padChar="0" padDir="LEFT" name="locationref"/>
3183     <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3184 9]" name="serial"/>
3185 </option>
3186     <option optionKey="8" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{8}\.[0-
3187 9]{4})\.[0-9]*" grammar="urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3188 locationref '.' serial" >
3189     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3190 length="1" padChar="0" padDir="LEFT" name="filter"/>
3191     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3192 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3193     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
3194 length="4" padChar="0" padDir="LEFT" name="locationref"/>
3195     <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3196 9]" name="serial"/>
3197 </option>
3198     <option optionKey="7" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{7}\.[0-
3199 9]{5})\.[0-9]*" grammar="urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3200 locationref '.' serial" >
3201     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3202 length="1" padChar="0" padDir="LEFT" name="filter"/>
3203     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3204 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3205     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
3206 length="5" padChar="0" padDir="LEFT" name="locationref"/>
3207     <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3208 9]" name="serial"/>
3209 </option>
3210     <option optionKey="6" pattern="urn:epc:tag:sgln-96:([0-7]{1})\.[0-9]{6}\.[0-
3211 9]{6})\.[0-9]*" grammar="urn:epc:tag:sgln-96:' filter '.' companyprefix '.'
3212 locationref '.' serial" >
3213     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3214 length="1" padChar="0" padDir="LEFT" name="filter"/>
3215     <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3216 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3217     <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
3218 length="6" padChar="0" padDir="LEFT" name="locationref"/>
3219     <field seq="4" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3220 9]" name="serial"/>
3221 </option>
3222 </level>
3223     <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:sgln" >
3224     <option optionKey="12" pattern="urn:epc:id:sgln:([0-9]{12})\.[0-9]{0}\.[0-9]*"
3225 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3226     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
3227 9]" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3228     <field seq="2" decimalMinimum="0" decimalMaximum="1" characterSet="[0-9]"
3229 length="0" padChar="0" padDir="LEFT" name="locationref"/>
3230     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3231 9]" name="serial"/>
3232 </option>

```



```

3233     <option optionKey="11" pattern="urn:epc:id:sgln:([0-9]{11})\.[0-9]{1}\.[0-9]*"
3234 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3235     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3236 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3237     <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"
3238 length="1" padChar="0" padDir="LEFT" name="locationref"/>
3239     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3240 9]*" name="serial"/>
3241     </option>
3242     <option optionKey="10" pattern="urn:epc:id:sgln:([0-9]{10})\.[0-9]{2}\.[0-9]*"
3243 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3244     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3245 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3246     <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"
3247 length="2" padChar="0" padDir="LEFT" name="locationref"/>
3248     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3249 9]*" name="serial"/>
3250     </option>
3251     <option optionKey="9" pattern="urn:epc:id:sgln:([0-9]{9})\.[0-9]{3}\.[0-9]*"
3252 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3253     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3254 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3255     <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"
3256 length="3" padChar="0" padDir="LEFT" name="locationref"/>
3257     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3258 9]*" name="serial"/>
3259     </option>
3260     <option optionKey="8" pattern="urn:epc:id:sgln:([0-9]{8})\.[0-9]{4}\.[0-9]*"
3261 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3262     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3263 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3264     <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
3265 length="4" padChar="0" padDir="LEFT" name="locationref"/>
3266     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3267 9]*" name="serial"/>
3268     </option>
3269     <option optionKey="7" pattern="urn:epc:id:sgln:([0-9]{7})\.[0-9]{5}\.[0-9]*"
3270 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3271     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3272 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3273     <field seq="2" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
3274 length="5" padChar="0" padDir="LEFT" name="locationref"/>
3275     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3276 9]*" name="serial"/>
3277     </option>
3278     <option optionKey="6" pattern="urn:epc:id:sgln:([0-9]{6})\.[0-9]{6}\.[0-9]*"
3279 grammar="urn:epc:id:sgln:' companyprefix '.' locationref '.' serial" >
3280     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
3281 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3282     <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]"
3283 length="6" padChar="0" padDir="LEFT" name="locationref"/>
3284     <field seq="3" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3285 9]*" name="serial"/>
3286     </option>
3287     </level>
3288     <level type="LEGACY" prefixMatch="gln" requiredParsingParameters="companyprefixlength"
3289     >
3290         <option optionKey="12" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln=' gln
3291         ';serial=' serial" >
3292             <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
3293 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3294             <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3295 9]*" name="serial"/>
3296             </option>
3297             <option optionKey="11" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3298             companyprefix locationref checkdigit ';serial=' serial" >
3299                 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
3300 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3301                 <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3302 9]*" name="serial"/>
3303                 </option>

```

```

3304     <option optionKey="10" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3305 companyprefix locationref checkdigit ';serial=' serial" >
3306     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3307 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3308     <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3309 9]*" name="serial"/>
3310 </option>
3311     <option optionKey="9" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3312 companyprefix locationref checkdigit ';serial=' serial" >
3313     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3314 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3315     <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3316 9]*" name="serial"/>
3317 </option>
3318     <option optionKey="8" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3319 companyprefix locationref checkdigit ';serial=' serial" >
3320     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3321 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3322     <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3323 9]*" name="serial"/>
3324 </option>
3325     <option optionKey="7" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3326 companyprefix locationref checkdigit ';serial=' serial" >
3327     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3328 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3329     <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3330 9]*" name="serial"/>
3331 </option>
3332     <option optionKey="6" pattern="gln=([0-9]{13});serial=([0-9]*)" grammar="'gln='
3333 companyprefix locationref checkdigit ';serial=' serial" >
3334     <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3335 9]*" length="13" padChar="0" padDir="LEFT" name="gln"/>
3336     <field seq="2" decimalMinimum="0" decimalMaximum="2199023255551" characterSet="[0-
3337 9]*" name="serial"/>
3338 </option>
3339     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="glnprefixremainder"
3340 characterSet="[0-9]*" length="12" function="SUBSTR(gln,0,12)/>
3341     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="locationref"
3342 characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,companyprefixlength)/>
3343     <rule type="EXTRACT" inputFormat="STRING" seq="3" newFieldName="companyprefix"
3344 characterSet="[0-9]*" function="SUBSTR(glnprefixremainder,0,companyprefixlength)/>
3345     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="glnprefix"
3346 characterSet="[0-9]*" length="12" function="CONCAT(companyprefix,locationref)/>
3347     <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
3348 characterSet="[0-9]*" length="1" function="GS1CHECKSUM(glnprefix)/>
3349 </level>
3350 </scheme>
3351 </epcTagDataTranslation>

```

3352 8.8. GRAI-64 Global Returnable Asset Identifier for 64-bit 3353 EPCs

```

3354 <?xml version="1.0" encoding="UTF-8"?>
3355 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
3356 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
3357 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
3358     <scheme name="GRAI-64" optionKey="companyprefixlength" tagLength="64" >
3359         <level type="BINARY" prefixMatch="00001010"
3360 requiredFormattingParameters="filter,taglength" >
3361             <option optionKey="12" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3362 grammar="'00001010' filter companyprefixindex assettype serial" >
3363                 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3364 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3365                 <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3366 bitLength="14" name="companyprefixindex"/>
3367                 <field seq="3" characterSet="[01]*" bitLength="20" length="0" padChar="0"
3368 padDir="LEFT" name="assettype"/>
3369                 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3370 bitLength="19" name="serial"/>

```

```

3371     </option>
3372     <option optionKey="11" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3373     grammar="'00001010' filter companyprefixindex assettype serial" >
3374         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3375         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3376         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3377         bitLength="14" name="companyprefixindex"/>
3378         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]*"
3379         bitLength="20" length="1" padChar="0" padDir="LEFT" name="assettype"/>
3380         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3381         bitLength="19" name="serial"/>
3382     </option>
3383     <option optionKey="10" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3384     grammar="'00001010' filter companyprefixindex assettype serial" >
3385         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3386         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3387         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3388         bitLength="14" name="companyprefixindex"/>
3389         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]*"
3390         bitLength="20" length="2" padChar="0" padDir="LEFT" name="assettype"/>
3391         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3392         bitLength="19" name="serial"/>
3393     </option>
3394     <option optionKey="9" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3395     grammar="'00001010' filter companyprefixindex assettype serial" >
3396         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3397         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3398         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3399         bitLength="14" name="companyprefixindex"/>
3400         <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]*"
3401         bitLength="20" length="3" padChar="0" padDir="LEFT" name="assettype"/>
3402         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3403         bitLength="19" name="serial"/>
3404     </option>
3405     <option optionKey="8" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3406     grammar="'00001010' filter companyprefixindex assettype serial" >
3407         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3408         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3409         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3410         bitLength="14" name="companyprefixindex"/>
3411         <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]*"
3412         bitLength="20" length="4" padChar="0" padDir="LEFT" name="assettype"/>
3413         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3414         bitLength="19" name="serial"/>
3415     </option>
3416     <option optionKey="7" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3417     grammar="'00001010' filter companyprefixindex assettype serial" >
3418         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3419         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3420         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3421         bitLength="14" name="companyprefixindex"/>
3422         <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]*"
3423         bitLength="20" length="5" padChar="0" padDir="LEFT" name="assettype"/>
3424         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3425         bitLength="19" name="serial"/>
3426     </option>
3427     <option optionKey="6" pattern="00001010([01]{3})([01]{14})([01]{20})([01]{19})"
3428     grammar="'00001010' filter companyprefixindex assettype serial" >
3429         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]*"
3430         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3431         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]*"
3432         bitLength="14" name="companyprefixindex"/>
3433         <field seq="3" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]*"
3434         bitLength="20" length="6" padChar="0" padDir="LEFT" name="assettype"/>
3435         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[01]*"
3436         bitLength="19" name="serial"/>
3437     </option>
3438     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="companyprefix"
3439     characterSet="[0-9]*"
3440     function="TABLELOOKUP(companyprefixindex, tdt64bitcpi, companyprefixindex, companyprefix)"
3441     tableURI="http://www.onsepc.com/ManagerTranslation.xml"

```

```

3442 tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
3443 tableParams="companyprefixindex"/>
3444 <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="companyprefixlength"
3445 characterSet="[0-9]*" function="LENGTH(companyprefix)"/>
3446 <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="companyprefixindex"
3447 characterSet="[0-9]*"
3448 function="TABLELOOKUP(companyprefix,tdt64bitcpi,companyprefix,companyprefixindex)"
3449 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
3450 tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index" tableParams="companyprefix"/>
3451 </level>
3452 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:grai-64"
3453 requiredFormattingParameters="filter,taglength" >
3454 <option optionKey="12" pattern="urn:epc:tag:grai-64:([0-7]{1})\\.([0-9]{12})\\.([0-
3455 9]{0})\\.([0-9]*)" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3456 '.' serial" >
3457 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
3458 length="1" padChar="0" padDir="LEFT" name="filter"/>
3459 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3460 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3461 <field seq="3" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT"
3462 name="assettype"/>
3463 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3464 name="serial"/>
3465 </option>
3466 <option optionKey="11" pattern="urn:epc:tag:grai-64:([0-7]{1})\\.([0-9]{11})\\.([0-
3467 9]{1})\\.([0-9]*)" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3468 '.' serial" >
3469 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
3470 length="1" padChar="0" padDir="LEFT" name="filter"/>
3471 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
3472 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3473 <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]*"
3474 length="1" padChar="0" padDir="LEFT" name="assettype"/>
3475 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3476 name="serial"/>
3477 </option>
3478 <option optionKey="10" pattern="urn:epc:tag:grai-64:([0-7]{1})\\.([0-9]{10})\\.([0-
3479 9]{2})\\.([0-9]*)" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3480 '.' serial" >
3481 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
3482 length="1" padChar="0" padDir="LEFT" name="filter"/>
3483 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
3484 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3485 <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]*"
3486 length="2" padChar="0" padDir="LEFT" name="assettype"/>
3487 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3488 name="serial"/>
3489 </option>
3490 <option optionKey="9" pattern="urn:epc:tag:grai-64:([0-7]{1})\\.([0-9]{9})\\.([0-
3491 9]{3})\\.([0-9]*)" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3492 '.' serial" >
3493 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
3494 length="1" padChar="0" padDir="LEFT" name="filter"/>
3495 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
3496 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3497 <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]*"
3498 length="3" padChar="0" padDir="LEFT" name="assettype"/>
3499 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3500 name="serial"/>
3501 </option>
3502 <option optionKey="8" pattern="urn:epc:tag:grai-64:([0-7]{1})\\.([0-9]{8})\\.([0-
3503 9]{4})\\.([0-9]*)" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3504 '.' serial" >
3505 <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
3506 length="1" padChar="0" padDir="LEFT" name="filter"/>
3507 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
3508 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3509 <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]*"
3510 length="4" padChar="0" padDir="LEFT" name="assettype"/>
3511 <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]*"
3512 name="serial"/>

```

```

3513     </option>
3514     <option optionKey="7" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{7}\.[0-
3515 9]{5})\.[0-9]*" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3516 '.' serial" >
3517         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3518 length="1" padChar="0" padDir="LEFT" name="filter"/>
3519         <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
3520 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3521         <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"
3522 length="5" padChar="0" padDir="LEFT" name="assettype"/>
3523         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3524 name="serial"/>
3525     </option>
3526     <option optionKey="6" pattern="urn:epc:tag:grai-64:([0-7]{1})\.[0-9]{6}\.[0-
3527 9]{6})\.[0-9]*" grammar="urn:epc:tag:grai-64:' filter '.' companyprefix '.' assettype
3528 '.' serial" >
3529         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
3530 length="1" padChar="0" padDir="LEFT" name="filter"/>
3531         <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
3532 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3533         <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
3534 length="6" padChar="0" padDir="LEFT" name="assettype"/>
3535         <field seq="4" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3536 name="serial"/>
3537     </option>
3538 </level>
3539 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:grai" >
3540     <option optionKey="12" pattern="urn:epc:id:grai:([0-9]{12})\.[0-9]{0}\.[0-9]*"
3541 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3542         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3543 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3544         <field seq="2" characterSet="[0-9]*" length="0" padChar="0" padDir="LEFT"
3545 name="assettype"/>
3546         <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3547 name="serial"/>
3548     </option>
3549     <option optionKey="11" pattern="urn:epc:id:grai:([0-9]{11})\.[0-9]{1}\.[0-9]*"
3550 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3551         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
3552 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3553         <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"
3554 length="1" padChar="0" padDir="LEFT" name="assettype"/>
3555         <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3556 name="serial"/>
3557     </option>
3558     <option optionKey="10" pattern="urn:epc:id:grai:([0-9]{10})\.[0-9]{2}\.[0-9]*"
3559 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3560         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
3561 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3562         <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"
3563 length="2" padChar="0" padDir="LEFT" name="assettype"/>
3564         <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3565 name="serial"/>
3566     </option>
3567     <option optionKey="9" pattern="urn:epc:id:grai:([0-9]{9})\.[0-9]{3}\.[0-9]*"
3568 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3569         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
3570 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3571         <field seq="2" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"
3572 length="3" padChar="0" padDir="LEFT" name="assettype"/>
3573         <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3574 name="serial"/>
3575     </option>
3576     <option optionKey="8" pattern="urn:epc:id:grai:([0-9]{8})\.[0-9]{4}\.[0-9]*"
3577 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3578         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
3579 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3580         <field seq="2" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"
3581 length="4" padChar="0" padDir="LEFT" name="assettype"/>
3582         <field seq="3" decimalMinimum="0" decimalMaximum="524287" characterSet="[0-9]"
3583 name="serial"/>

```

[illegible]

```

    <rule type="EXTRACT" inputFormat="STRING" seq="4" newFieldName="companyprefix"
characterSet="[0-9]" function="SUBSTR(graiprefixremainder,0,companyprefixlength)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="graiprefix"
characterSet="[0-9]" length="13" function="CONCAT(0,companyprefix,assettype)"/>
    <rule type="FORMAT" inputFormat="STRING" seq="2" newFieldName="checkdigit"
characterSet="[0-9]" length="1" function="GS1CHECKSUM(graiprefix)"/>
  </level>
</scheme>
</epcTagDataTranslation>

```

8.9. GRAI-96 Global Returnable Asset Identifier for 96-bit EPCs

```

<?xml version="1.0" encoding="UTF-8"?>
<epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
xmlns:xsi="http://www.w3.org/2001/XMLSchema"
xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
  <scheme name="GRAI-96" optionKey="companyprefixlength" tagLength="96" >
    <level type="BINARY" prefixMatch="00110011"
requiredFormattingParameters="filter,taglength" >
      <option optionKey="12" pattern="00110011([01]{3})000([01]{40})([01]{4})([01]{38})"
grammar="'00110011' filter '000' companyprefix assettype serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]"
bitLength="40" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
        <field seq="3" characterSet="[01]" bitLength="4" length="0" padChar="0"
padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"
bitLength="38" name="serial"/>
      </option>
      <option optionKey="11" pattern="00110011([01]{3})001([01]{37})([01]{7})([01]{38})"
grammar="'00110011' filter '001' companyprefix assettype serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]"
bitLength="37" length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[01]"
bitLength="7" length="1" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"
bitLength="38" name="serial"/>
      </option>
      <option optionKey="10" pattern="00110011([01]{3})010([01]{34})([01]{10})([01]{38})"
grammar="'00110011' filter '010' companyprefix assettype serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]"
bitLength="34" length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[01]"
bitLength="10" length="2" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"
bitLength="38" name="serial"/>
      </option>
      <option optionKey="9" pattern="00110011([01]{3})011([01]{30})([01]{14})([01]{38})"
grammar="'00110011' filter '011' companyprefix assettype serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
        <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[01]"
bitLength="30" length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
        <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[01]"
bitLength="14" length="3" padChar="0" padDir="LEFT" name="assettype"/>
        <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"
bitLength="38" name="serial"/>
      </option>
      <option optionKey="8" pattern="00110011([01]{3})100([01]{27})([01]{17})([01]{38})"
grammar="'00110011' filter '100' companyprefix assettype serial" >
        <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

3720     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"*
3721 bitLength="27" length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3722     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[01]"*
3723 bitLength="17" length="4" padChar="0" padDir="LEFT" name="assettype"/>
3724     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"*
3725 bitLength="38" name="serial"/>
3726 </option>
3727 <option optionKey="7" pattern="00110011([01]{3})101([01]{24})([01]{20})([01]{38})"
3728 grammar="'00110011' filter '101' companyprefix assettype serial" >
3729     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3730 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3731     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"*
3732 bitLength="24" length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3733     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[01]"*
3734 bitLength="20" length="5" padChar="0" padDir="LEFT" name="assettype"/>
3735     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"*
3736 bitLength="38" name="serial"/>
3737 </option>
3738 <option optionKey="6" pattern="00110011([01]{3})110([01]{20})([01]{24})([01]{38})"
3739 grammar="'00110011' filter '110' companyprefix assettype serial" >
3740     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3741 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3742     <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"*
3743 bitLength="20" length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3744     <field seq="3" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"*
3745 bitLength="24" length="6" padChar="0" padDir="LEFT" name="assettype"/>
3746     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[01]"*
3747 bitLength="38" name="serial"/>
3748 </option>
3749 </level>
3750 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:grai-96"
3751 requiredFormattingParameters="filter,taglength" >
3752     <option optionKey="12" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[([0-9]{12})\.[([0-
3753 9]{0})\.[([0-9]*)" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3754 '.' serial" >
3755         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3756 length="1" padChar="0" padDir="LEFT" name="filter"/>
3757         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
3758 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3759         <field seq="3" characterSet="[0-9]"* length="0" padChar="0" padDir="LEFT"
3760 name="assettype"/>
3761         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3762 9]"* name="serial"/>
3763     </option>
3764     <option optionKey="11" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[([0-9]{11})\.[([0-
3765 9]{1})\.[([0-9]*)" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3766 '.' serial" >
3767         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3768 length="1" padChar="0" padDir="LEFT" name="filter"/>
3769         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
3770 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3771         <field seq="3" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
3772 length="1" padChar="0" padDir="LEFT" name="assettype"/>
3773         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3774 9]"* name="serial"/>
3775     </option>
3776     <option optionKey="10" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[([0-9]{10})\.[([0-
3777 9]{2})\.[([0-9]*)" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3778 '.' serial" >
3779         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3780 length="1" padChar="0" padDir="LEFT" name="filter"/>
3781         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
3782 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3783         <field seq="3" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"*
3784 length="2" padChar="0" padDir="LEFT" name="assettype"/>
3785         <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3786 9]"* name="serial"/>
3787     </option>
3788     <option optionKey="9" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[([0-9]{9})\.[([0-
3789 9]{3})\.[([0-9]*)" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3790 '.' serial" >

```



```

3791     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3792 length="1" padChar="0" padDir="LEFT" name="filter"/>
3793     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
3794 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
3795     <field seq="3" decimalMinimum="0" decimalMaximum="999" characterSet="[0-9]"*
3796 length="3" padChar="0" padDir="LEFT" name="assettype"/>
3797     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3798 9]"* name="serial"/>
3799 </option>
3800 <option optionKey="8" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{8})\.[0-
3801 9]{4})\.[0-9]" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3802 '.' serial" >
3803     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3804 length="1" padChar="0" padDir="LEFT" name="filter"/>
3805     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
3806 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
3807     <field seq="3" decimalMinimum="0" decimalMaximum="9999" characterSet="[0-9]"*
3808 length="4" padChar="0" padDir="LEFT" name="assettype"/>
3809     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3810 9]"* name="serial"/>
3811 </option>
3812 <option optionKey="7" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{7})\.[0-
3813 9]{5})\.[0-9]" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3814 '.' serial" >
3815     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3816 length="1" padChar="0" padDir="LEFT" name="filter"/>
3817     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
3818 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
3819     <field seq="3" decimalMinimum="0" decimalMaximum="99999" characterSet="[0-9]"*
3820 length="5" padChar="0" padDir="LEFT" name="assettype"/>
3821     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3822 9]"* name="serial"/>
3823 </option>
3824 <option optionKey="6" pattern="urn:epc:tag:grai-96:([0-7]{1})\.[0-9]{6})\.[0-
3825 9]{6})\.[0-9]" grammar="urn:epc:tag:grai-96:' filter '.' companyprefix '.' assettype
3826 '.' serial" >
3827     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
3828 length="1" padChar="0" padDir="LEFT" name="filter"/>
3829     <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"*
3830 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
3831     <field seq="3" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"*
3832 length="6" padChar="0" padDir="LEFT" name="assettype"/>
3833     <field seq="4" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3834 9]"* name="serial"/>
3835 </option>
3836 </level>
3837 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:grai" >
3838     <option optionKey="12" pattern="urn:epc:id:grai:([0-9]{12})\.[0-9]{0})\.[0-9]"*
3839 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3840     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-
3841 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
3842     <field seq="2" characterSet="[0-9]"* length="0" padChar="0" padDir="LEFT"
3843 name="assettype"/>
3844     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3845 9]"* name="serial"/>
3846 </option>
3847 <option optionKey="11" pattern="urn:epc:id:grai:([0-9]{11})\.[0-9]{1})\.[0-9]"*
3848 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3849     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]"*
3850 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
3851     <field seq="2" decimalMinimum="0" decimalMaximum="9" characterSet="[0-9]"*
3852 length="1" padChar="0" padDir="LEFT" name="assettype"/>
3853     <field seq="3" decimalMinimum="0" decimalMaximum="274877906943" characterSet="[0-
3854 9]"* name="serial"/>
3855 </option>
3856 <option optionKey="10" pattern="urn:epc:id:grai:([0-9]{10})\.[0-9]{2})\.[0-9]"*
3857 grammar="urn:epc:id:grai:' companyprefix '.' assettype '.' serial" >
3858     <field seq="1" decimalMinimum="0" decimalMaximum="9999999999999" characterSet="[0-9]"*
3859 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
3860     <field seq="2" decimalMinimum="0" decimalMaximum="99" characterSet="[0-9]"*
3861 length="2" padChar="0" padDir="LEFT" name="assettype"/>

```


[illegible]

```

3962 <?xml version="1.0" encoding="UTF-8"?>
3963 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
3964 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
3965 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
3966   <scheme name="GIAI-64" optionKey="companyprefixlength" tagLength="64" >
3967     <level type="BINARY" prefixMatch="00001011"
3968     requiredFormattingParameters="filter,taglength" >
3969       <option optionKey="12" pattern="00001011([01]{3})([01]{14})([01]{39})"
3970       grammar="'00001011' filter companyprefixindex indasassetref" >
3971         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3972         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3973         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
3974         bitLength="14" name="companyprefixindex"/>
3975         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
3976         bitLength="39" length="12" padChar="0" padDir="LEFT" name="indasassetref"/>
3977       </option>
3978       <option optionKey="11" pattern="00001011([01]{3})([01]{14})([01]{39})"
3979       grammar="'00001011' filter companyprefixindex indasassetref" >
3980         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3981         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3982         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
3983         bitLength="14" name="companyprefixindex"/>
3984         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
3985         bitLength="39" length="13" padChar="0" padDir="LEFT" name="indasassetref"/>
3986       </option>
3987       <option optionKey="10" pattern="00001011([01]{3})([01]{14})([01]{39})"
3988       grammar="'00001011' filter companyprefixindex indasassetref" >
3989         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3990         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
3991         <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
3992         bitLength="14" name="companyprefixindex"/>
3993         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
3994         bitLength="39" length="14" padChar="0" padDir="LEFT" name="indasassetref"/>
3995       </option>
3996       <option optionKey="9" pattern="00001011([01]{3})([01]{14})([01]{39})"
3997       grammar="'00001011' filter companyprefixindex indasassetref" >
3998         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
3999         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

4000     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
4001 bitLength="14" name="companyprefixindex"/>
4002     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
4003 bitLength="39" length="15" padChar="0" padDir="LEFT" name="indassetref"/>
4004     </option>
4005     <option optionKey="8" pattern="00001011([01]{3})([01]{14})([01]{39})"
4006 grammar="'00001011' filter companyprefixindex indassetref" >
4007     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
4008 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4009     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
4010 bitLength="14" name="companyprefixindex"/>
4011     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
4012 bitLength="39" length="16" padChar="0" padDir="LEFT" name="indassetref"/>
4013     </option>
4014     <option optionKey="7" pattern="00001011([01]{3})([01]{14})([01]{39})"
4015 grammar="'00001011' filter companyprefixindex indassetref" >
4016     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
4017 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4018     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
4019 bitLength="14" name="companyprefixindex"/>
4020     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
4021 bitLength="39" length="17" padChar="0" padDir="LEFT" name="indassetref"/>
4022     </option>
4023     <option optionKey="6" pattern="00001011([01]{3})([01]{14})([01]{39})"
4024 grammar="'00001011' filter companyprefixindex indassetref" >
4025     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"*
4026 bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4027     <field seq="2" decimalMinimum="0" decimalMaximum="16383" characterSet="[01]"*
4028 bitLength="14" name="companyprefixindex"/>
4029     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[01]"*
4030 bitLength="39" length="18" padChar="0" padDir="LEFT" name="indassetref"/>
4031     </option>
4032     <rule type="EXTRACT" inputFormat="STRING" seq="1" newFieldName="companyprefix"
4033 characterSet="[0-9]"*
4034 function="TABLELOOKUP(companyprefixindex, tdt64bitcpi, companyprefixindex, companyprefix)"
4035 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
4036 tableXPath="/GEPC64Table/entry[@index='$1']/@companyPrefix"
4037 tableParams="companyprefixindex"/>
4038     <rule type="EXTRACT" inputFormat="STRING" seq="2" newFieldName="companyprefixlength"
4039 characterSet="[0-9]"* function="LENGTH(companyprefix)"/>
4040     <rule type="FORMAT" inputFormat="STRING" seq="1" newFieldName="companyprefixindex"
4041 characterSet="[0-9]"*
4042 function="TABLELOOKUP(companyprefix, tdt64bitcpi, companyprefix, companyprefixindex)"
4043 tableURI="http://www.onsepc.com/ManagerTranslation.xml"
4044 tableXPath="/GEPC64Table/entry[@companyPrefix='$1']/@index" tableParams="companyprefix"/>
4045     </level>
4046     <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:giai-64"
4047 requiredFormattingParameters="filter, taglength" >
4048     <option optionKey="12" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[([0-9]{12})]\.[([0-
4049 9]{12})" grammar="urn:epc:tag:giai-64: filter '.' companyprefix '.' indassetref" >
4050     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
4051 length="1" padChar="0" padDir="LEFT" name="filter"/>
4052     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4053 9]"* length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
4054     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4055 9]"* length="12" padChar="0" padDir="LEFT" name="indassetref"/>
4056     </option>
4057     <option optionKey="11" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[([0-9]{11})]\.[([0-
4058 9]{13})" grammar="urn:epc:tag:giai-64: filter '.' companyprefix '.' indassetref" >
4059     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
4060 length="1" padChar="0" padDir="LEFT" name="filter"/>
4061     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"*
4062 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
4063     <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4064 9]"* length="13" padChar="0" padDir="LEFT" name="indassetref"/>
4065     </option>
4066     <option optionKey="10" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[([0-9]{10})]\.[([0-
4067 9]{14})" grammar="urn:epc:tag:giai-64: filter '.' companyprefix '.' indassetref" >
4068     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"*
4069 length="1" padChar="0" padDir="LEFT" name="filter"/>

```

```

4070         <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
4071 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
4072         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4073 9]" length="14" padChar="0" padDir="LEFT" name="indassetref"/>
4074     </option>
4075     <option optionKey="9" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{9})\.[0-
4076 9]{15}" grammar="urn:epc:tag:giai-64:' filter '.' companyprefix '.' indassetref" >
4077         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
4078 length="1" padChar="0" padDir="LEFT" name="filter"/>
4079         <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
4080 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
4081         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4082 9]" length="15" padChar="0" padDir="LEFT" name="indassetref"/>
4083     </option>
4084     <option optionKey="8" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{8})\.[0-
4085 9]{16}" grammar="urn:epc:tag:giai-64:' filter '.' companyprefix '.' indassetref" >
4086         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
4087 length="1" padChar="0" padDir="LEFT" name="filter"/>
4088         <field seq="2" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]"
4089 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
4090         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4091 9]" length="16" padChar="0" padDir="LEFT" name="indassetref"/>
4092     </option>
4093     <option optionKey="7" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{7})\.[0-
4094 9]{17}" grammar="urn:epc:tag:giai-64:' filter '.' companyprefix '.' indassetref" >
4095         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
4096 length="1" padChar="0" padDir="LEFT" name="filter"/>
4097         <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
4098 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
4099         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4100 9]" length="17" padChar="0" padDir="LEFT" name="indassetref"/>
4101     </option>
4102     <option optionKey="6" pattern="urn:epc:tag:giai-64:([0-7]{1})\.[0-9]{6})\.[0-
4103 9]{18}" grammar="urn:epc:tag:giai-64:' filter '.' companyprefix '.' indassetref" >
4104         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]"
4105 length="1" padChar="0" padDir="LEFT" name="filter"/>
4106         <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]"
4107 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
4108         <field seq="3" decimalMinimum="0" decimalMaximum="549755813887" characterSet="[0-
4109 9]" length="18" padChar="0" padDir="LEFT" name="indassetref"/>
4110     </option>
4111 </level>
4112 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:giai" >
4113     <option optionKey="12" pattern="urn:epc:id:giai:([0-9]{12})\.[0-9]{12}"
4114 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4115         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4116 9]" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
4117         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4118 9]" length="12" padChar="0" padDir="LEFT" name="indassetref"/>
4119     </option>
4120     <option optionKey="11" pattern="urn:epc:id:giai:([0-9]{11})\.[0-9]{13}"
4121 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4122         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
4123 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
4124         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
4125 9]" length="13" padChar="0" padDir="LEFT" name="indassetref"/>
4126     </option>
4127     <option optionKey="10" pattern="urn:epc:id:giai:([0-9]{10})\.[0-9]{14}"
4128 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4129         <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]"
4130 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
4131         <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999" characterSet="[0-
4132 9]" length="14" padChar="0" padDir="LEFT" name="indassetref"/>
4133     </option>
4134     <option optionKey="9" pattern="urn:epc:id:giai:([0-9]{9})\.[0-9]{15}"
4135 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4136         <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]"
4137 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
4138         <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999999" characterSet="[0-
4139 9]" length="15" padChar="0" padDir="LEFT" name="indassetref"/>
4140     </option>

```



```

4209 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
4210 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
4211 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
4212   <scheme name="GIAI-96" optionKey="companyprefixlength" tagLength="96" >
4213     <level type="BINARY" prefixMatch="00110100"
4214       requiredFormattingParameters="filter,taglength" >
4215       <option optionKey="12" pattern="00110100([01]{3})000([01]{40})([01]{42})"
4216       grammar="'00110100' filter '000' companyprefix indassetref" >
4217         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4218         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4219         <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"
4220         bitLength="40" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
4221         <field seq="3" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"
4222         bitLength="42" length="12" padChar="0" padDir="LEFT" name="indassetref"/>
4223       </option>
4224       <option optionKey="11" pattern="00110100([01]{3})001([01]{37})([01]{45})"
4225       grammar="'00110100' filter '001' companyprefix indassetref" >
4226         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4227         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4228         <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"
4229         bitLength="37" length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
4230         <field seq="3" decimalMinimum="0" decimalMaximum="999999999999"
4231         characterSet="[01]" bitLength="45" length="13" padChar="0" padDir="LEFT"
4232         name="indassetref"/>
4233       </option>
4234       <option optionKey="10" pattern="00110100([01]{3})010([01]{34})([01]{48})"
4235       grammar="'00110100' filter '010' companyprefix indassetref" >
4236         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4237         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4238         <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"
4239         bitLength="34" length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
4240         <field seq="3" decimalMinimum="0" decimalMaximum="999999999999"
4241         characterSet="[01]" bitLength="48" length="14" padChar="0" padDir="LEFT"
4242         name="indassetref"/>
4243       </option>
4244       <option optionKey="9" pattern="00110100([01]{3})011([01]{30})([01]{52})"
4245       grammar="'00110100' filter '011' companyprefix indassetref" >
4246         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4247         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4248         <field seq="2" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[01]"
4249         bitLength="30" length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
4250         <field seq="3" decimalMinimum="0" decimalMaximum="99999999999999"
4251         characterSet="[01]" bitLength="52" length="15" padChar="0" padDir="LEFT"
4252         name="indassetref"/>
4253       </option>
4254       <option optionKey="8" pattern="00110100([01]{3})100([01]{27})([01]{55})"
4255       grammar="'00110100' filter '100' companyprefix indassetref" >
4256         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4257         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4258         <field seq="2" decimalMinimum="0" decimalMaximum="99999999" characterSet="[01]"
4259         bitLength="27" length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
4260         <field seq="3" decimalMinimum="0" decimalMaximum="9999999999999999"
4261         characterSet="[01]" bitLength="55" length="16" padChar="0" padDir="LEFT"
4262         name="indassetref"/>
4263       </option>
4264       <option optionKey="7" pattern="00110100([01]{3})101([01]{24})([01]{58})"
4265       grammar="'00110100' filter '101' companyprefix indassetref" >
4266         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4267         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4268         <field seq="2" decimalMinimum="0" decimalMaximum="9999999" characterSet="[01]"
4269         bitLength="24" length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
4270         <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999"
4271         characterSet="[01]" bitLength="58" length="17" padChar="0" padDir="LEFT"
4272         name="indassetref"/>
4273       </option>
4274       <option optionKey="6" pattern="00110100([01]{3})110([01]{20})([01]{62})"
4275       grammar="'00110100' filter '110' companyprefix indassetref" >
4276         <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[01]"
4277         bitLength="3" length="1" padChar="0" padDir="LEFT" name="filter"/>
4278         <field seq="2" decimalMinimum="0" decimalMaximum="999999" characterSet="[01]"
4279         bitLength="20" length="6" padChar="0" padDir="LEFT" name="companyprefix"/>

```

```

4280     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999"
4281 characterSet="[01]*" bitLength="62" length="18" padChar="0" padDir="LEFT"
4282 name="indassetref"/>
4283 </option>
4284 </level>
4285 <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:giai-96"
4286 requiredFormattingParameters="filter,taglength" >
4287   <option optionKey="12" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{12})\.([0-
4288 9]{12})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4289     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4290 length="1" padChar="0" padDir="LEFT" name="filter"/>
4291     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4292 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
4293     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4294 9]*" length="12" padChar="0" padDir="LEFT" name="indassetref"/>
4295   </option>
4296   <option optionKey="11" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{11})\.([0-
4297 9]{13})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4298     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4299 length="1" padChar="0" padDir="LEFT" name="filter"/>
4300     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4301 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
4302     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4303 9]*" length="13" padChar="0" padDir="LEFT" name="indassetref"/>
4304   </option>
4305   <option optionKey="10" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{10})\.([0-
4306 9]{14})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4307     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4308 length="1" padChar="0" padDir="LEFT" name="filter"/>
4309     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4310 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
4311     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4312 9]*" length="14" padChar="0" padDir="LEFT" name="indassetref"/>
4313   </option>
4314   <option optionKey="9" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{9})\.([0-
4315 9]{15})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4316     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4317 length="1" padChar="0" padDir="LEFT" name="filter"/>
4318     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4319 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
4320     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999" characterSet="[0-
4321 9]*" length="15" padChar="0" padDir="LEFT" name="indassetref"/>
4322   </option>
4323   <option optionKey="8" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{8})\.([0-
4324 9]{16})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4325     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4326 length="1" padChar="0" padDir="LEFT" name="filter"/>
4327     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4328 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
4329     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999"
4330 characterSet="[0-9]*" length="16" padChar="0" padDir="LEFT" name="indassetref"/>
4331   </option>
4332   <option optionKey="7" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{7})\.([0-
4333 9]{17})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4334     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4335 length="1" padChar="0" padDir="LEFT" name="filter"/>
4336     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4337 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
4338     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999"
4339 characterSet="[0-9]*" length="17" padChar="0" padDir="LEFT" name="indassetref"/>
4340   </option>
4341   <option optionKey="6" pattern="urn:epc:tag:giai-96:([0-7]{1})\.([0-9]{6})\.([0-
4342 9]{18})" grammar="urn:epc:tag:giai-96:' filter '.' companyprefix '.' indassetref" >
4343     <field seq="1" decimalMinimum="0" decimalMaximum="7" characterSet="[0-7]*"
4344 length="1" padChar="0" padDir="LEFT" name="filter"/>
4345     <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4346 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
4347     <field seq="3" decimalMinimum="0" decimalMaximum="999999999999999999"
4348 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="indassetref"/>
4349   </option>
4350 </level>

```



```

4351 <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:giai" >
4352 <option optionKey="12" pattern="urn:epc:id:giai:([0-9]{12})\.([0-9]{12})"
4353 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4354 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4355 9]*" length="12" padChar="0" padDir="LEFT" name="companyprefix"/>
4356 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4357 9]*" length="12" padChar="0" padDir="LEFT" name="indassetref"/>
4358 </option>
4359 <option optionKey="11" pattern="urn:epc:id:giai:([0-9]{11})\.([0-9]{13})"
4360 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4361 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-9]*"
4362 length="11" padChar="0" padDir="LEFT" name="companyprefix"/>
4363 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999" characterSet="[0-
4364 9]*" length="13" padChar="0" padDir="LEFT" name="indassetref"/>
4365 </option>
4366 <option optionKey="10" pattern="urn:epc:id:giai:([0-9]{10})\.([0-9]{14})"
4367 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4368 <field seq="1" decimalMinimum="0" decimalMaximum="9999999999" characterSet="[0-9]*"
4369 length="10" padChar="0" padDir="LEFT" name="companyprefix"/>
4370 <field seq="2" decimalMinimum="0" decimalMaximum="99999999999999" characterSet="[0-
4371 9]*" length="14" padChar="0" padDir="LEFT" name="indassetref"/>
4372 </option>
4373 <option optionKey="9" pattern="urn:epc:id:giai:([0-9]{9})\.([0-9]{15})"
4374 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4375 <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
4376 length="9" padChar="0" padDir="LEFT" name="companyprefix"/>
4377 <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999" characterSet="[0-
4378 9]*" length="15" padChar="0" padDir="LEFT" name="indassetref"/>
4379 </option>
4380 <option optionKey="8" pattern="urn:epc:id:giai:([0-9]{8})\.([0-9]{16})"
4381 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4382 <field seq="1" decimalMinimum="0" decimalMaximum="999999999" characterSet="[0-9]*"
4383 length="8" padChar="0" padDir="LEFT" name="companyprefix"/>
4384 <field seq="2" decimalMinimum="0" decimalMaximum="9999999999999999"
4385 characterSet="[0-9]*" length="16" padChar="0" padDir="LEFT" name="indassetref"/>
4386 </option>
4387 <option optionKey="7" pattern="urn:epc:id:giai:([0-9]{7})\.([0-9]{17})"
4388 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4389 <field seq="1" decimalMinimum="0" decimalMaximum="9999999" characterSet="[0-9]*"
4390 length="7" padChar="0" padDir="LEFT" name="companyprefix"/>
4391 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999999"
4392 characterSet="[0-9]*" length="17" padChar="0" padDir="LEFT" name="indassetref"/>
4393 </option>
4394 <option optionKey="6" pattern="urn:epc:id:giai:([0-9]{6})\.([0-9]{18})"
4395 grammar="urn:epc:id:giai:' companyprefix '.' indassetref" >
4396 <field seq="1" decimalMinimum="0" decimalMaximum="999999" characterSet="[0-9]*"
4397 length="6" padChar="0" padDir="LEFT" name="companyprefix"/>
4398 <field seq="2" decimalMinimum="0" decimalMaximum="999999999999999999"
4399 characterSet="[0-9]*" length="18" padChar="0" padDir="LEFT" name="indassetref"/>
4400 </option>
4401 </level>
4402 <level type="LEGACY" prefixMatch="giai="
4403 requiredParsingParameters="companyprefixlength" >
4404 <option optionKey="12" pattern="giai=([0-9]{13,30})" grammar="'giai=' companyprefix
4405 indassetref" >
4406 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999999999999999"
4407 characterSet="[0-9]*" name="giai"/>
4408 </option>
4409 <option optionKey="11" pattern="giai=([0-9]{12,30})" grammar="'giai=' companyprefix
4410 indassetref" >
4411 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999999999999999"
4412 characterSet="[0-9]*" name="giai"/>
4413 </option>
4414 <option optionKey="10" pattern="giai=([0-9]{11,30})" grammar="'giai=' companyprefix
4415 indassetref" >
4416 <field seq="1" decimalMinimum="0" decimalMaximum="999999999999999999999999999999"
4417 characterSet="[0-9]*" name="giai"/>
4418 </option>
4419 <option optionKey="9" pattern="giai=([0-9]{10,30})" grammar="'giai=' companyprefix
4420 indassetref" >

```

[illegible]

```

4450 <?xml version="1.0" encoding="UTF-8"?>
4451 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
4452 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
4453 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
4454   <scheme name="USD00-64" optionKey="1" tagLength="64">
4455     <level type="BINARY" prefixMatch="11001110" requiredFormattingParameters="taglength">
4456       <option optionKey="1" pattern="11001110([01]{2})([01]{30})([01]{24})"
4457       grammar="11001110 filter cageordodaac serial">
4458         <field seq="1" decimalMinimum="0" decimalMaximum="3" characterSet="[01]"
4459         bitLength="2" name="filter"/>
4460         <field seq="2" characterSet="[01]" compaction="6-bit" length="5" padChar=" "
4461         padDir="LEFT" bitLength="30" name="cageordodaac"/>
4462         <field seq="3" decimalMinimum="0" decimalMaximum="16777215" characterSet="[01]"
4463         bitLength="24" name="serial"/>
4464       </option>
4465     </level>
4466     <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:usdod-64"
4467     requiredFormattingParameters="taglength">
4468       <option optionKey="1" pattern="urn:epc:tag:usdod-64:([0-9])\.[0-9 A-HJ-NP-
4469       Z]{5})\.[0-9+]" grammar="urn:epc:tag:usdod-64: filter '.' cageordodaac '.' serial">
4470         <field seq="1" decimalMinimum="0" decimalMaximum="3" characterSet="[0-3]"
4471         name="filter"/>
4472         <field seq="2" characterSet="[0-9 A-HJ-NP-Z]" name="cageordodaac"/>
4473         <field seq="3" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]"
4474         name="serial"/>
4475       </option>
4476     </level>
4477     <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:usdod">
4478       <option optionKey="1" pattern="urn:epc:id:usdod:([0-9 A-HJ-NP-Z]{5})\.[0-9+]"
4479       grammar="urn:epc:id:usdod: cageordodaac '.' serial">
4480         <field seq="1" characterSet="[0-9 A-HJ-NP-Z]" name="cageordodaac"/>
4481         <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]"
4482         name="serial"/>
4483       </option>
4484     </level>
4485     <level type="LEGACY" prefixMatch="cageordodaac">
4486       <option optionKey="1" pattern="cageordodaac([0-9 A-HJ-NP-Z]{5});serial=([0-9+)"
4487       grammar="cageordodaac=' cageordodaac ';serial=' serial">

```

```

4488         <field seq="1" characterSet="[0-9 A-HJ-NP-Z]*" name="cageordodaac"/>
4489         <field seq="2" decimalMinimum="0" decimalMaximum="16777215" characterSet="[0-9]*"
4490         name="serial"/>
4491     </option>
4492 </level>
4493 </scheme>
4494 </epcTagDataTranslation>
4495

```

4496 **8.13. USDOD-96** **DEPARTMENT OF DEFENCE for 96-bit** 4497 **EPCs**

```

4498 <?xml version="1.0" encoding="UTF-8"?>
4499 <epcTagDataTranslation version="0.04" date="2005-04-18T16:05Z" epcTDSVersion="1.1r1.27"
4500 xmlns:xsi="http://www.w3.org/2001/XMLSchema"
4501 xsi:noNamespaceSchemaLocation="EpcTagDataTranslation.xsd">
4502     <scheme name="USDOD-96" optionKey="1" tagLength="96" >
4503         <level type="BINARY" prefixMatch="00101111" requiredFormattingParameters="taglength" >
4504             <option optionKey="1" pattern="00101111([01]{4})([01]{48})([01]{36})"
4505             grammar="'00101111' filter cageordodaac serial" >
4506                 <field seq="1" decimalMinimum="0" decimalMaximum="15" characterSet="[01]*"
4507                 bitLength="4" name="filter"/>
4508                 <field seq="2" characterSet="[01]*" compaction="8-bit" padChar=" " padDir="LEFT"
4509                 length="6" bitLength="48" name="cageordodaac"/>
4510                 <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[01]*"
4511                 bitLength="36" name="serial"/>
4512             </option>
4513         </level>
4514         <level type="TAG_ENCODING" prefixMatch="urn:epc:tag:usdod-96"
4515         requiredFormattingParameters="taglength" >
4516             <option optionKey="1" pattern="urn:epc:tag:usdod-96:([0-9])\.[0-9 A-HJ-NP-
4517             Z]{5,6})\.[0-9]*" grammar="'urn:epc:tag:usdod-96:' filter '.' cageordodaac '.' serial"
4518             >
4519                 <field seq="1" decimalMinimum="0" decimalMaximum="15" characterSet="[0-9]*"
4520                 name="filter"/>
4521                 <field seq="2" characterSet="[0-9 A-HJ-NP-Z]*" name="cageordodaac"/>
4522                 <field seq="3" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]*"
4523                 name="serial"/>
4524             </option>
4525         </level>
4526         <level type="PURE_IDENTITY" prefixMatch="urn:epc:id:usdod" >
4527             <option optionKey="1" pattern="urn:epc:id:usdod:([0-9 A-HJ-NP-Z]{5,6})\.[0-9]*"
4528             grammar="'urn:epc:id:usdod:' cageordodaac '.' serial" >
4529                 <field seq="1" characterSet="[0-9 A-HJ-NP-Z]*" name="cageordodaac"/>
4530                 <field seq="2" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]*"
4531                 name="serial"/>
4532             </option>
4533         </level>
4534         <level type="LEGACY" prefixMatch="cageordodaac" >
4535             <option optionKey="1" pattern="cageordodaac([0-9 A-HJ-NP-Z]{5,6});serial=([0-9]+)"
4536             grammar="'cageordodaac=' cageordodaac ';' serial" >
4537                 <field seq="1" characterSet="[0-9 A-HJ-NP-Z]*" name="cageordodaac"/>
4538                 <field seq="2" decimalMinimum="0" decimalMaximum="68719476735" characterSet="[0-9]*"
4539                 name="serial"/>
4540             </option>
4541         </level>
4542     </scheme>
4543 </epcTagDataTranslation>

```

9. Glossary (non-normative)

This section provides a non-normative summary of terms used within this specification. For normative definitions of these terms, please consult the relevant sections of the document.

Term	Meaning
[Numbering/Coding] Scheme	A well-defined method of assigning an identification code to an object / shipment / location / transaction
Serialised	Provides a unique serial number for each unique object referenced using that coding scheme
GTIN	Global Trade Item Number – used to identify traded objects and services.
SSCC	Serial Shipping Container Code – provides a globally unique reference number for each shipment
GLN	Global Location Number – used to identify physical locations but also legal and organizational entities and departments
GRAI	Global Returnable Asset Identifier – used to identify returnable assets such as pallets and crates, gas cylinders, etc.
GIAI	Global Individual Asset Identifier – used to identify assets owned by an organisation, which are not being traded – often used for tracking inventory of high value equipment
GID	General Identifier – original hierarchical structure proposed for EPC by Auto-ID Centre. GID is a generic scheme, not specifically aligned with any particular legacy coding scheme.
Levels of Representation	The way in which the identifier is represented. Examples of different types of representation include sequences of binary digits (bits), sequences of numeric or alphanumeric characters, as well as Uniform Resource Identifiers (URIs)
Input Value	The identifier to be translated. The format in which it is expressed is the Inbound Representation.
Inbound representation	The way in which the identifier is supplied to the translation software. This may be auto-detected from the input value.
Outbound representation	The way in which the output from the translation software should be expressed. This must be specified by the client.
Binary	A sequence of binary digits or bits, consisting of only the digits '0' or '1'

Term	Meaning
Non-Binary Form	An integer, numeric or alphanumeric character string when not expressed in the corresponding binary format
URI / URN	A Uniform Resource Identifier / Uniform Resource Name – a string that uniquely identifies any particular object. Unlike a URL (Uniform Resource Locator) which may change when a web page moves from one website to another, the URI is intended to be a permanent reference, fixed for all time – even if the underlying binding to a particular website address changes. The URI is therefore at a higher level of abstraction than a URL. Currently most web browser technology will only resolve URLs – but not URIs.
Tag-Encoding URI	A URI format which encodes the physical tag length and fast-filter values in addition to the information encoded in the pure-identity URI. Intended for low-level applications – e.g. sorting machines, tag writers, etc.
Pure-Identity URI	A more abstract URI format that provides each object with a unique identity but conveys no information regarding the physical limitations of the tag used to deliver that EPC. <i>If an object is tagged with either a 64-bit tag or a 96-bit tag, then although the binary representation and tag-encoding URIs will differ, the pure-identity URI will be the same. Intended for use by high-level applications which are not concerned with writing to tags nor sorting on packaging level.</i>
Legacy Coding	Existing numbering/coding schemes already in use. Examples include GTIN, SSCC, GLN, GRAI, GIAI from the GS1 family – but also many other numbering schemes, such as VIN (Vehicle Identification Number) – or those managed by IATA for the aviation industry or the US DOD for defence.
Physical Level[s]	Representations where the encoding conveys information about the physical tag length (number of bits) and/or the packaging/classification level of the object. Specifically, the binary representation and tag-encoding URI.
Identity Level[s]	Higher-level representations that say nothing about the physical tag length, nor include explicit information about the packaging/classification level. Specifically the pure-identity URI and legacy coding levels of representation
Supplied parameters	Parameters that shall be supplied in addition to the input value, mainly because the input value itself lacks specific information required for constructing the output.

Term	Meaning
Options	Variations to handle variable-length data partitions, such as those resulting from the variable-length Company Prefix in the GS1 family of coding schemes. Where multiple options are specified, the same number of options should be specified for each level of representation and translation should always translate from the matching option within the inbound level to the corresponding option within the outbound level.
Regular Expression Pattern	A notation for representing sub-patterns of particular groups of characters to match
ABNF Grammar	Augmented Backus-Naur Form. Defined in RFC 2234. [http://www.ietf.org/rfc/rfc2234.txt] Notation indicating how the result can be expressed through a concatenation of fixed literal values and values of variable fields, whose values are previously determined.
[Fast] Filter	A number which is used to conveniently select only EPCs of a particular packaging level or classification – e.g. a filter within a smart reader may be configured to report only the cases and pallets – but not all of the items within those cases. The fast filter value may also be used for filtering and sorting.
Header	A binary EPC prefix which indicates the coding scheme and usually also the tag length. Headers of 2 bits and 8 bits are defined in the EPC Tag Data Standards specification
Field	The variable elements of the EPC in any of its representations – each partition or field has a logical role, such as identifying the responsible company (e.g. the manufacturer of a trade item) or the object class or SKU. Tag Data Translation software uses the regular expression pattern to extract values for each field. These may be temporarily stored in variables or an associative array (key-value lookup table) until they are later required for substitution into the outbound format.

Term	Meaning
Rules	There are already a number of requirements to perform various string manipulations and other calculations in order to comply with the current TDS specification. Neither the regular expression patterns nor the ABNF grammar contain any embedded inline functions. Instead, additional fields are embedded and a separate list of rules are provided, in order to define how their values should be derived from fields whose values are already known. The rules also indicate the context and running order in which they should be executed, namely by specifying the scheme, level and stage of execution (Extract or Format) and the running order as an integer index, with functions executed in ascending order of the sequence number indicated by the seq attribute
Prefix Match	The Prefix Match is a substring which is used to determine the scheme of the inbound string. This is merely a method of optimizing the performance of translation software by limiting the number of pattern-match tests that are required, since the translation software only attempts full pattern matching and processing for the options of those schemes/levels whose Prefix Match matches at the start of the input value.
OptionKey	The OptionKey is used to identify the appropriate option to use where multiple variations are specified to deal with partitions of variable length. A default strategy may be to simply iterate through all the possible options and find only one where the format string matches the inbound string. However, this approach fails when multiple options match the inbound value. In this case, the translation software can use the enumerated value of the OptionKey to select the appropriate option to use. Each option entry is numbered – and each level specifies (via the name of a field) the appropriate option to choose. For example for the GS1 codes, the level element always specifies that the OptionKey="companyprefixlength" , so for a Company Prefix of '0037000', then field "companyprefixlength" would be specified as 7 via the supplied parameters and therefore Option #7 would be chosen for both the inbound and outbound levels.
Encoding	A conversion process towards the binary representation, i.e in the direction: Legacy code → Pure-identity URI → Tag-encoding URI → Binary

Term	Meaning
Decoding	A conversion process away from the binary representation, i.e in the direction: Binary → Tag-encoding URI → Pure-identity URI → Legacy code → ONS hostname
Built-In Functions	Functions that should be supported by all implementations of the tag data translation software, irrespective of the programming language in which the software was actually written. See Table 6.
TDT XML Markup	A well-defined machine-readable structured packet of data that represents the patterns, grammar, rules, and field constraints for each identifier coding scheme. Tag data translation software should periodically receive updated versions or patches of the XML markup tables, which it can then use to update its own internal set of rules for performing the conversions, whether this is done at run-time or compile-time. We envisage that the XML Data Table should be freely downloadable and should ideally use human-readable tagnames. Furthermore, it should be possible to use XML transformation technologies such as XSLT to render it into a suitably formatted human-readable table for use in revised versions of the Tag Data Standards specification. Rendering the tables for the specification from the XML Data Table as the master table should avoid any inconsistencies being introduced between the TDS specification and the master table used by the translation software.
[EPC] [Tag Data] Translation Software	A piece of software that performs conversions between different representations of the EPC within any given coding scheme. The translation software may be a library module or object which may be accessed by / embedded within any technology component in the EPC Network technology stack. It may also be implemented as a standalone service, such as an interactive web page form or a web service for automated batch-processing of conversions.
EPC Tag Data Validation Software	Software which need not perform any translation but may nevertheless make use of the Tag Data Translation markup files in order to validate that an EPC in any of its representations conforms to a valid format.

Term	Meaning
EPC Network [Technology] Stack	This consists of several architectural building blocks in order to connect physical objects with information systems. The technology stack includes: EPC – the Electronic Product Code Tags and Readers Filtering and Collection middleware Object Name Service (ONS) EPC Information Service (EPCIS).
Checksum / Check Digit	A number that is computed algorithmically from other digits in a numerical code in order to perform a very basic check of the integrity of the number; if the check digit supplied does not correspond to the check digit calculated from the other digits, then the number may have been corrupted. The check digit is in a way analogous to a message digest of a data packet or software package – except that message digests tend to be more robust since they consist of strings of several characters and hence many more possible permutations than a single check digit 0-9, with the result that there is a much smaller probability that a corrupted number or data packet will product the same message digest than that it will fortuitously produce a valid check digit. The algorithm for computing the check digit for EAN.UCC GS1 coding schemes is specified at http://www.ean-int.org/cdcalcul.html ISO 7064 is a standard specifying a generic framework for check digit calculations.
[EAN.UCC] Company Prefix	A number allocated by the EAN.UCC which uniquely specifies a unique company – often the manufacturer of a trade item
Company Prefix Index	An integer used to obtain the full EAN.UCC Company Prefix via a lookup table, keyed on the smaller integer number of the Company Prefix Index. This is used with the 64-bit schemes in order to allocate a larger range of bits for the remaining data partitions. The Company Prefix Index is tabulated in XML and comma-separated value formats at http://www.onsepc.com

4548

10. References

- 4549
- 4550
- 4551 TDS - EPCglobal Tag Data Standards
- 4552 ONS - Object Name Service
- 4553 See <http://www.epcglobalinc.org/under 'Standards'>
- 4554
- 4555 GTIN – Global Trade Item Number
- 4556 GLN – Global Location Number
- 4557 SSCC – Serial Shipping Container Code
- 4558 GRAI – Global Returnable Asset Identifier
- 4559 GIAI – Global Individual Asset Identifier
- 4560 GS1 (formerly EAN UCC Company Prefix)
- 4561 GS1 Check Digit Calculation
- 4562 See <http://www.gs1.org> under 'The EAN.UCC System' > 'Identification'
- 4563
- 4564 US DOD / CAGE and DODAAC codes in passive tags
- 4565 See <http://www.acq.osd.mil/log/rfid/> under 'Passive Tag Data'
- 4566
- 4567 NAPTR – Naming Authority Pointer records
- 4568 See RFC2915 at <http://www.ietf.org/rfc/rfc2915.txt?number=2915>
- 4569
- 4570 PCRE – Perl-Compliant Regular Expressions
- 4571 See <http://www.pcre.org>
- 4572
- 4573 ABNF – Augmented Backus-Naur Form
- 4574 See RFC2234 at <http://www.ietf.org/rfc/rfc2234.txt?number=2234>
- 4575
- 4576 URI – Uniform Resource Identifiers
- 4577 See RFC2396 at <http://www.ietf.org/rfc/rfc2396.txt?number=2234>
- 4578
- 4579 CGI – Common Gateway Interface
- 4580 See <http://hoohoo.ncsa.uiuc.edu/cgi/>

4581
4582 UML – Unified Modelling Language
4583 See <http://www.uml.org/>
4584
4585 ISO AFI – Application Family Identifier
4586 See ISO/IEC 15693 and ISO/IEC 15961 and 15962
4587
4588 EPCglobal UHF Generation 2 Protocol
4589 See <http://www.epcglobalinc.org>
4590
4591 XML DOM (Document Object Model) and XPath
4592 See <http://www.w3.org/TR/xpath>